US005970125A

United States Patent [19]

Smith et al.

[11] **Patent Number: 5,970,125**

[45] **Date of Patent: Oct. 19, 1999**

[54] **METHOD, SYSTEM AND APPARATUS FOR CAUSING CUSTOMER PREMISES EQUIPMENT TO AUTOMATICALLY CALL A TELECOMMUNICATIONS SERVER**

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[73] Assignee: **Nortel Networks Corporation**, Montreal, Canada

[21] Appl. No.: **08/933,752**

[22] Filed: **Sep. 23, 1997**

[51] **Int. Cl.**[6] **H04M 11/00**

[52] **U.S. Cl.** **379/93.25**; 379/142

[58] **Field of Search** 379/90.01, 92.01, 379/92.03–93.08, 93.14, 93.17, 93.24–93.27, 100.08, 100.14–100.17, 142, 110.01, 354–357

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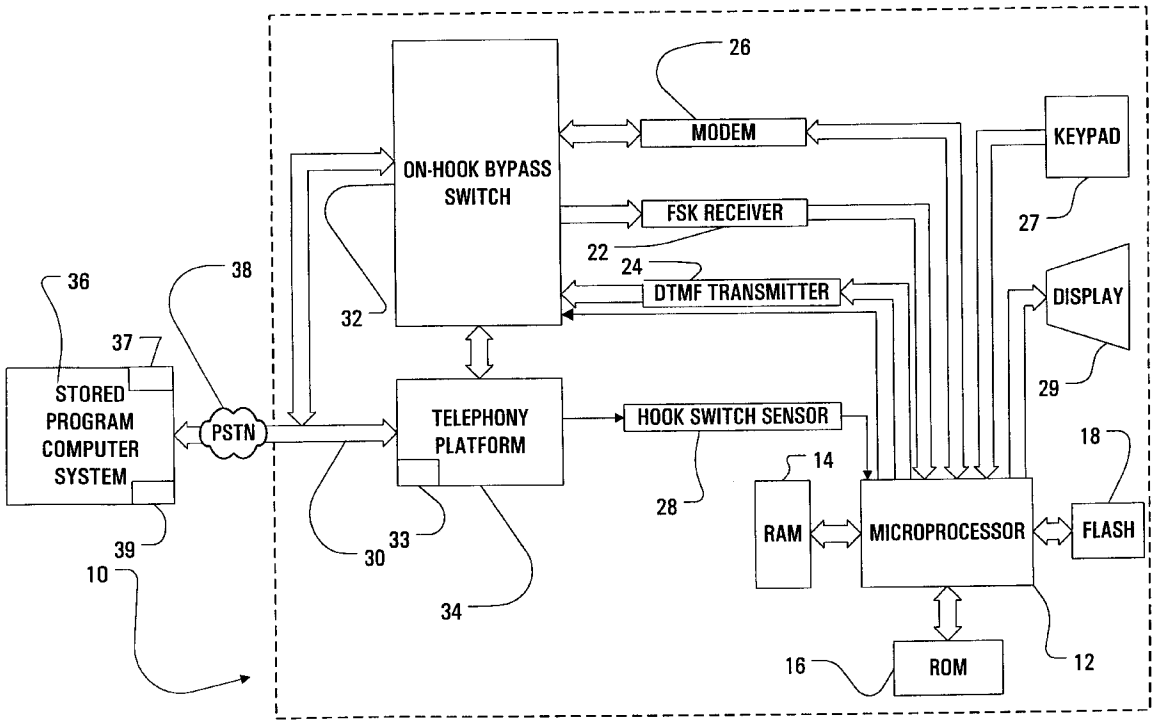
Primary Examiner—Wing F. Chan

[57] **ABSTRACT**

A system and method for providing messaging services at customer premises equipment. The system includes a messaging platform for receiving and storing messages, the messaging platform including a transmitter for receiving information request messages and a transmitter for transmitting information and customer premises equipment. The customer premises equipment includes a transmitter for transmitting an information request message from the CPE to the messaging platform and a receiver for receiving information from the messaging platform in response to the information request message.

30 Claims, 35 Drawing Sheets



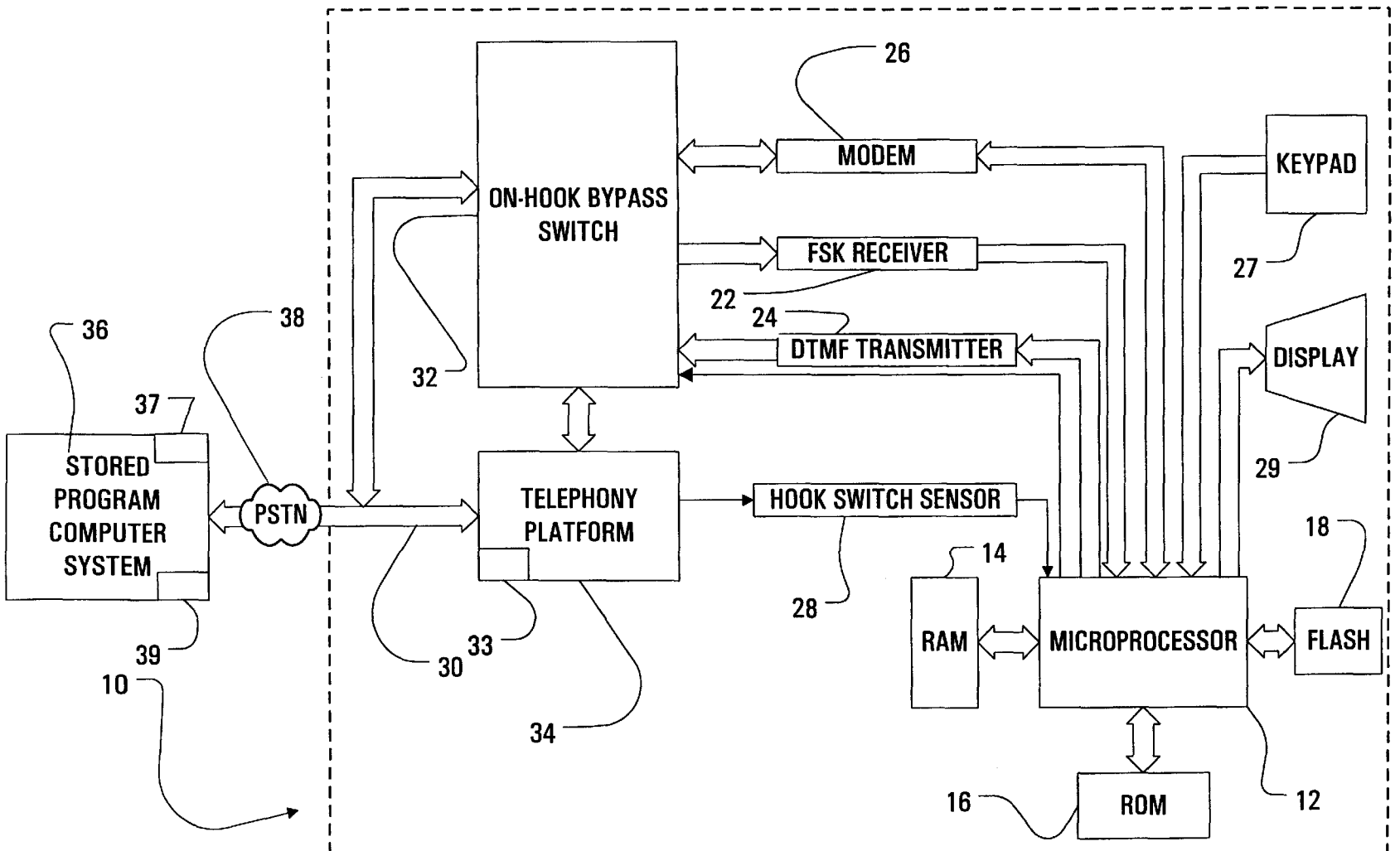

ON-HOOK BYPASS SWITCH
MODEM
FSK RECEIVER
DTMF TRANSMITTER
KEYPAD
DISPLAY
STORED PROGRAM COMPUTER SYSTEM
PSTN
TELEPHONY PLATFORM
HOOK SWITCH SENSOR
RAM
MICROPROCESSOR
FLASH
ROM
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Figure 1

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FAMILIES OF EXTENDED ADSI STA SIGNALS										
SIGNAL BYTE 1 (42)	SIGNAL BYTE 2 (44)	SIGNAL BYTE 3 (46)	SIGNAL BYTE 4 (48)	SIGNAL BYTE 5 (50)	SIGNAL BYTE 6 (52)	SIGNAL BYTE 7 (62)	SIGNAL BYTE 8	...	SIGNAL BYTE 255	FAMILY FUNCTION
STA PARAMETER	DATA LENGTH	APPLICATION NAME				ACTION CODE	ACTION CODE DATA BYTE 1	...	ACTION CODE DATA BYTE 248	
151D (54)	$XX	"Q"	"U"	"E"	"R"	$XX	$XX	...	$XX	SPCS QUERIES CPE MODES
151D (55)	$XX	"F"	"O"	"N"	"E"	$XX	$XX	...	$XX	SPCS CONTROLS CPE
151D (56)	$XX	"M"	"E"	"S"	"G"	$XX	$XX	...	$XX	(1) SPCS AND CPE EXCHANGE MESSAGE DATA (2) SPCS MAINTAINS CPE'S CRC, IRN, AND MSR CONFIGURATION
151D (58)	$XX	"S"	"C"	"R"	"I"	$XX	$XX	...	$XX	SPCS MAINTAINS CPE'S SCRIPTS
151D (60)	$XX	"P"	"O"	"L"	"L"	$XX	$XX	...	$XX	SPCS MAINTAINS CPE'S MDP AND SPP CONFIGURATION
151D (61)	$XX	"R"	"E"	"M"	"I"	$XX	$XX	...	$XX	SPCS MAINTAINS CPE'S CRC EVENTS

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Figure 2

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SUBSET OF EXTENDED STA CODES BEGINNING WITH: <151d> <STA SIGNAL LENGTH> <"Q"> <"U"> <"E"> <"R">			
ACTION CODE	OPERATION	DATA BYTE	DATA
$81	QUERY: WHAT DIAL-UP MODE WILL BE USED?	NULL	

Figure 3

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SUBSET OF EXTENDED STA CODES BEGINNING WITH: <151d> <STA SIGNAL LENGTH> <"F"> <"O"> <"N"> <"E">			
ACTION CODE	OPERATION	DATA BYTE	DATA
$01	COMMAND: ISSUE DTMF "B" AND HANG UP LINE	NULL	

Figure 4

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SUBSET OF EXTENDED STA CODES BEGINNING WITH: <151d> <STA SIGNAL LENGTH> <"M"> <"E"> <"S"> <"G">				
ACTION CODE	OPERATION	DATA BYTE	DATA	
$18	CONFIGURE IRN CRC MSR	BYTES 1 - 12	PRIMARY MESSAGE SENDING SERVER TELEPHONE # (BCD)	74
		BYTE 13	DELIMITER (BINARY) ($FF)	76
		BYTES 14 - 25	BACKUP MESSAGE SENDING SERVER TELEPHONE # (BCD)	78
		BYTE 26	DELIMITER (BINARY) ($FF)	80
		BYTE 27	NUMBER OF RETRIES (BINARY)	82
		BYTE 28	DELAY BETWEEN RETRIES (BINARY)	84
		BYTES 29 - 89	DISPLAY MESSAGE (ASCII)	86
		BYTE 90	DELIMITER (BINARY) ($FF)	88
		BYTES 91 - 94	CURRENT SECURITY CODE (BINARY)	90
		BYTE 95	DELIMITER (BINARY) ($FF)	92
		BYTES 96 - 103	NEW SECURITY CODE TWICE (OPTIONAL)	94
		BYTE 104	DELIMITER (BINARY) ($FF)	96

Figure 5

98

SUBSET OF EXTENDED STA CODES BEGINNING WITH: <151d> <STA SIGNAL LENGTH> <"S"> <"C"> <"R"> <"I">				
ACTION CODE	OPERATION	DATA BYTE	DATA	
99 $01	COMMENCE NON-INTERACTIVE FDM SESSION WITHIN CURRENT SDC SESSION	BYTES 1 - 4	CPE - ID (BINARY)	101
100 $06	CONFIGURE FSU	BYTES 1 - 4	FDM NUMBER (BINARY)	102
		BYTES 5 - 8	SECURITY CODE (BINARY)	104
		BYTES 9 - 12	TIME TO CALL SERVER (BCD) (MM:DD:HH:MM)	106
		BYTE 13	DELIMITER (BINARY) ($FF)	108
		BYTES 14 - 26	PRIMARY FSU SERVER TELEPHONE # (BCD)	110
		BYTE 27	DELIMITER (BINARY) ($FF)	112
		BYTES 28 - 39	BACKUP FSU SERVER TELEPHONE # (BCD)	114
		BYTE 40	DELIMITER (BINARY) ($FF)	116
		BYTE 41	NUMBER OF RETRIES (BINARY)	118
		BYTE 42	DELAY BETWEEN RETRIES (BINARY)	120
		BYTES 43 - 102	DISPLAY MESSAGE (ASCII)	122
		BYTE 103	DELIMITER (BINARY) ($FF)	124

Figure 6

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SUBSET OF EXTENDED STA CODES BEGINNING WITH:
<151d> <STA SIGNAL LENGTH> <"P"> <"O"> <"L"> <"L">

ACTION CODE	OPERATION	DATA BYTE	DATA	
$01	CONFIGURE MDP	BYTES 1 - 12	PRIMARY MESSAGING SERVER TELEPHONE # (BCD)	130
		BYTE 13	DELIMITER (BINARY) ($FF)	132
		BYTES 14 - 25	BACKUP MESSAGING SERVER TELEPHONE # (BCD)	134
		BYTE 26	DELIMITER (BINARY) ($FF)	136
		BYTES 27 - 28	POLL TIME 1 (BCD) (HH:MM)	138
		BYTES 29 - 30	POLL TIME 2 (BCD) (HH:MM)	140
		BYTES 31 - 32	POLL TIME 3 (BCD) (HH:MM)	142
		BYTES 33 - 34	POLL TIME 4 (BCD) (HH:MM)	144
		BYTES 35 - 36	POLL TIME 5 (BCD) (HH:MM)	146
		BYTE 37	DELIMITER (BINARY) ($FF)	148
		BYTE 38	NUMBER OF RETRIES (BINARY)	150
		BYTE 39	DELAY BETWEEN RETRIES (BINARY)	152
		BYTES 40 - 99	DISPLAY MESSAGE (ASCII)	154
		BYTE 100	DELIMITER (BINARY) ($FF)	156
$02	CONFIGURE SPP	BYTES 1 - 12	PRIMARY PAGE RESPONSE SERVER TELEPHONE # (BCD)	160
		BYTE 13	DELIMITER (BINARY) ($FF)	162
		BYTES 14 - 25	BACKUP PAGE RESPONSE SERVER TELEPHONE # (BCD)	164
		BYTE 26	DELIMITER (BINARY) ($FF)	166
		BYTE 27	DELAY BEFORE 1ST POLL (BINARY)	168
		BYTE 28	DELAY BETWEEN RETRIES (BINARY)	170
		BYTE 29	MAXIMUM NUMBER OF RETRIES (BINARY)	172
		BYTES 30 - 89	DISPLAY MESSAGE (ASCII)	174
		BYTE 90	DELIMITER (BINARY) ($FF)	176

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Figure 7

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SUBSET OF EXTENDED STA CODES BEGINNING WITH: <151d> <STA SIGNAL LENGTH> <"R"> <"E"> <"M"> <"I">				
ACTION CODE	OPERATION	DATA BYTE	DATA	
$01	DOWNLOAD CRC EVENT	BYTES 1 - 4	EVENT TIME (BCD) (MM:DD:HH:MM)	131
		BYTES 5 - 8	REMINDER TIME (BCD) (MM:DD:HH:MM)	133
		BYTE 9	REPEATING STATUS FLAG (BINARY)	135
		BYTES 10 - 69	DISPLAY MESSAGE (ASCII)	137
		BYTE 70	DELIMITER (BINARY) ($FF)	139
		BYTES 71 - 82	PAGER NUMBER RECEIVING REMINDER (BCD)	141

129

Figure 8

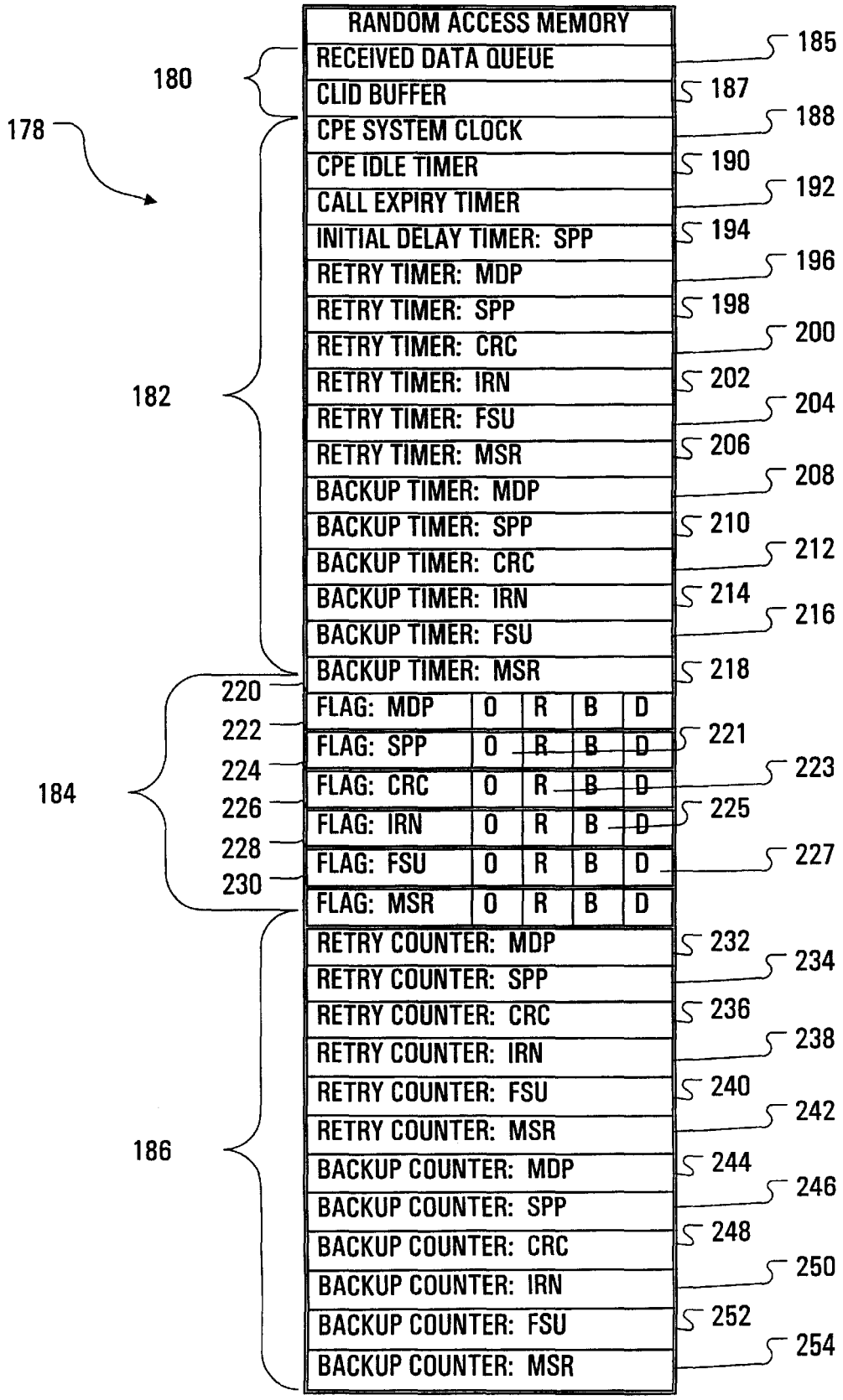
RANDOM ACCESS MEMORY
RECEIVED DATA QUEUE
CLID BUFFER
CPE SYSTEM CLOCK
CPE IDLE TIMER
CALL EXPIRY TIMER
INITIAL DELAY TIMER: SPP
RETRY TIMER: MDP
RETRY TIMER: SPP
RETRY TIMER: CRC
RETRY TIMER: IRN
RETRY TIMER: FSU
RETRY TIMER: MSR
BACKUP TIMER: MDP
BACKUP TIMER: SPP
BACKUP TIMER: CRC
BACKUP TIMER: IRN
BACKUP TIMER: FSU
BACKUP TIMER: MSR
FLAG: MDP O R B D
FLAG: SPP O R B D
FLAG: CRC O R B D
FLAG: IRN O R B D
FLAG: FSU O R B D
FLAG: MSR O R B D
RETRY COUNTER: MDP
RETRY COUNTER: SPP
RETRY COUNTER: CRC
RETRY COUNTER: IRN
RETRY COUNTER: FSU
RETRY COUNTER: MSR
BACKUP COUNTER: MDP
BACKUP COUNTER: SPP
BACKUP COUNTER: CRC
BACKUP COUNTER: IRN
BACKUP COUNTER: FSU
BACKUP COUNTER: MSR
178
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Figure 9

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	FLASH ALLOCATION		
	SEGMENT NAME	SEGMENT CHARACTERISTICS	
258	PRIMARY MESSAGE SERVER TELEPHONE #	24 DIGITS (12 BCD BYTES)	270
	BACKUP MESSAGE SERVER TELEPHONE #	24 DIGITS (12 BCD BYTES)	272
	FIRST POLL TIME	3 BINARY BYTES (HH:MM:SS)	274
	SECOND POLL TIME	3 BINARY BYTES (HH:MM:SS)	276
	THIRD POLL TIME	3 BINARY BYTES (HH:MM:SS)	278
	FOURTH POLL TIME	3 BINARY BYTES (HH:MM:SS)	280
	FIFTH POLL TIME	3 BINARY BYTES (HH:MM:SS)	282
	NUMBER OF RETRY POLLS	1 BINARY BYTE (0 ≤ BUFFER VALUE ≤ 10)	284
	RETRY POLL DELAY TIME	1 BINARY BYTE (REPRESENTING 10 SECOND INTERVALS)	286
	DISPLAY MESSAGE	60 ASCII BYTES	288
260	PRIMARY PAGE RESPONSE SERVER TELEPHONE #	24 DIGITS (12 BCD BYTES)	290
	BACKUP PAGE RESPONSE SERVER TELEPHONE #	24 DIGITS (12 BCD BYTES)	292
	INITIAL POLL DELAY	1 BINARY BYTE (REPRESENTING 10 SECOND INTERVALS)	294
	RETRY POLL DELAY TIME	1 BINARY BYTE (REPRESENTING 10 SECOND INTERVALS)	296
	NUMBER OF RETRY POLLS	1 BINARY BYTE (0 ≤ BUFFER VALUE ≤ 15)	298
	DISPLAY MESSAGE	60 ASCII BYTES	300
262	PRIMARY MESSAGE SENDING SERVER TELEPHONE #	24 DIGITS (12 BCD BYTES)	302
	BACKUP MESSAGE SENDING SERVER TELEPHONE #	24 DIGITS (12 BCD BYTES)	304
	NUMBER OF RETRY POLLS	1 BINARY BYTE (0 ≤ BUFFER VALUE ≤ 15)	306
	RETRY POLL DELAY TIME	1 BINARY BYTE (REPRESENTING 10 SECOND INTERVALS)	308
	DISPLAY MESSAGE	60 ASCII BYTES	310
	SECURITY CODE	4 BYTE BINARY	312

To Figure 10B

Figure 10A

To Figure 10A

264	PRIMARY FSU SERVER TELEPHONE #	24 DIGITS (12 BCD BYTES)	314
	BACKUP FSU SERVER TELEPHONE #	24 DIGITS (12 BCD BYTES)	316
	NUMBER RETRY POLLS	1 BINARY BYTE (0 ≤ BUFFER VALUE ≤ 10)	318
	RETRY POLL DELAY TIME	1 BINARY BYTE (REPRESENTING 10 SECOND INTERVALS)	320
	NEXT TIME TO CALL SERVER	5 BINARY BYTES (MM:DD:HH:MM:SS)	322
	DISPLAY MESSAGE	60 ASCI BYTES	324
	SECURITY CODE	4 BYTE BINARY	326
266	POLL PROXIMITY THRESHOLD	1 BINARY BYTE (REPRESENTING 1 SECOND INTERVALS)	327
	CPE IDLE TIME THRESHOLD	1 BINARY BYTE (REPRESENTING 1 SECOND INTERVALS)	328
	CALL TIME-OUT THRESHOLD	1 BINARY BYTE (REPRESENTING 1 SECOND INTERVALS)	329
	FEATURE DISABLE ERROR THRESHOLD	1 BINARY BYTE (REPRESENTING NUMBER OF FAILURES)	330
	NUMBER OF MDP FAILURES	1 BINARY BYTE	332
	NUMBER OF SPP FAILURES	1 BINARY BYTE	334
	NUMBER OF IRN FAILURES	1 BINARY BYTE	336
	NUMBER OF CRC FAILURES	1 BINARY BYTE	338
	NUMBER OF FSU FAILURES	1 BINARY BYTE	340
	NUMBER OF MSR FAILURES	1 BINARY BYTE	342
	CPE ID	4 BINARY BYTE	339
	REGULATORY PARAMETER LIMITS	ARRAY (PARAMETER/MAX VALUE/MIN VALUE)	345
	ERROR LOG ENTRY # 1	TIME/SUBROUTINE/LOCATION/DELIMITER	343
	***	***	
	ERROR LOG ENTRY # n	TIME/SUBROUTINE/LOCATION/DELIMITER	343

To Figure 10C

Figure 10B

To Figure 10B

268	344	CALENDER TIME OF EVENT 1	4 BINARY BYTES (MM:DD:HH:MM)
		TIME TO SEND REMINDER 1	4 BINARY BYTES (MM:DD:HH:MM)
		REPEATING STATUS FLAG 1	1 BINARY BYTE (DAILY, WEEKLY, MONTHLY,YEARLY, ONCE)
		REMINDER PAGER NUMBER 1	24 DIGITS (12 BCD BYTES)
		DISPLAY MESSAGE 1	60 ASCII BYTES
		***	***
	344	CALENDER TIME OF EVENT "n"	4 BINARY BYTES (MM:DD:HH:MM)
		TIME TO SEND REMINDER "n"	4 BINARY BYTES (MM:DD:HH:MM)
		REPEATING STATUS FLAG "n"	1 BINARY BYTE (DAILY, WEEKLY, MONTHLY,YEARLY, ONCE)
		REMINDER PAGER NUMBER "n"	24 DIGITS (12 BCD BYTES)
		DISPLAY MESSAGE "n"	60 ASCII BYTES
	346	IRN CALLER ID 1	24 DIGITS (12 BCD BYTES)
		IRN PAGER NUMBER 1	24 DIGITS (12 BCD BYTES)
		IRN MESSAGE TO DISPLAY 1	60 ASCII BYTES
		***	***
	346	IRN CALLER ID n	24 DIGITS (12 BCD BYTES)
		IRN PAGER NUMBER n	24 DIGITS (12 BCD BYTES)
		IRN MESSAGE TO DISPLAY n	60 ASCII BYTES

348 350 352 349 354 348 350 352 349 354 351 353 355 351 353 355

Figure 10C

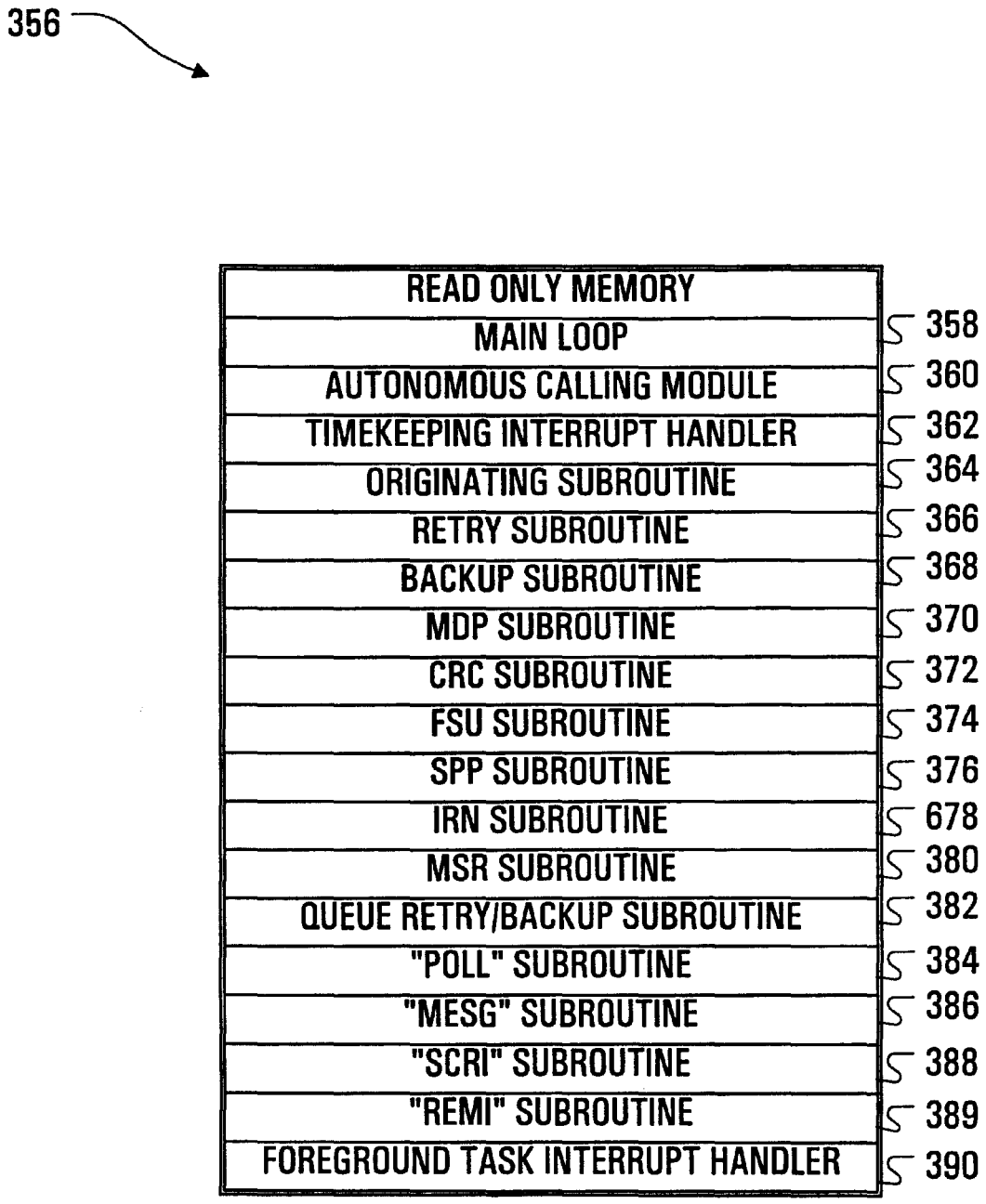
356
READ ONLY MEMORY
MAIN LOOP
358
AUTONOMOUS CALLING MODULE
360
TIMEKEEPING INTERRUPT HANDLER
362
ORIGINATING SUBROUTINE
364
RETRY SUBROUTINE
366
BACKUP SUBROUTINE
368
MDP SUBROUTINE
370
CRC SUBROUTINE
372
FSU SUBROUTINE
374
SPP SUBROUTINE
376
IRN SUBROUTINE
678
MSR SUBROUTINE
380
QUEUE RETRY/BACKUP SUBROUTINE
382
"POLL" SUBROUTINE
384
"MESG" SUBROUTINE
386
"SCRI" SUBROUTINE
388
"REMI" SUBROUTINE
389
FOREGROUND TASK INTERRUPT HANDLER
390

Figure 11

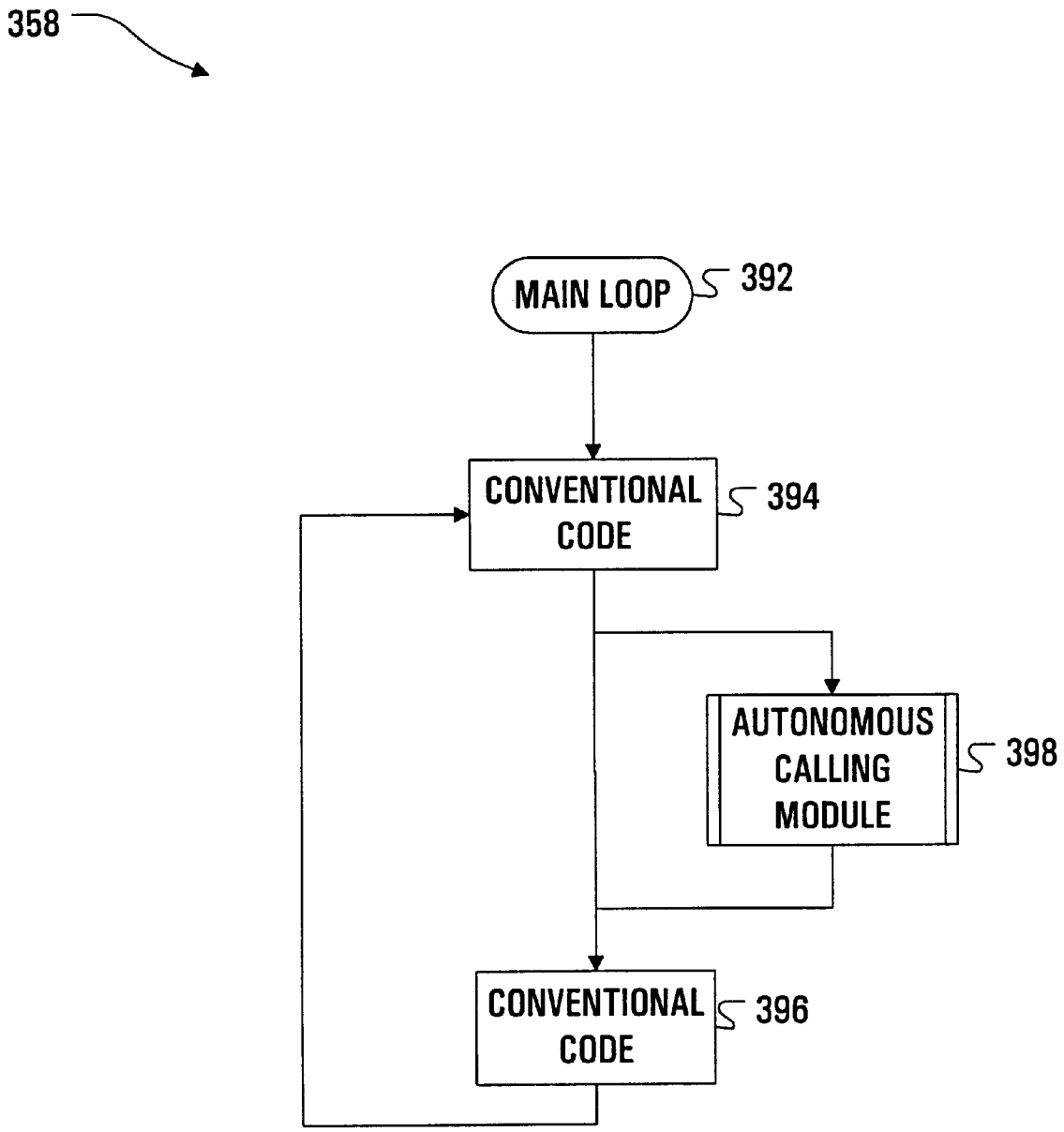
358
MAIN LOOP
392
CONVENTIONAL CODE
394
AUTONOMOUS CALLING MODULE
398
CONVENTIONAL CODE
396

Figure 12

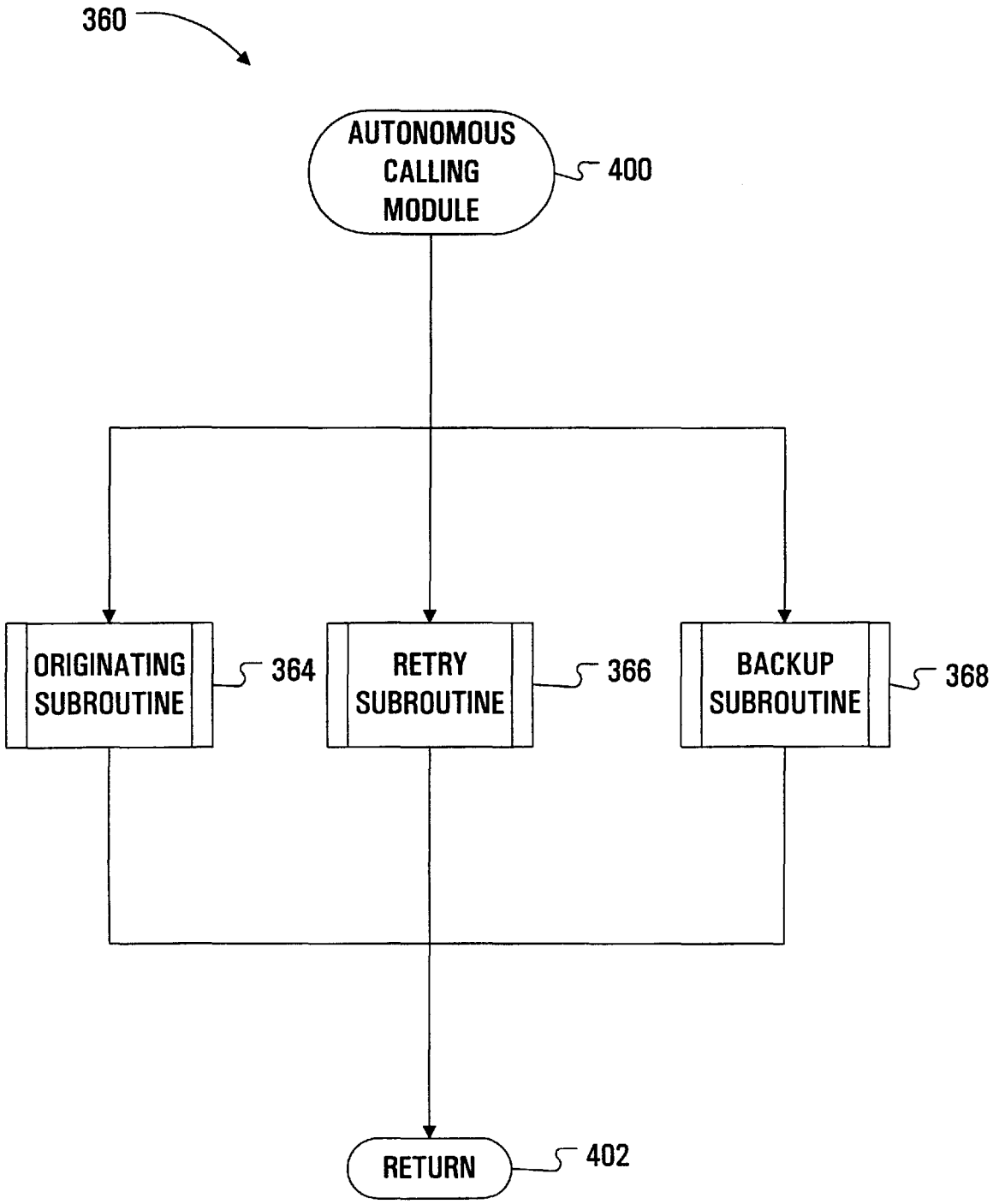
360
AUTONOMOUS CALLING MODULE
400
ORIGINATING SUBROUTINE
364
RETRY SUBROUTINE
366
BACKUP SUBROUTINE
368
RETURN
402

Figure 13

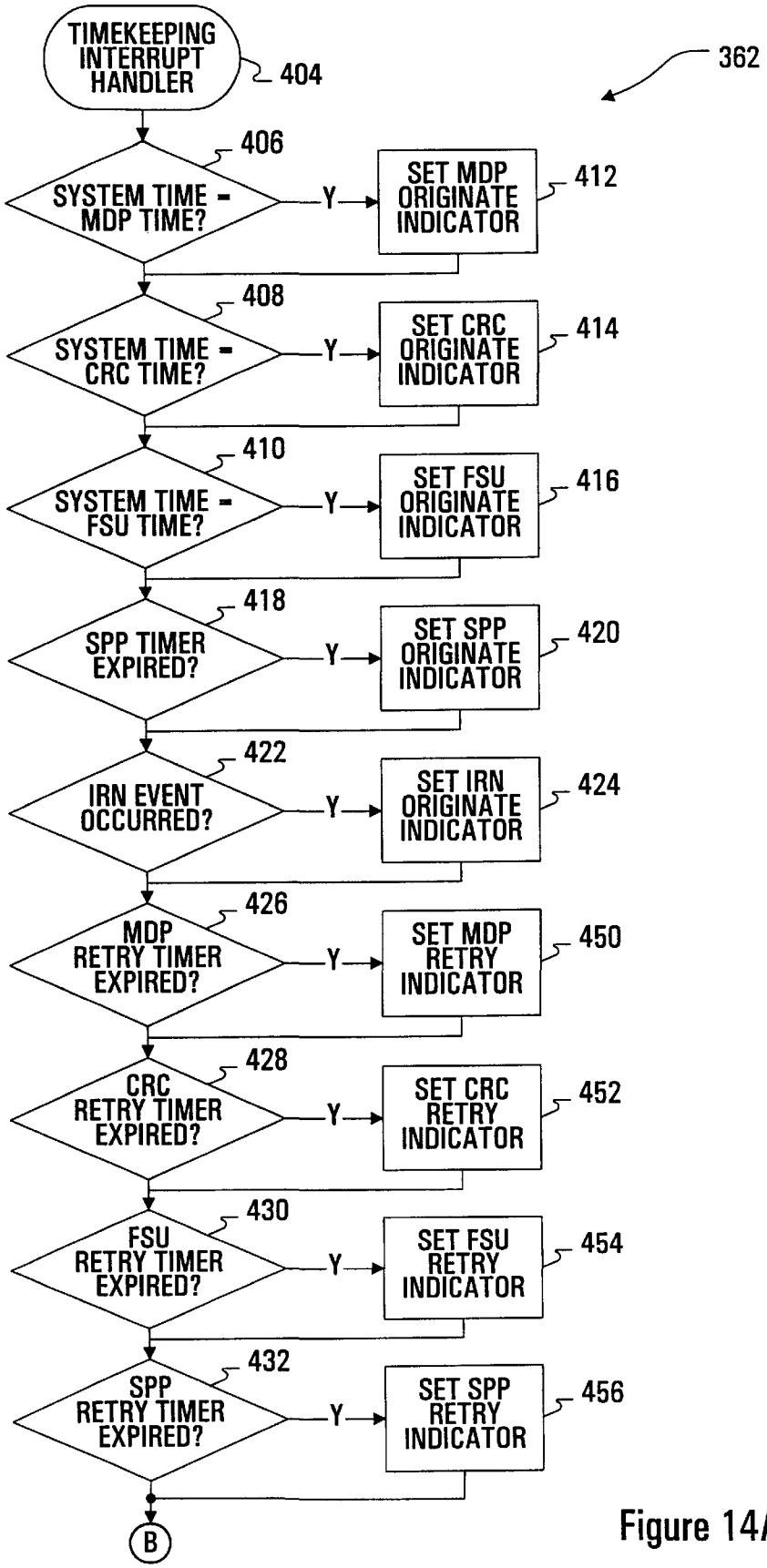
TIMEKEEPING INTERRUPT HANDLER
404
362
406
SYSTEM TIME = MDP TIME?
Y
SET MDP ORIGINATE INDICATOR
412
408
SYSTEM TIME = CRC TIME?
Y
SET CRC ORIGINATE INDICATOR
414
410
SYSTEM TIME = FSU TIME?
Y
SET FSU ORIGINATE INDICATOR
416
418
SPP TIMER EXPIRED?
Y
SET SPP ORIGINATE INDICATOR
420
422
IRN EVENT OCCURRED?
Y
SET IRN ORIGINATE INDICATOR
424
426
MDP RETRY TIMER EXPIRED?
Y
SET MDP RETRY INDICATOR
450
428
CRC RETRY TIMER EXPIRED?
Y
SET CRC RETRY INDICATOR
452
430
FSU RETRY TIMER EXPIRED?
Y
SET FSU RETRY INDICATOR
454
432
SPP RETRY TIMER EXPIRED?
Y
SET SPP RETRY INDICATOR
456
B

Figure 14A

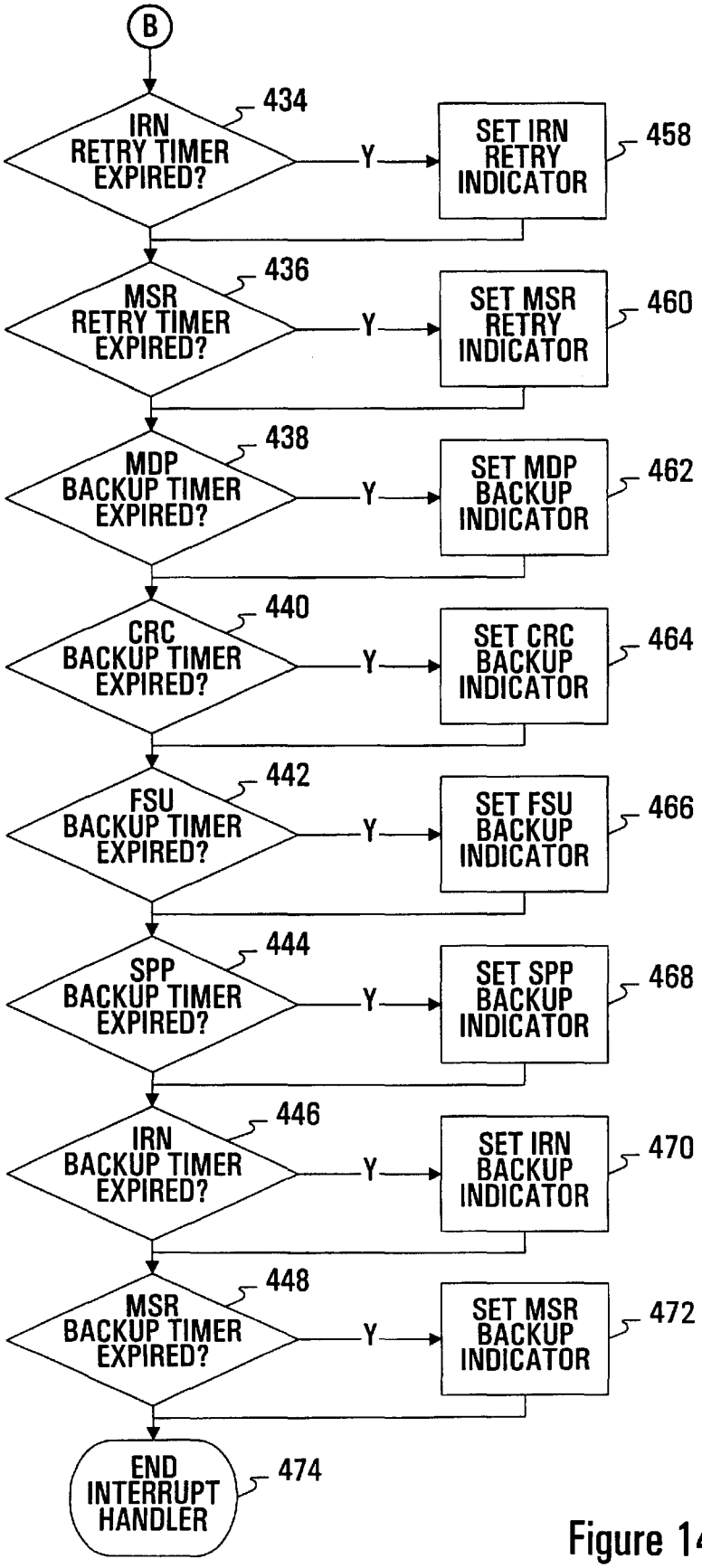
B
IRN RETRY TIMER EXPIRED?
434
Y
SET IRN RETRY INDICATOR
458
MSR RETRY TIMER EXPIRED?
436
Y
SET MSR RETRY INDICATOR
460
MDP BACKUP TIMER EXPIRED?
438
Y
SET MDP BACKUP INDICATOR
462
CRC BACKUP TIMER EXPIRED?
440
Y
SET CRC BACKUP INDICATOR
464
FSU BACKUP TIMER EXPIRED?
442
Y
SET FSU BACKUP INDICATOR
466
SPP BACKUP TIMER EXPIRED?
444
Y
SET SPP BACKUP INDICATOR
468
IRN BACKUP TIMER EXPIRED?
446
Y
SET IRN BACKUP INDICATOR
470
MSR BACKUP TIMER EXPIRED?
448
Y
SET MSR BACKUP INDICATOR
472
END INTERRUPT HANDLER
474

Figure 14B

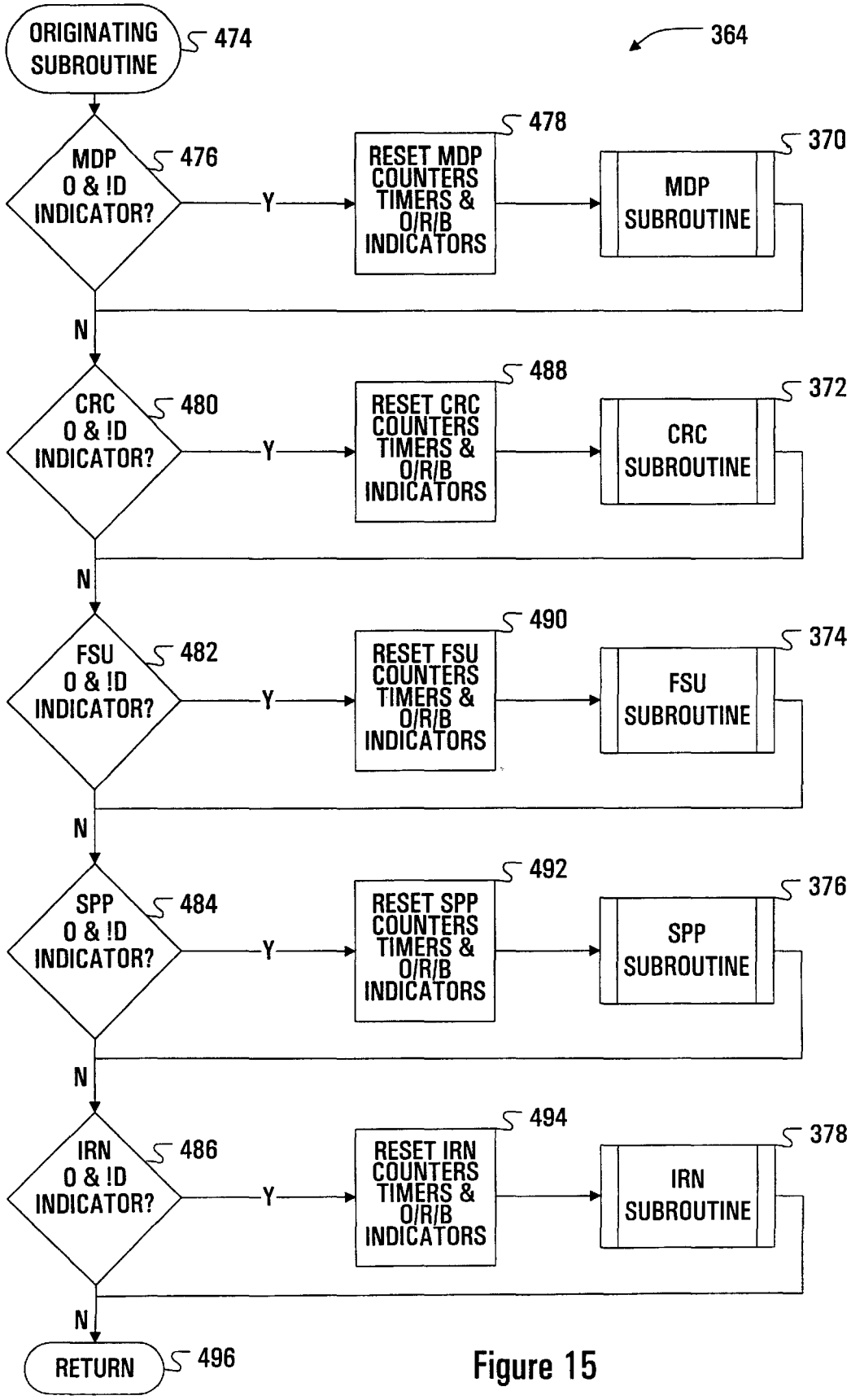
ORIGINATING SUBROUTINE
474
364
MDP O & !D INDICATOR?
476
Y
RESET MDP COUNTERS TIMERS & O/R/B INDICATORS
478
MDP SUBROUTINE
370
N
CRC O & !D INDICATOR?
480
Y
RESET CRC COUNTERS TIMERS & O/R/B INDICATORS
488
CRC SUBROUTINE
372
N
FSU O & !D INDICATOR?
482
Y
RESET FSU COUNTERS TIMERS & O/R/B INDICATORS
490
FSU SUBROUTINE
374
N
SPP O & !D INDICATOR?
484
Y
RESET SPP COUNTERS TIMERS & O/R/B INDICATORS
492
SPP SUBROUTINE
376
N
IRN O & !D INDICATOR?
486
Y
RESET IRN COUNTERS TIMERS & O/R/B INDICATORS
494
IRN SUBROUTINE
378
N
RETURN
496

Figure 15

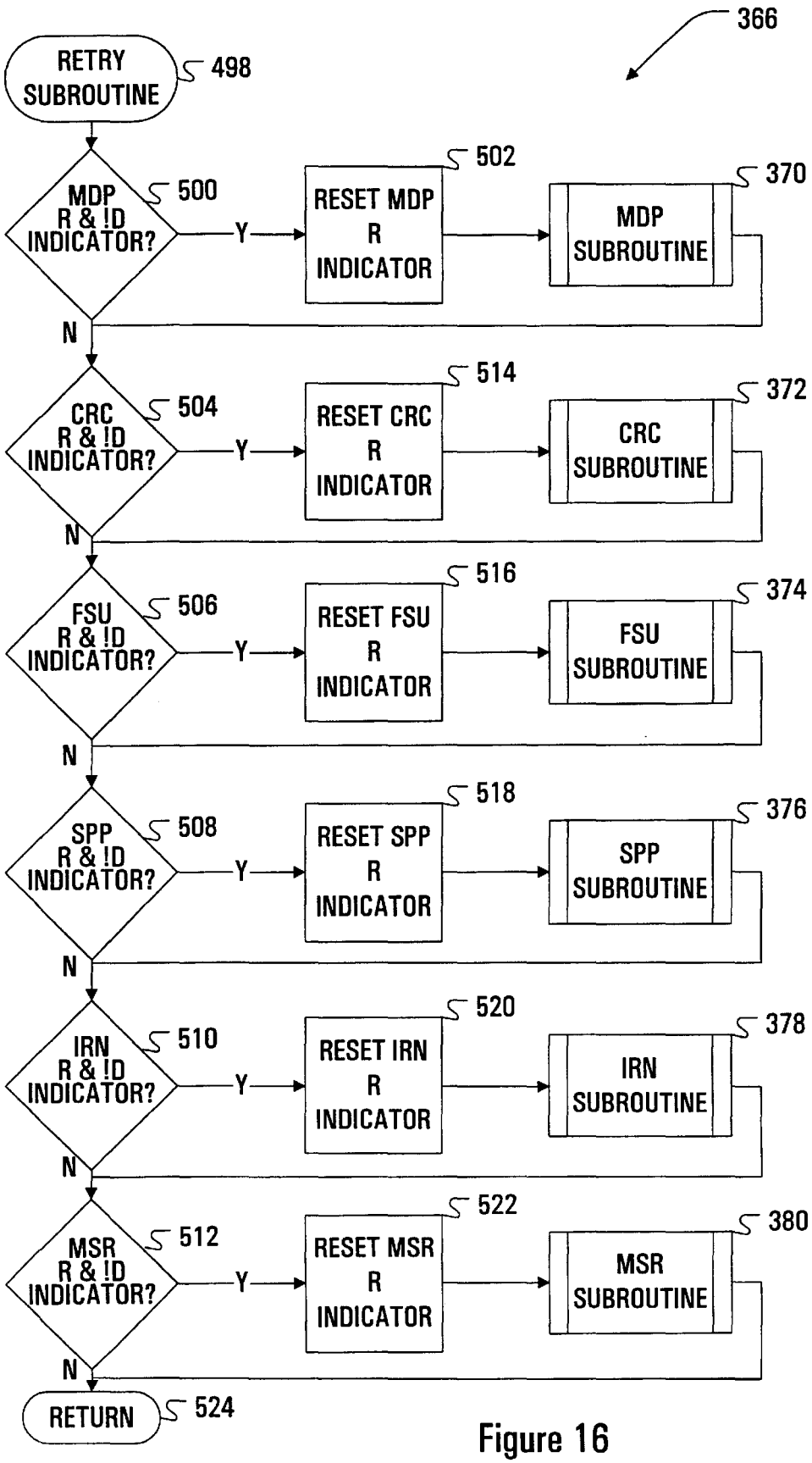
366
RETRY SUBROUTINE
498
MDP R & !D INDICATOR?
500
Y
N
RESET MDP R INDICATOR
502
MDP SUBROUTINE
370
CRC R & !D INDICATOR?
504
RESET CRC R INDICATOR
514
CRC SUBROUTINE
372
FSU R & !D INDICATOR?
506
RESET FSU R INDICATOR
516
FSU SUBROUTINE
374
SPP R & !D INDICATOR?
508
RESET SPP R INDICATOR
518
SPP SUBROUTINE
376
IRN R & !D INDICATOR?
510
RESET IRN R INDICATOR
520
IRN SUBROUTINE
378
MSR R & !D INDICATOR?
512
RESET MSR R INDICATOR
522
MSR SUBROUTINE
380
RETURN
524

Figure 16

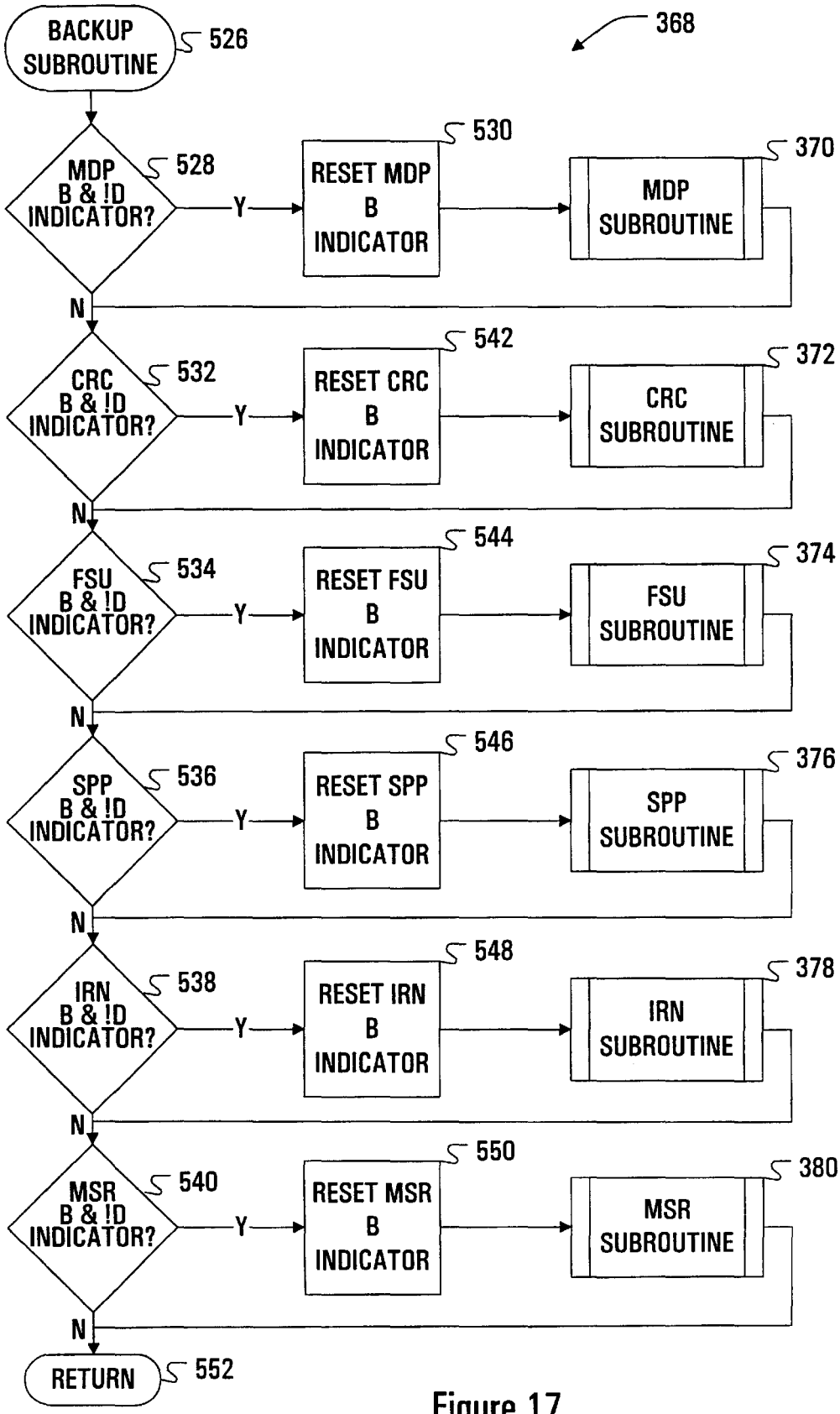
BACKUP SUBROUTINE
526
368
MDP B & !D INDICATOR?
528
Y
N
RESET MDP B INDICATOR
530
MDP SUBROUTINE
370
CRC B & !D INDICATOR?
532
RESET CRC B INDICATOR
542
CRC SUBROUTINE
372
FSU B & !D INDICATOR?
534
RESET FSU B INDICATOR
544
FSU SUBROUTINE
374
SPP B & !D INDICATOR?
536
RESET SPP B INDICATOR
546
SPP SUBROUTINE
376
IRN B & !D INDICATOR?
538
RESET IRN B INDICATOR
548
IRN SUBROUTINE
378
MSR B & !D INDICATOR?
540
RESET MSR B INDICATOR
550
MSR SUBROUTINE
380
RETURN
552

Figure 17

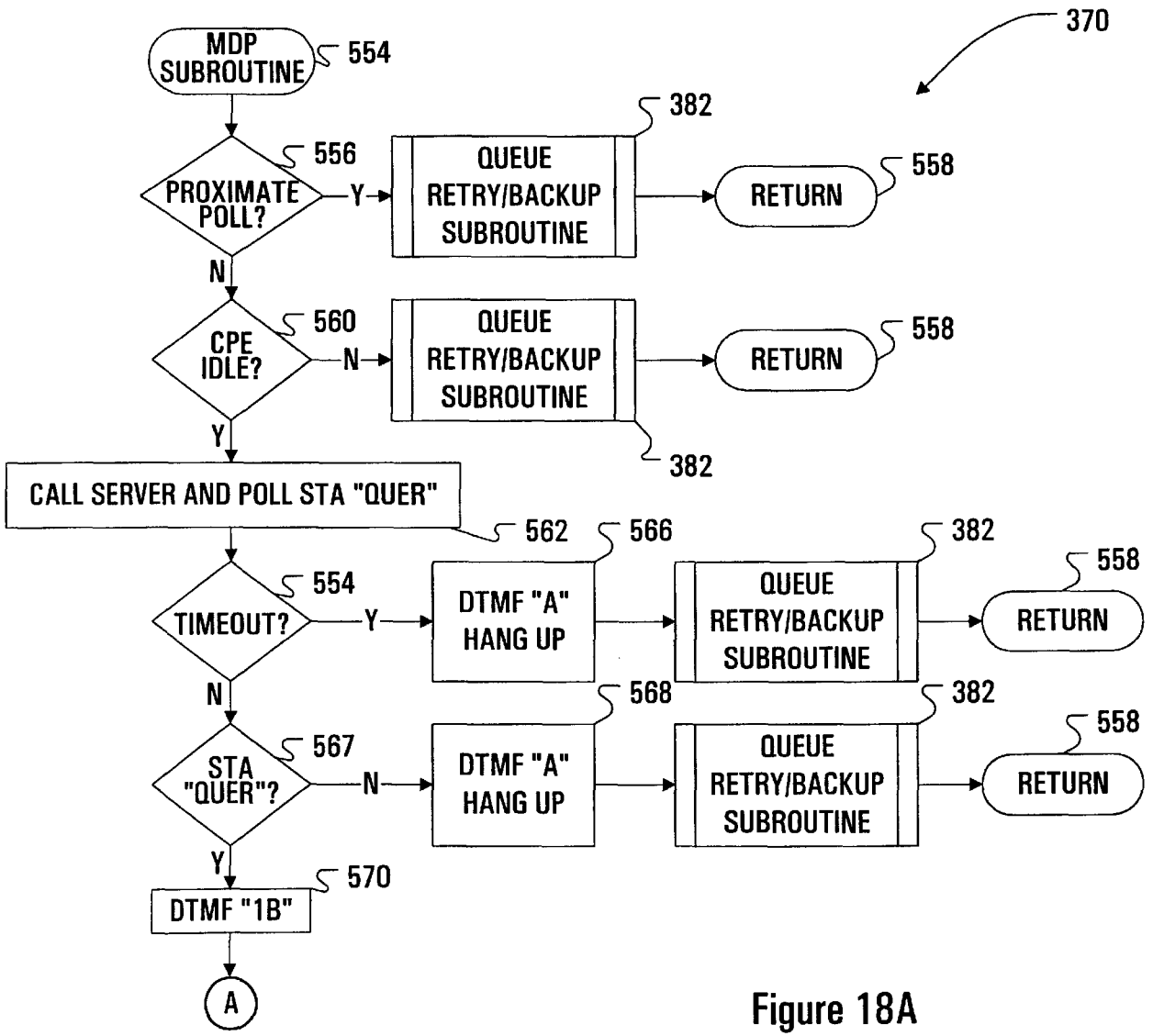
370
MDP SUBROUTINE
554
PROXIMATE POLL?
556
Y
N
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
558
CPE IDLE?
560
N
Y
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
558
CALL SERVER AND POLL STA "QUER"
562
TIMEOUT?
554
Y
N
DTMF "A" HANG UP
566
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
558
STA "QUER"?
567
N
Y
DTMF "A" HANG UP
568
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
558
DTMF "1B"
570
A

Figure 18A

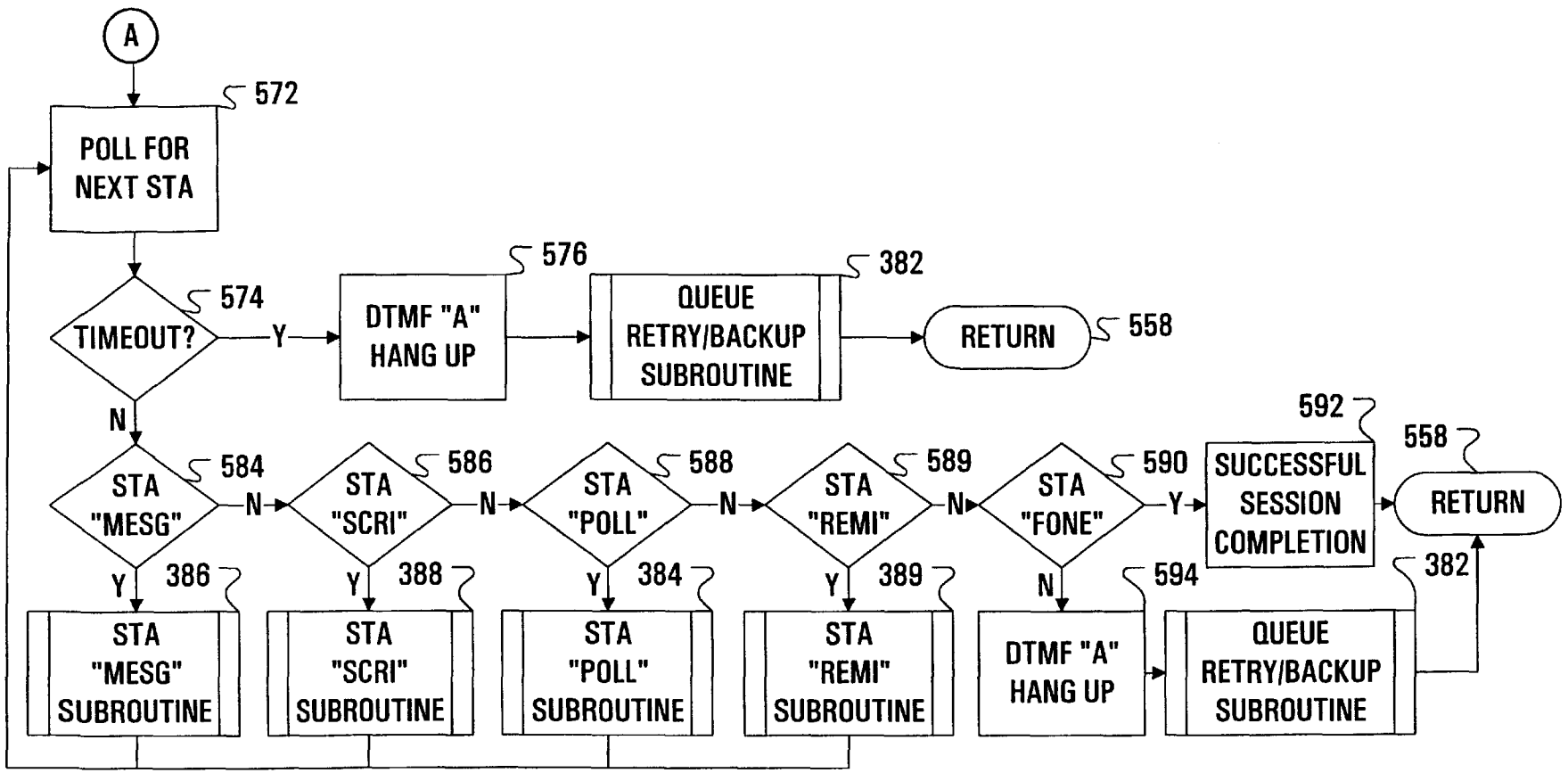
A
572
POLL FOR NEXT STA
574
TIMEOUT?
Y
576
DTMF "A" HANG UP
382
QUEUE RETRY/BACKUP SUBROUTINE
558
RETURN
N
584
STA "MESG"
N
586
STA "SCRI"
N
588
STA "POLL"
N
589
STA "REMI"
N
590
STA "FONE"
Y
592
SUCCESSFUL SESSION COMPLETION
558
RETURN
Y
386
STA "MESG" SUBROUTINE
Y
388
STA "SCRI" SUBROUTINE
Y
384
STA "POLL" SUBROUTINE
Y
389
STA "REMI" SUBROUTINE
N
594
DTMF "A" HANG UP
QUEUE RETRY/BACKUP SUBROUTINE
382

Figure 18B

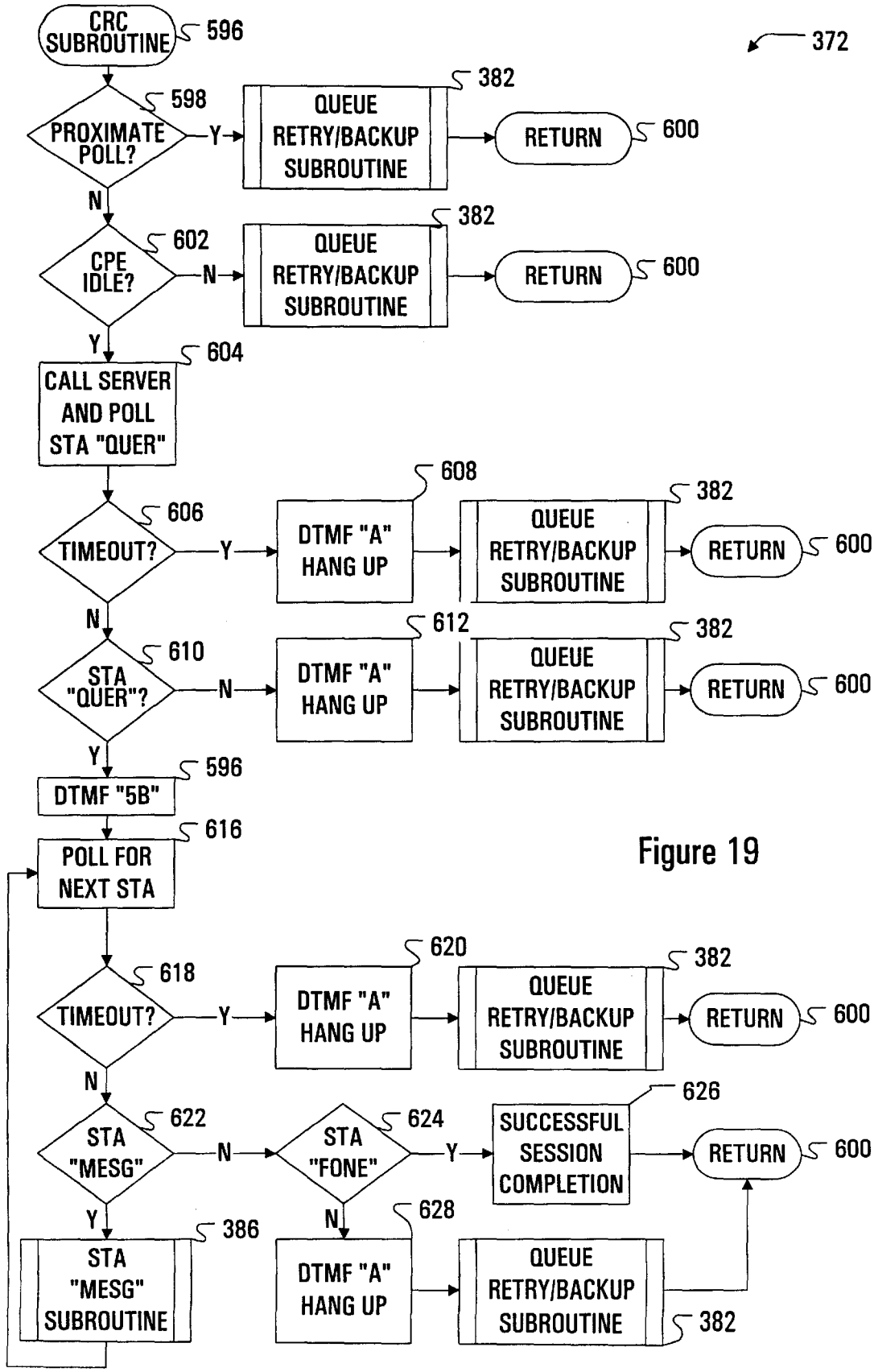
CRC SUBROUTINE
596
372
PROXIMATE POLL?
598
Y
N
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
600
CPE IDLE?
602
N
Y
CALL SERVER AND POLL STA "QUER"
604
TIMEOUT?
606
DTMF "A" HANG UP
608
STA "QUER"?
610
DTMF "A" HANG UP
612
DTMF "5B"
596
POLL FOR NEXT STA
616
TIMEOUT?
618
DTMF "A" HANG UP
620
STA "MESG"
622
STA "FONE"
624
SUCCESSFUL SESSION COMPLETION
626
STA "MESG" SUBROUTINE
386
DTMF "A" HANG UP
628

Figure 19

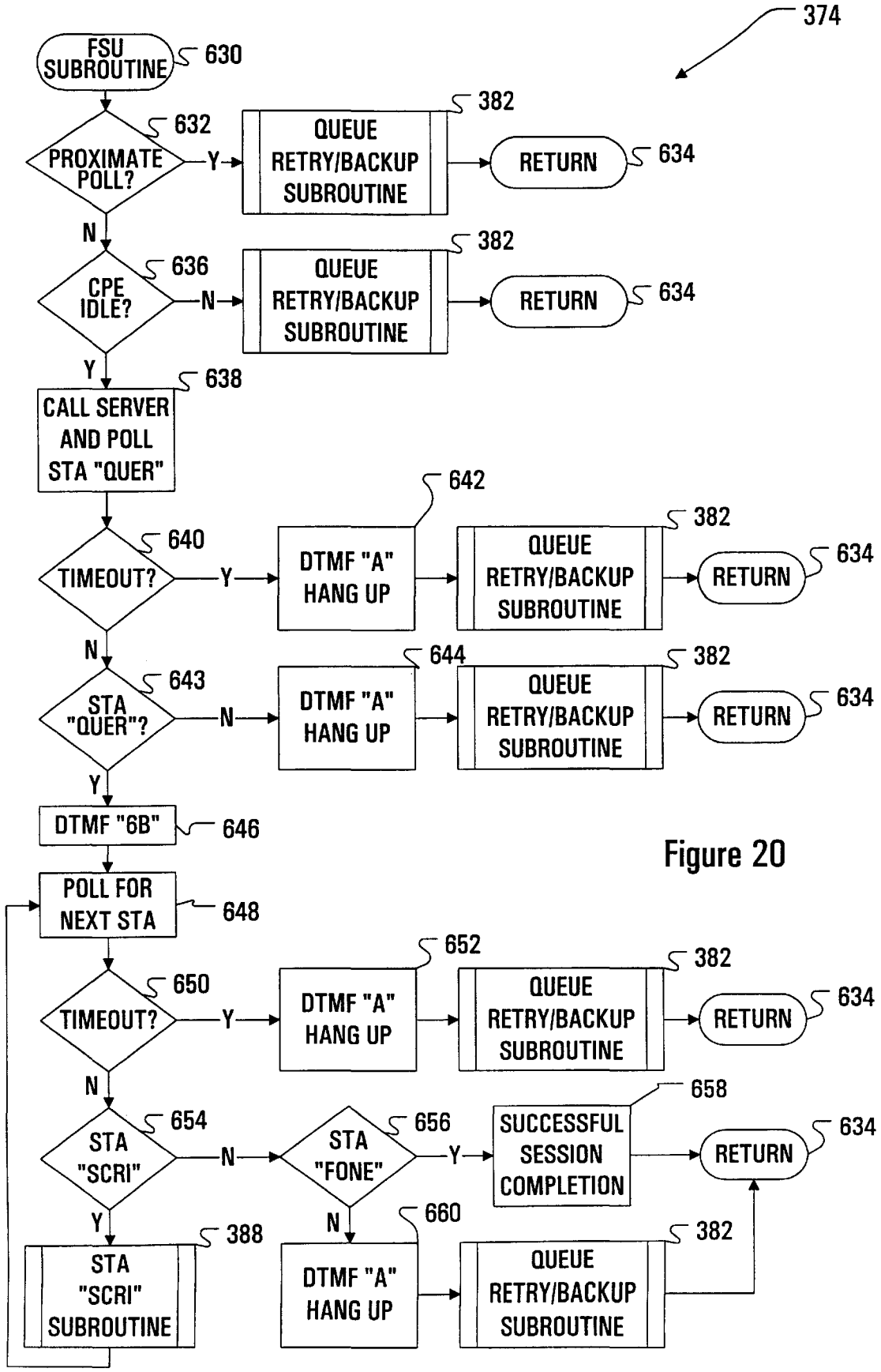
374
FSU SUBROUTINE
630
PROXIMATE POLL?
632
Y
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
634
N
CPE IDLE?
636
N
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
634
Y
CALL SERVER AND POLL STA "QUER"
638
TIMEOUT?
640
Y
DTMF "A" HANG UP
642
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
634
N
STA "QUER"?
643
N
DTMF "A" HANG UP
644
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
634
Y
DTMF "6B"
646
POLL FOR NEXT STA
648
TIMEOUT?
650
Y
DTMF "A" HANG UP
652
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
634
N
STA "SCRI"
654
N
STA "FONE"
656
Y
SUCCESSFUL SESSION COMPLETION
658
RETURN
634
Y
STA "SCRI" SUBROUTINE
388
N
DTMF "A" HANG UP
660
QUEUE RETRY/BACKUP SUBROUTINE
382

Figure 20

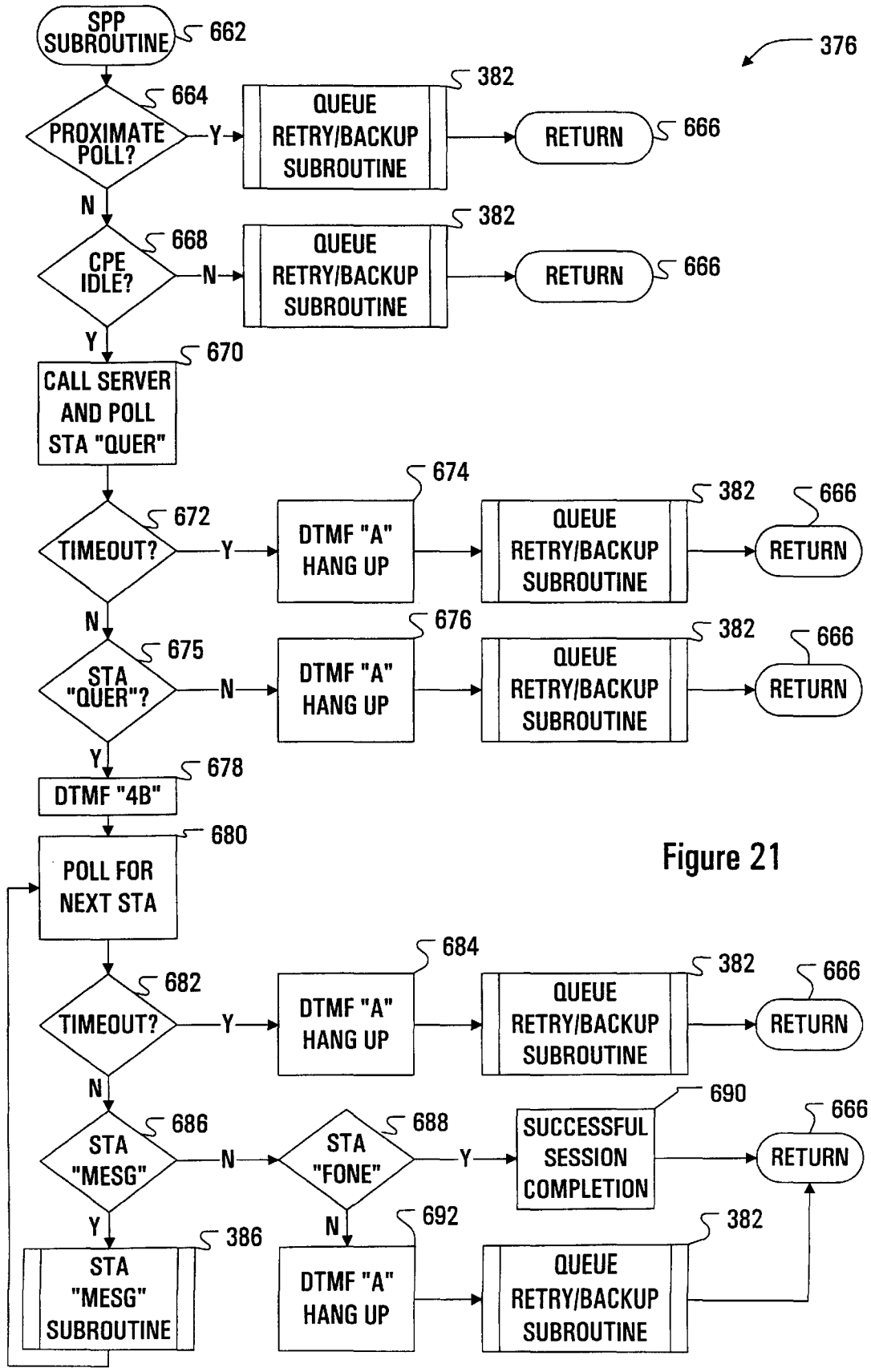
SPP SUBROUTINE
662
376
PROXIMATE POLL?
664
Y
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
666
N
CPE IDLE?
668
N
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
666
Y
CALL SERVER AND POLL STA "QUER"
670
TIMEOUT?
672
Y
DTMF "A" HANG UP
674
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
666
N
STA "QUER"?
675
N
DTMF "A" HANG UP
676
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
666
Y
DTMF "4B"
678
POLL FOR NEXT STA
680
TIMEOUT?
682
Y
DTMF "A" HANG UP
684
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
666
N
STA "MESG"
686
N
STA "FONE"
688
Y
SUCCESSFUL SESSION COMPLETION
690
RETURN
666
Y
STA "MESG" SUBROUTINE
386
N
DTMF "A" HANG UP
692
QUEUE RETRY/BACKUP SUBROUTINE
382

Figure 21

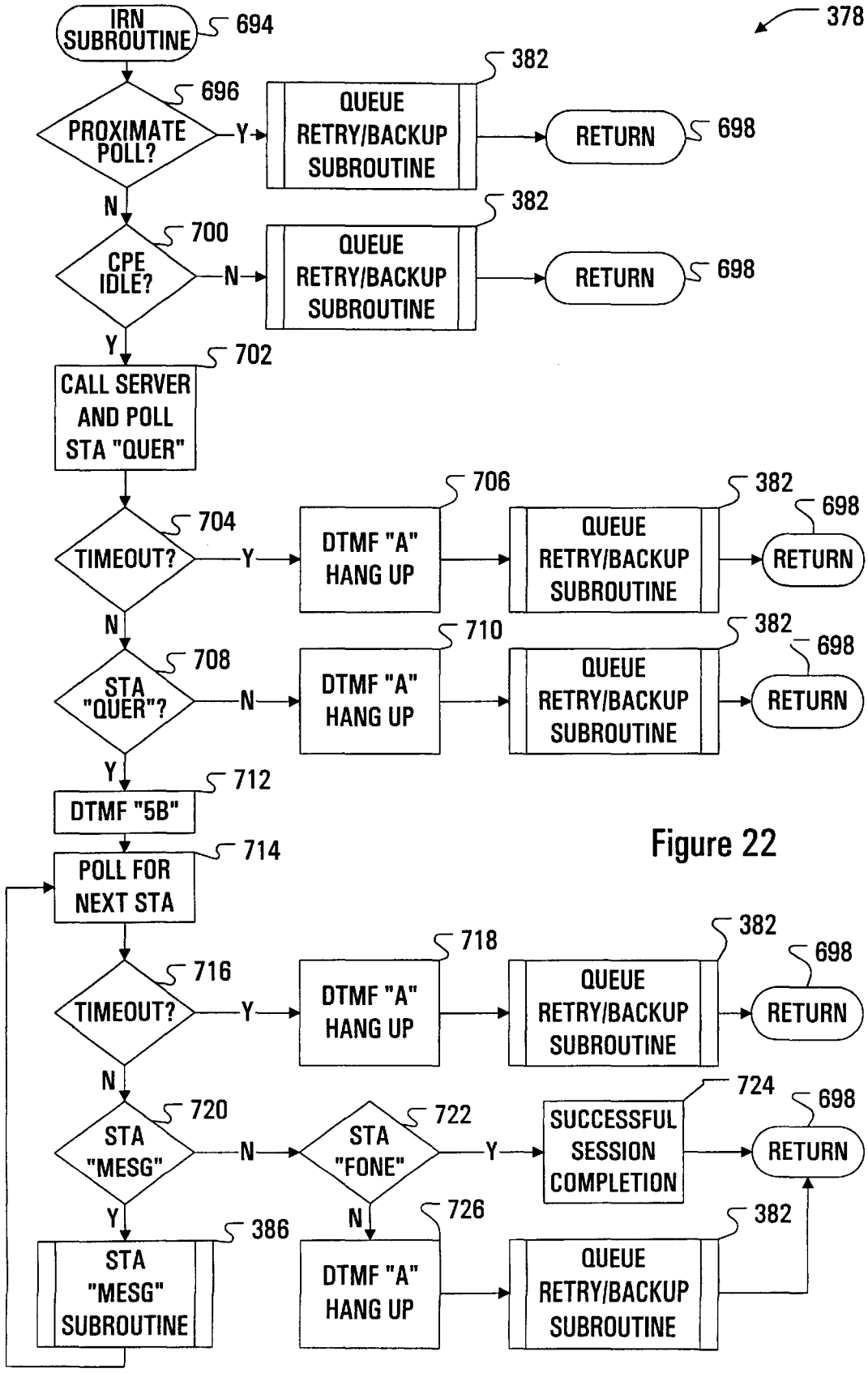
378
IRN SUBROUTINE
694
PROXIMATE POLL?
696
Y
N
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
698
CPE IDLE?
700
N
Y
CALL SERVER AND POLL STA "QUER"
702
TIMEOUT?
704
DTMF "A" HANG UP
706
STA "QUER"?
708
710
DTMF "5B"
712
POLL FOR NEXT STA
714
TIMEOUT?
716
718
STA "MESG"
720
STA "FONE"
722
SUCCESSFUL SESSION COMPLETION
724
STA "MESG" SUBROUTINE
386
726

Figure 22

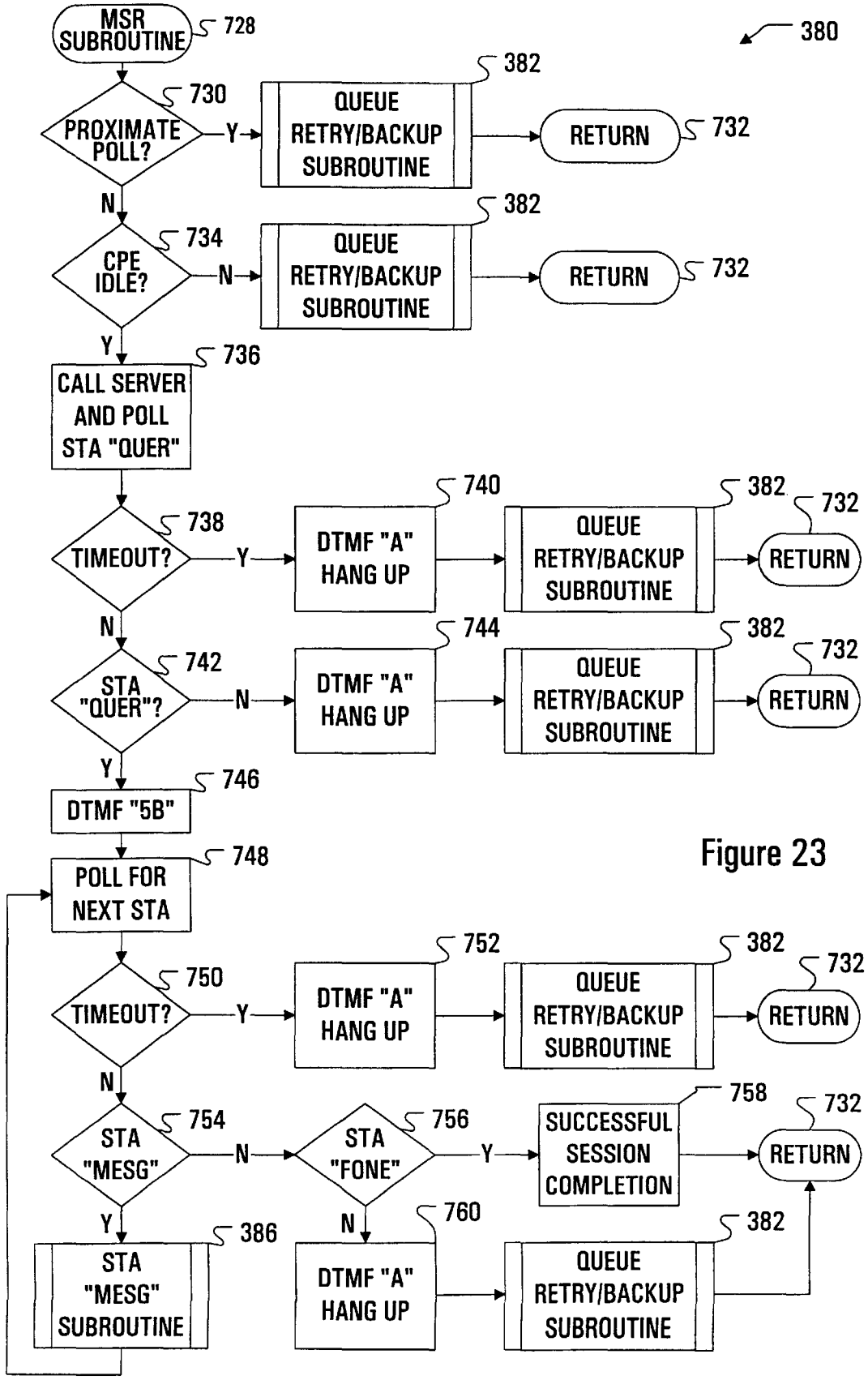
MSR SUBROUTINE
728
380
PROXIMATE POLL?
730
Y
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
732
N
CPE IDLE?
734
N
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
732
Y
CALL SERVER AND POLL STA "QUER"
736
TIMEOUT?
738
Y
DTMF "A" HANG UP
740
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
732
N
STA "QUER"?
742
N
DTMF "A" HANG UP
744
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
732
Y
DTMF "5B"
746
POLL FOR NEXT STA
748
TIMEOUT?
750
Y
DTMF "A" HANG UP
752
QUEUE RETRY/BACKUP SUBROUTINE
382
RETURN
732
N
STA "MESG"
754
N
STA "FONE"
756
Y
SUCCESSFUL SESSION COMPLETION
758
RETURN
732
Y
STA "MESG" SUBROUTINE
386
N
DTMF "A" HANG UP
760
QUEUE RETRY/BACKUP SUBROUTINE
382

Figure 23

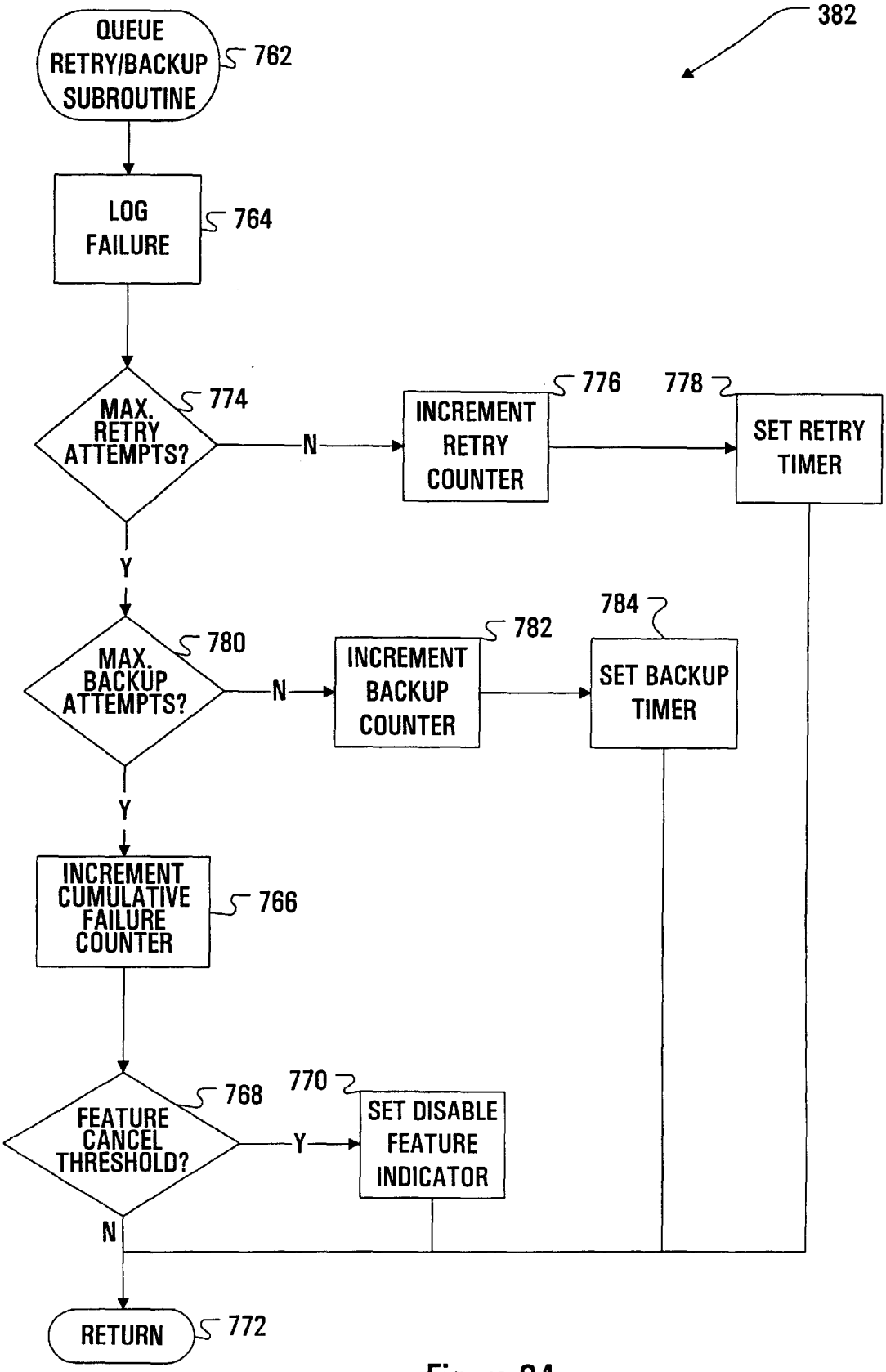
382
QUEUE RETRY/BACKUP SUBROUTINE
762
LOG FAILURE
764
MAX. RETRY ATTEMPTS?
774
N
INCREMENT RETRY COUNTER
776
778
SET RETRY TIMER
Y
MAX. BACKUP ATTEMPTS?
780
N
INCREMENT BACKUP COUNTER
782
784
SET BACKUP TIMER
Y
INCREMENT CUMULATIVE FAILURE COUNTER
766
FEATURE CANCEL THRESHOLD?
768
Y
770
SET DISABLE FEATURE INDICATOR
N
RETURN
772

Figure 24

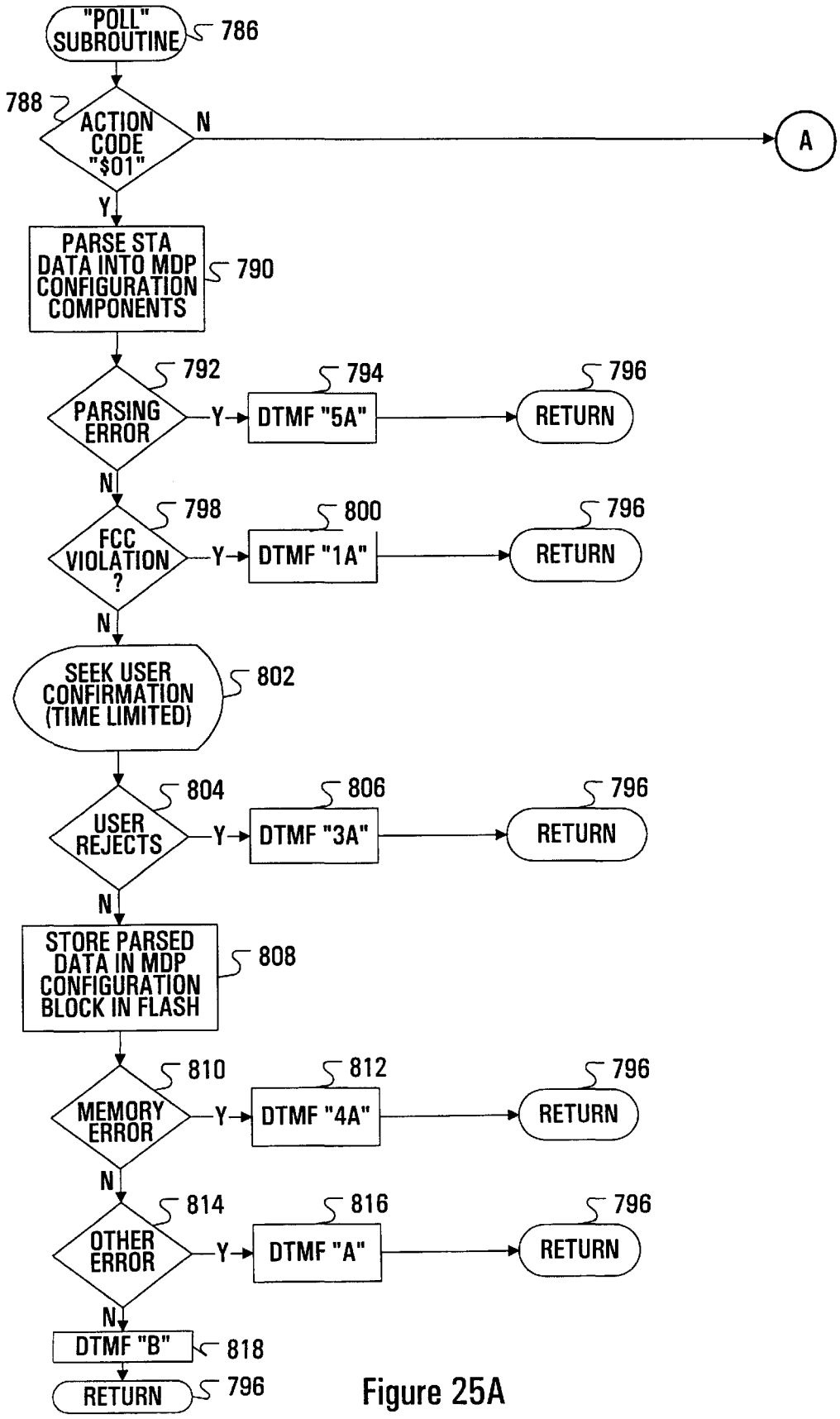
"POLL" SUBROUTINE
786
788
ACTION CODE "$01"
N
A
Y
PARSE STA DATA INTO MDP CONFIGURATION COMPONENTS
790
792
PARSING ERROR
Y
794
DTMF "5A"
796
RETURN
N
798
FCC VIOLATION ?
Y
800
DTMF "1A"
796
RETURN
N
SEEK USER CONFIRMATION (TIME LIMITED)
802
804
USER REJECTS
Y
806
DTMF "3A"
796
RETURN
N
STORE PARSED DATA IN MDP CONFIGURATION BLOCK IN FLASH
808
810
MEMORY ERROR
Y
812
DTMF "4A"
796
RETURN
N
814
OTHER ERROR
Y
816
DTMF "A"
796
RETURN
N
DTMF "B"
818
RETURN
796

Figure 25A

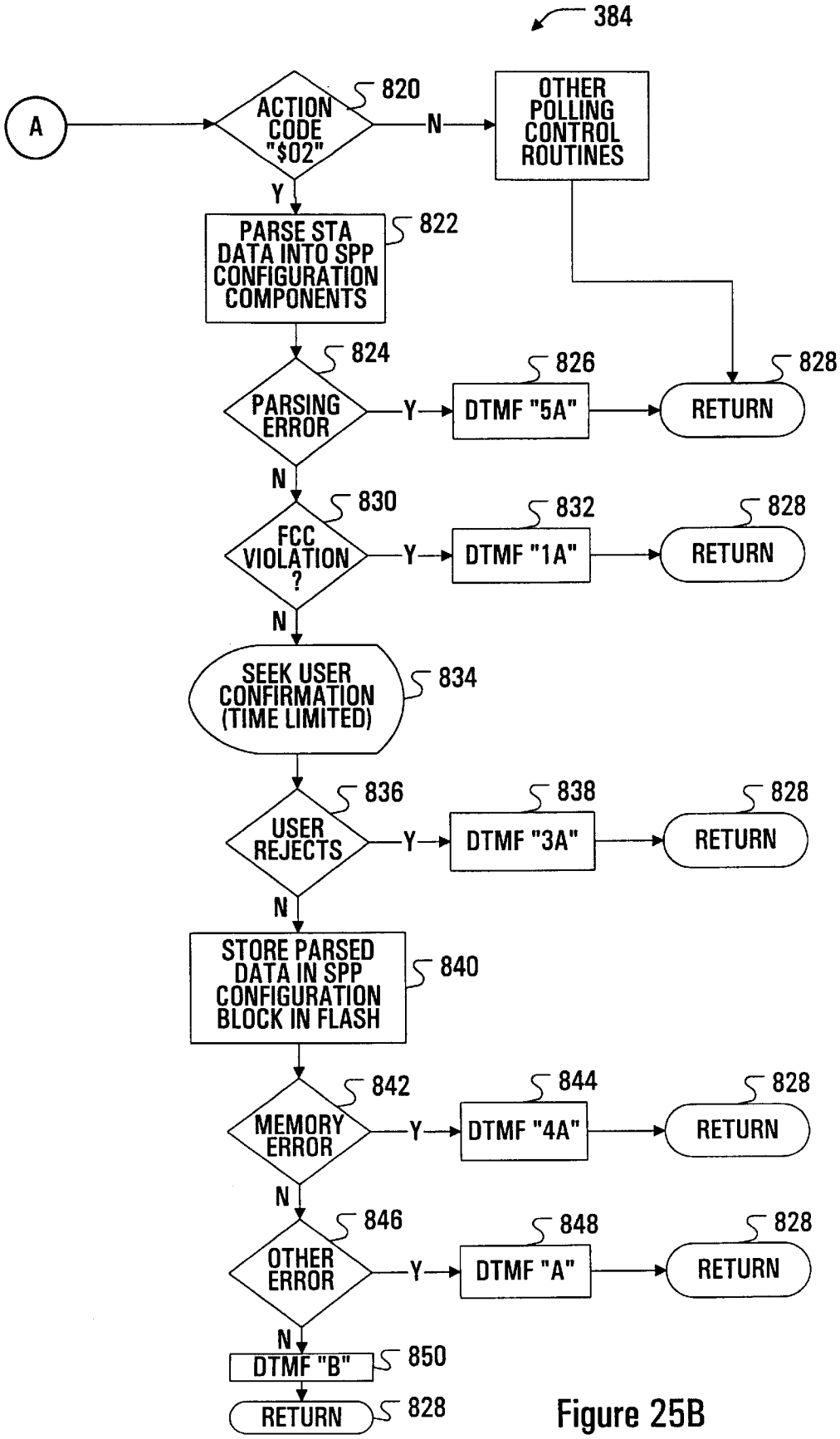
384
A
ACTION CODE "$02"
820
N
OTHER POLLING CONTROL ROUTINES
Y
PARSE STA DATA INTO SPP CONFIGURATION COMPONENTS
822
PARSING ERROR
824
Y
DTMF "5A"
826
RETURN
828
N
FCC VIOLATION ?
830
Y
DTMF "1A"
832
RETURN
828
N
SEEK USER CONFIRMATION (TIME LIMITED)
834
USER REJECTS
836
Y
DTMF "3A"
838
RETURN
828
N
STORE PARSED DATA IN SPP CONFIGURATION BLOCK IN FLASH
840
MEMORY ERROR
842
Y
DTMF "4A"
844
RETURN
828
N
OTHER ERROR
846
Y
DTMF "A"
848
RETURN
828
N
DTMF "B"
850
RETURN
828

Figure 25B

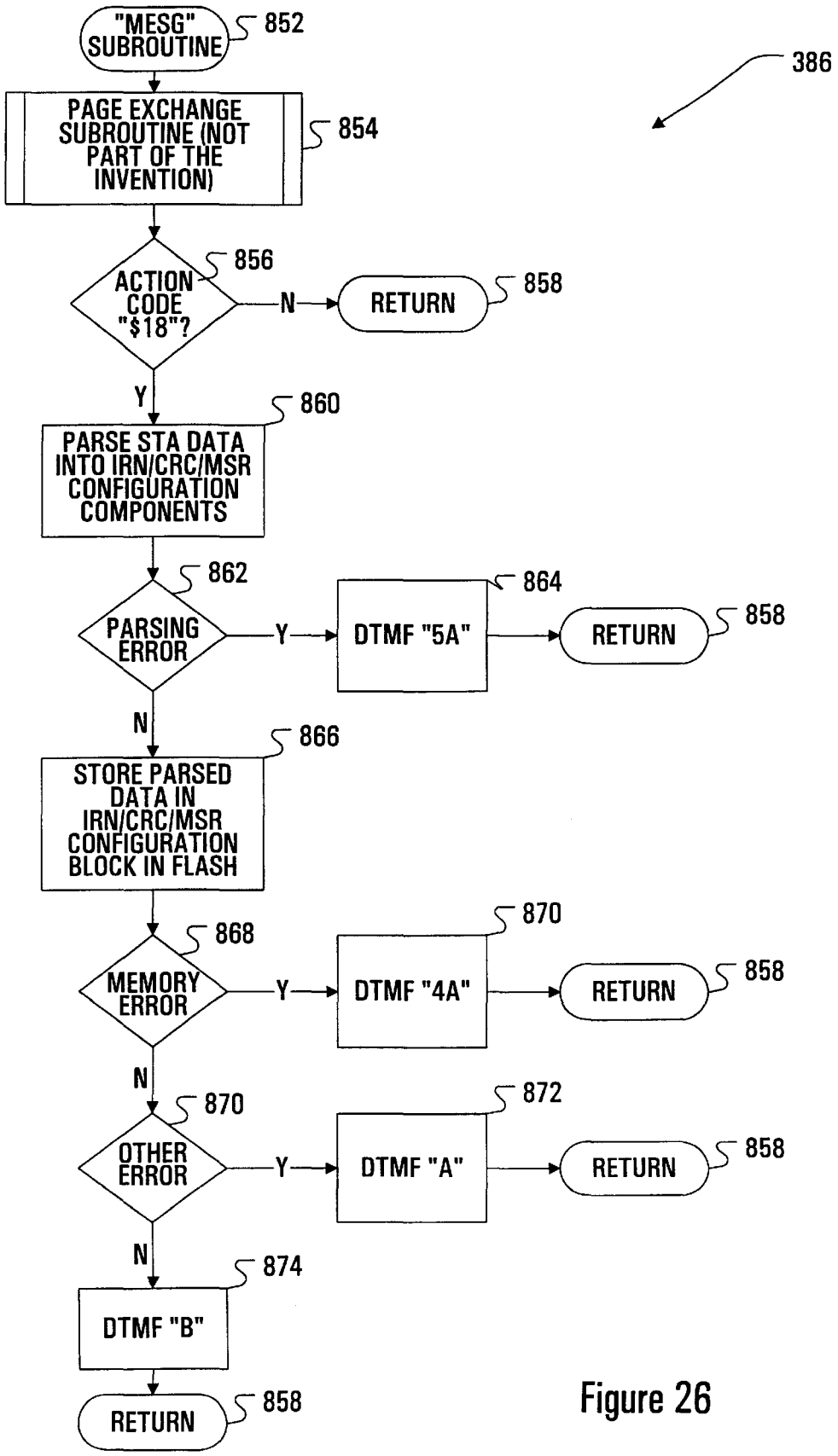
"MESG" SUBROUTINE
852
386
PAGE EXCHANGE SUBROUTINE (NOT PART OF THE INVENTION)
854
ACTION CODE "$18"?
856
N
RETURN
858
Y
PARSE STA DATA INTO IRN/CRC/MSR CONFIGURATION COMPONENTS
860
PARSING ERROR
862
Y
DTMF "5A"
864
RETURN
858
N
STORE PARSED DATA IN IRN/CRC/MSR CONFIGURATION BLOCK IN FLASH
866
MEMORY ERROR
868
Y
DTMF "4A"
870
RETURN
858
N
OTHER ERROR
870
Y
DTMF "A"
872
RETURN
858
N
DTMF "B"
874
RETURN
858

Figure 26

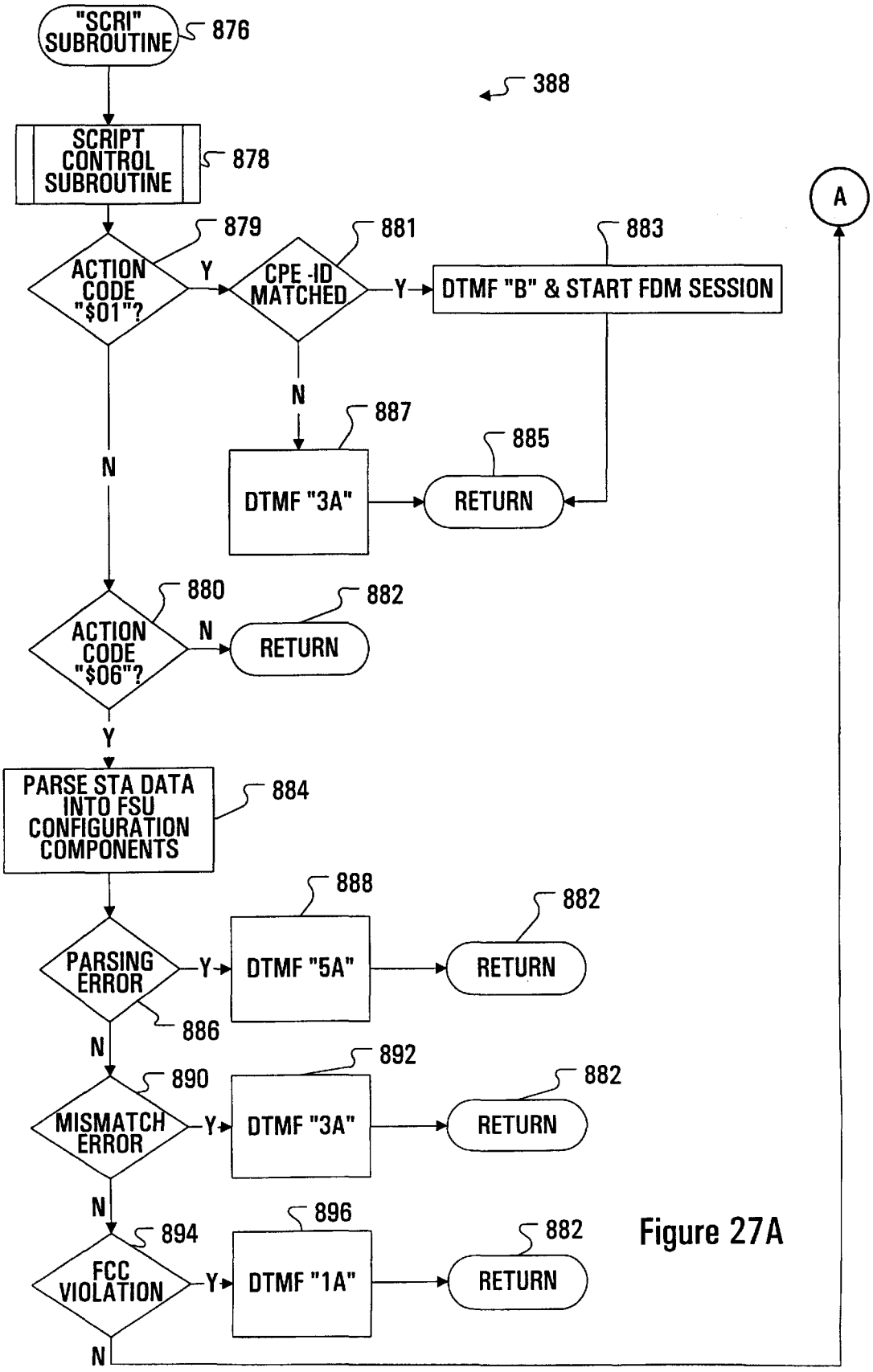
"SCRI" SUBROUTINE
876
388
SCRIPT CONTROL SUBROUTINE
878
A
879
881
883
ACTION CODE "$01"?
Y
CPE -ID MATCHED
Y
DTMF "B" & START FDM SESSION
N
887
885
N
DTMF "3A"
RETURN
880
882
ACTION CODE "$06"?
N
RETURN
Y
PARSE STA DATA INTO FSU CONFIGURATION COMPONENTS
884
888
882
PARSING ERROR
Y
DTMF "5A"
RETURN
886
N
892
890
882
MISMATCH ERROR
Y
DTMF "3A"
RETURN
N
896
894
882
FCC VIOLATION
Y
DTMF "1A"
RETURN
N

Figure 27A

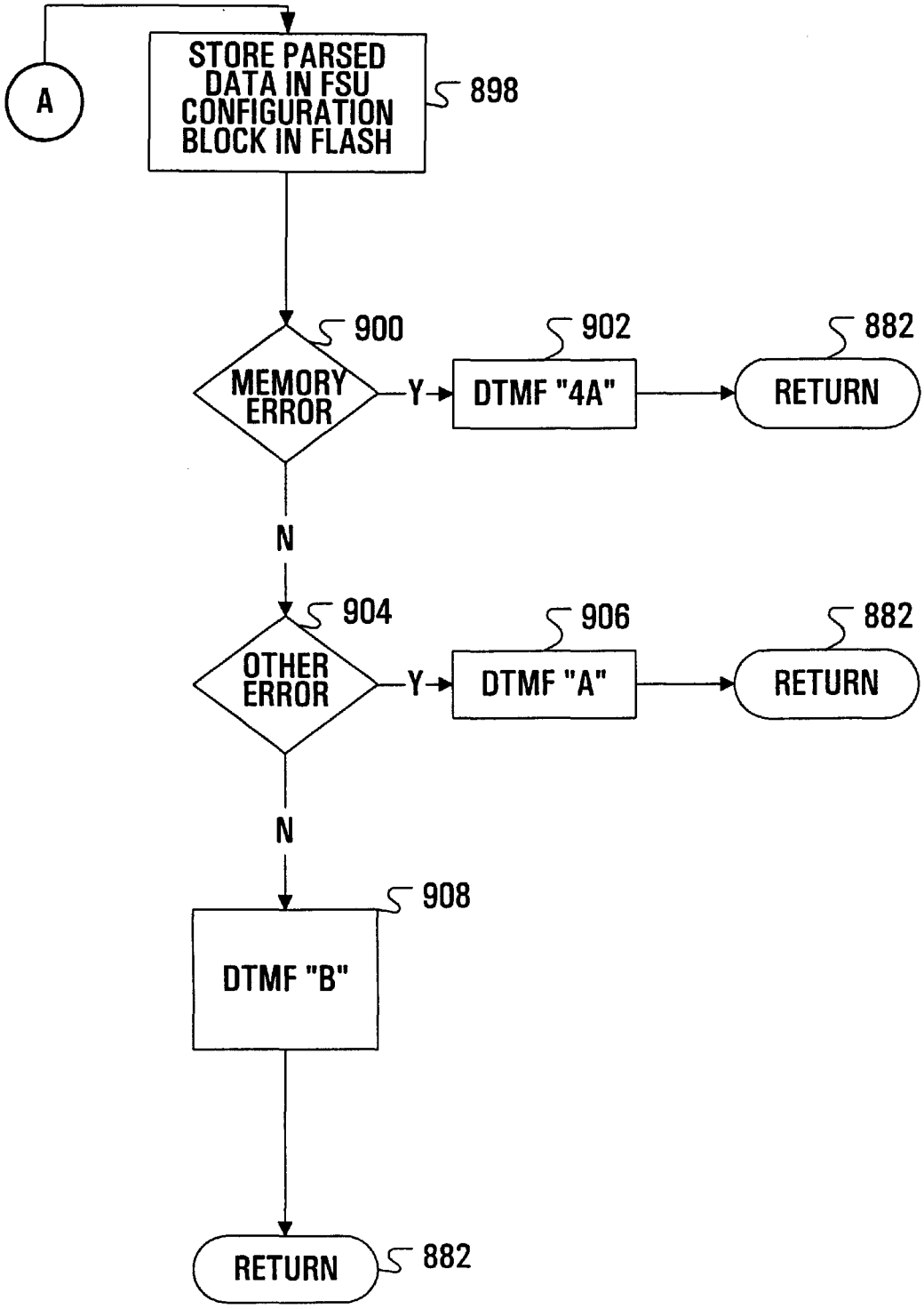
A
STORE PARSED DATA IN FSU CONFIGURATION BLOCK IN FLASH
898
900
MEMORY ERROR
Y
902
DTMF "4A"
882
RETURN
N
904
OTHER ERROR
Y
906
DTMF "A"
882
RETURN
N
908
DTMF "B"
RETURN
882

Figure 27B

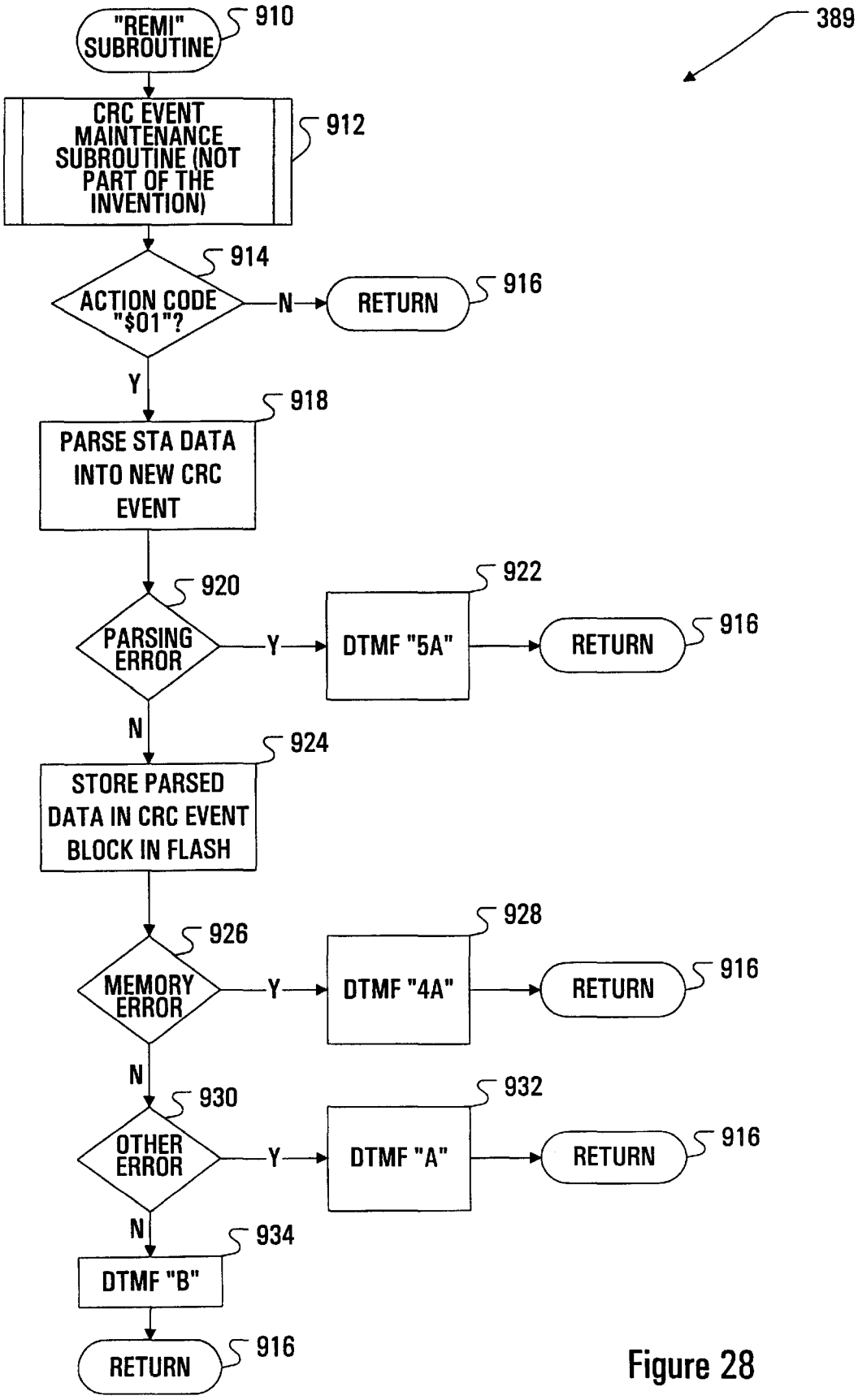
389
"REMI" SUBROUTINE
910
CRC EVENT MAINTENANCE SUBROUTINE (NOT PART OF THE INVENTION)
912
914
ACTION CODE "$01"?
N
RETURN
916
Y
918
PARSE STA DATA INTO NEW CRC EVENT
920
PARSING ERROR
Y
922
DTMF "5A"
RETURN
916
N
924
STORE PARSED DATA IN CRC EVENT BLOCK IN FLASH
926
MEMORY ERROR
Y
928
DTMF "4A"
RETURN
916
N
930
OTHER ERROR
Y
932
DTMF "A"
RETURN
916
N
934
DTMF "B"
RETURN
916

Figure 28

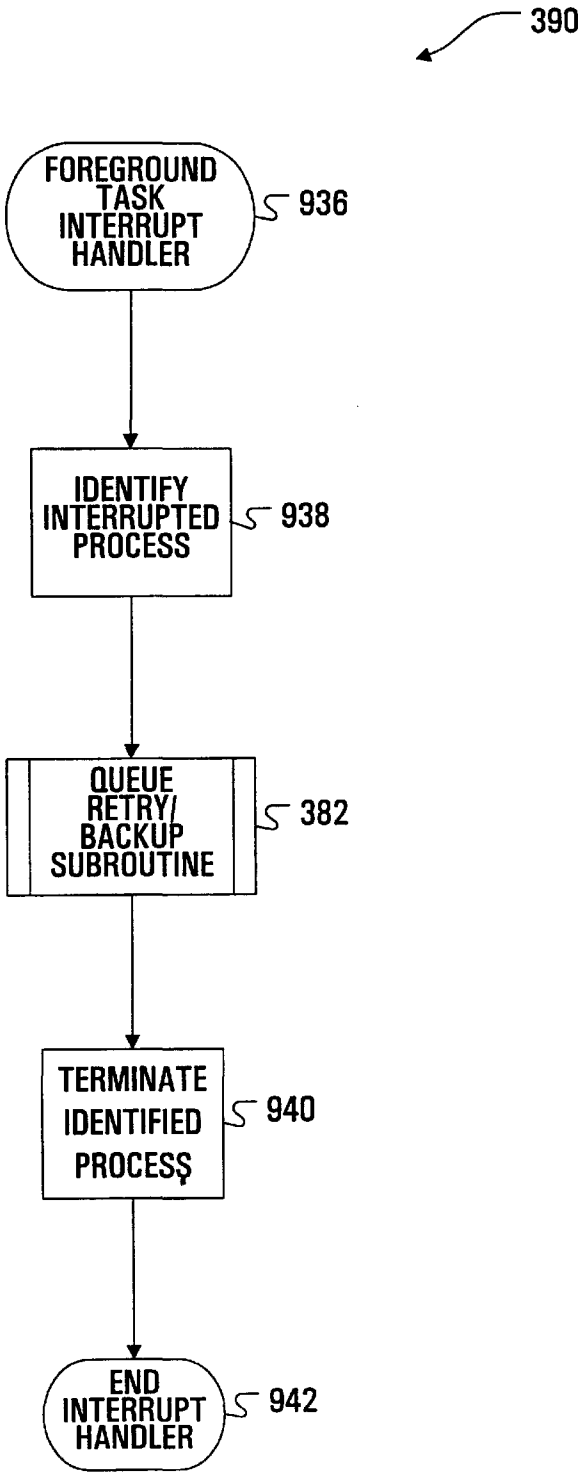
390
FOREGROUND TASK INTERRUPT HANDLER
936
IDENTIFY INTERRUPTED PROCESS
938
QUEUE RETRY/ BACKUP SUBROUTINE
382
TERMINATE IDENTIFIED PROCESS
940
END INTERRUPT HANDLER
942

Figure 29

METHOD, SYSTEM AND APPARATUS FOR CAUSING CUSTOMER PREMISES EQUIPMENT TO AUTOMATICALLY CALL A TELECOMMUNICATIONS SERVER

BACKGROUND OF THE INVENTION

This invention relates to customer premises equipment (CFE) that is enabled to autonomously call a telecommunications server, including a Stored Program Computer System (SPCS).

As CPEs are built with increased computational power, it has become desirable for them to communicate regularly with an SPCS server both to exchange data and to update their local configuration and programming scripts. Such communication is most sensibly conducted over the pre-existing Public Switched Telephone Network (PSTN) to which both the CPE and the SPCS are connected.

It is desirable that communication between the CPE and the SPCS not unduly interfere with the CPE's conventional use as a telephone. For this reason, it would be inappropriate for the SPCS to initiate a call to the CPE because, in the absence of a CPE ring-suppression mechanism, the CPE would ring and thereby misleadingly summon its user. It is therefore desirable for the CPE to autonomously call the SPCS on a regular basis or in response to specific events.

SUMMARY OF THE INVENTION

In accordance with one aspect of the invention, there is provided a method of receiving information at customer premises equipment. The method includes the steps of transmitting an information request message from the CPE to a messaging platform and receiving information from the messaging platform in response to the information request message.

In accordance with another aspect of the invention, there is provided a method of providing messaging services at customer premises equipment. The method includes the steps of transmitting an information request message from the CPE to a messaging platform and receiving information from the messaging platform in response to the information request message.

In accordance with another aspect of the invention, there is provided an apparatus for receiving information at customer premises equipment. The apparatus includes a transmitter for transmitting an information request message from the CPE to a messaging platform and a receiver for receiving information from the messaging platform in response to the information request message.

Preferably, the transmitter is operable to receive at least one stimulus and transmit the request message in response to the stimulus.

In one embodiment, the apparatus includes a timer tor identifying a pre-defined time, the timer being operable to stimulate the transmitter to transmit the message at a pre-defined time.

In one embodiment, the apparatus includes a timer for identifying pre-defined intervals, the timer being operable to stimulate the transmitter to transmit the message at the pre-defined intervals.

In one embodiment, the apparatus includes a CLID receiver for receiving CLID information and a processor for determining whether or not the CLID information meets a pre-defined criteria and if so stimulating the transmitter to transmit the message.

Preferably, the transmitter includes a dialler and the dialler is operable to dial a first predefined telephone number. Preferably, the dialler is operable to transmit the information request using DTMP signals and the dialler is operable to dial at least one alternate telephone number when the first pre-defined telephone number is unavailable.

Preferably, the receiver is operable to receive the requested information according to the ADSI protocol.

In accordance with another aspect of the invention, there is provided an apparatus for providing messaging services at customer premises equipment. The apparatus includes a transmitter for transmitting an information request message from the CPE to a messaging platform and a receiver for receiving information from the messaging platform in response to the information request message.

In accordance with another aspect of the invention, there is provided a system for providing messaging services at customer premises equipment. The system includes a messaging platform for receiving and storing messages, the messaging platform including a receiver for receiving information request messages and a transmitter for transmitting information and customer premises equipment. The customer premises equipment includes a transmitter for transmitting an information request message from the CPE to the messaging platform and a receiver for receiving information from the messaging platform in response to the information request message.

Preferably, the apparatus includes a public telephone network, the CPE and the messaging platform being operable to communicate over the public telephone network.

In one embodiment, the information includes an e-mail message or a page.

Preferably, the transmitter in the messaging platform includes a FSK transmitter for transmitting the information in an FSK format to the CPE and preferably, the receiver in the CPE includes an FSK receiver for receiving the information in an PSK format.

Preferably, the information is provided to the CPE in accordance with an ADSI protocol extension.

Preferably, the transmitter at the CPE includes a DTMF transmitter and the receiver at the messaging platform includes a DTMF receiver, the information request message being transmitted to the messaging platform in DTMF signals.

In one sense, the present invention is directed to a CPE that autonomously calls a selection of SPCS servers in response to six types of stimulus.

In Multiple Daily Polling (MDP) the CPE logs into its main SPCS according to a regular schedule, both to exchange message data and to update local configuration parameters and programming script.

In Sent Page Polling (SPP), the CPE logs into a paging system SPCS to determine whether a page that the CPE had previously transmitted has been read and responded to yet.

In Calendar Reminder Calling (CRC), the CPE logs into a message upload SPCS to transmit to a particular subscriber a reminder page regarding an upcoming event.

In Intelligent Remote Notification (IRN), the CPE logs into a message upload SPCS to transmit a page to a particular subscriber confirming that the CPE has received a telephone call from a predefined telephone number.

In Feature Download Management (FDM) Slot Updating (FSU), the CPE logs into a script SPCS to commence an FDM session to update its programming scripts.

In Message Sending Retrying (MSR), the CPE logs into a message upload SPCS to attempt to upload a page that the CPE user had previously tried to upload manually without success.

An autonomous calling system must overcome certain challenges. The SPCS will be busy some of the times that the CPE, calls. Or a communication channel problem might develop during a connect session. The present invention provides for the CPR to retry unsuccessful calls and to call backup SPCSs when connecting to a primary SPCS has been continuously problematic.

The present invention further provides that the CPE will terminate an ongoing autonomous call when a user engages the CPE through actions that include lifting the handsct and operating the keypad. Where the user has recently operated the CPE, the invention provides that autonomous calls be delayed until sufficient idle time has passed to indicate that the user no longer requires the CPE for his own use. These provisions will sometimes require that the CPE retry autonomous calls at a later time.

The mechanism for achieving this functionality includes extensions to the core Analogue Display Services Interface (ADSI) protocol through which an SPCS transmits information for display at the CPE. In particular, certain new Switch To Application (STA) codes are defined to control and reconfigurc the CPE under the direction of the SPCS.

BRIEF DESCRIPTION OF THE DRAWINGS

These and other features, advantages, and aspects of the present invention will become better understood with reference to the following description, appended claims and accompanying drawings in which:

FIG. **1** is a block diagram of a customer premises equipment (CPE) apparatus according to a first aspect of the invention, the apparatus including circuitry enabling it to autonomously connect to a stored program computer system (SPCS) server through a public switched telephone network;

FIG. **2** is a tabular representation of families of Analog Display Services Interface (ADSI) switch to application (STA) code extensions that facilitate autonomous connection of the CPE to the SPCS;

FIG. **3** is a tabular representation of ADSI STA codes that comprise a subset of a "QUER" family of ADSI STA code extensions illustrated in FIG. **2**; FIG. **4** is a tabular representation of ADSI STA codes that comprise a subset of a "FONE" family of ADST STA code extensions illustrated in FIG. **2**;

FIG. **5** is a tabular representation of ADSI STA codes that comprise a subset of a "MESG" family of ADSI STA code extensions illustrated in FIG. **2**;

FIG. **6** is a tabular representation of ADSI STA codes that comprise a subset of a "SCRI" family of ADSI STA code extensions illustrated in FIG. **2**;

FIG. **7** is a tabular representation of ADSI STA codes that comprise a subset of a "POLL" family of ADSI STA code extensions illustrated in FIG. **2**;

FIG. **8** is a tabular re-presentation of ADSI STA codes that comprise a subset of a "REMI" family of ADSI STA code extensions illustrated in FIG. **2**;

FIG. **9** is a tabular representation of the organization of a segment of a Read only Memory (RAM) in the CPE illustrated in FIG. **1**;

FIG. **10**A–C are a tabular representation of the organization of a segment of a non-volatile FLASH memory in the CPE illustrated in FIG. **1**;

FIG. **11** is a tabular representation of programs encoded in a segment of a Read Only Memory (ROM) in the CPE illustrated in FIG. **1**;

FIG. **12** is a flowchart of a Main Program loop encoded in the ROM as illustrated in FIG. **11**;

FIG. **13** is a flowchart of an Autonomous Calling Module encoded in the ROM as illustrated in FIG. **11**;

FIG. **14**A and B are a flowchart of a Timkeeping Interrupt Handler encoded in the ROM as illustrated in FIG. **11**;

FIG. **15** is a flowchart of an Originating Subroutine encoded in the ROM as illustrated in FIG. **10**;

FIG. **16** is a flowchart of a Retry Subroutine encoded in the ROM as illustrated in FIG. **11**;

FIG. **17** is a flowchart of a Backup Subroutine encoded in the ROM as illustrated in FIG. **11**;

FIG. **18**A and B are a flowchart of an MDP Subroutine encoded in the ROM as illustrated in FIG. **11**;

FIG. **19** is a flowchart of a CRC Subroutine encoded in the ROM as illustrated in FIG. **11**;

FIG. **20** is a flowchart of an FSU Subroutine encoded in the ROM as illustrated in FIG. **11**;

FIG. **21** is a flowchart of an SPP Subroutine encoded in the ROM as illustrated in FIG. **11**;

FIG. **22** is a flowchart of an IRN Subroutine encoded in the ROM as illustrated in FIG. **11**;

FIG. **23** is a flowchart of an MSR Subroutine encoded in the ROM as illustrated in FIG. **11**;

FIG. **24** is a flowchart of a Queue Retry/Backup Subroutine encoded in the ROM as illustrated in FIG. **11**;

FIG. **25**A and B are a flowchart of a POLL Subroutine encoded in the ROM as illustrated in FIG. **11**;

FIG. **26** is a flowchart of an MESG Subroutine encoded in the ROM as illustrated in FIG. **11**;

FIG. **27**A and B are a flowchart of an SCRI Subroutine encoded in the ROM as illustrated in FIG. **11**;

FIG. **28** is a flowchart of an REMI Subroutine encoded in the ROM as illustrated in FIG. **11**; and

FIG. **29** is a flowchart of a Foreground Task Interrupt Handler encoded in the ROM as illustrated in FIG. **11**.

DETAILED DESCRIPTION

FIG. **1**

Referring to FIG. **1**, a customer premises equipment (CPE) apparatus according to a first embodiment of the invention is generally illustrated at **10**. A conventional tip and ring loop **30** connects the CPE **10** to a public switched telephone network (PSTN) **38**. A stored program computer system (SPCS) **36** is connected to the CPE **10** through the PSTN **38** and the tip and ring loop **30**. The CPE **10** is enabled to autonomously call the SPCS **36** to exchange page messages as well as to download configuration parameters and programming scripts. This autonomous connection ensures both that page messages to and from the user of the CPE are regularly delivered and that the CPE **10** itself is maintained properly configured.

The SPCS **36** runs a messaging platform and has an FSK transmitter **37** for transmitting information to the CPE and has a DTMF receiver **39** for receiving information request messages from the CPE.

The CPE **10** includes a microprocessor **12**, and memory including Random Access Memory (RAM) **14**, Read Only Memory (ROM) **16**, and non-volatile random access memory **18**, in this embodiment, Flash EEPROM (FLASH), all connected to be controlled by and to exchange data with the microprocessor **12**. It has been found that the Mitsubishi 37702 16-bit microprocessor having 2 K of RAM and 32 K of ROM on-chip and having conventional address and data buses is suitable as the microprocessor **12**. It has also been found that AMD 29040 Flash EEPROM is suitable as the FLASH **18**. The RAM **14** and the ROM **16** are conventional.

The CPR **10** is further connected to control and exchange data with a Frequency Shift Key (FSK) receiver **22**, a Dual Tone Multi-Frequency (DTMF) transmitter **24**, a modem **26**, a display **25** and a keypad **27**, each of which has a designated address space. The microprocessor **12** is operable to cooperate with the DTMP transmitter to act as a dialler for dialling telephone numbers and is operable to present messages at the display **25** and to receive user instructions at the keypad **27**.

A conventional hook switch sensor **28** is connected to provide an off-hook signal to the microprocessor **12** to indicate whether or not the CPE handset (not shown) is on or off hook. A telephony platform **34** and an off-hook bypass switch **32** are connected in parallel to the tip and ring loop **30**. Moreover, the off-hook bypass switch **32** is connected indirectly to the tip and ring loop **30** through the telephony platform **34**.

The FSK receiver **22**, the DTMF transmitter **24**, and the modem **26** are each connected to the tip and ring loop **30** via the off-hook bypass switch **32** which, under the direction of the microprocessor **12** as informed by the hook switch sensor **28**, routes this connection either directly to the network side of the tip and ring loop **30** or indirectly to the tip and ring loop **30** through the telephony platform **34**. The on-hook bypass switch **32** therefore enables the microprocessor **12** and the SPCS **36** to communicate at all times regardless of the hook state of the CPE **10**

In most situations, information from the SPCS **36** is conveyed to the microprocessor **12** via the FSK receiver **22** and information from the microprocessor **12** is conveyed to the SPCS **36** via the DTMF transmitter **24**. For large uploads, the SPCS **36** may be connected to communicate at higher speed with the microprocessor **12** via the modem **26**.

The core Analog Display Services Interface (ADSI) standard specifies a mechanism for presenting SPCS **36** information at the CPE **10**. The SPCS **36** and the CPE **10** both comply with the standard ADSI protocol and the SPCS **36** is enabled to transmit messages for presentation at the CPE **10** on the display **25**. Despite this limited scope, the standard ADSI protocol does provide a foundation on which to build additional functionality. This foundation is the Switch To Application (STA) code. Where both SPCS **36** and CPE **10** are programmed to interpret a particular set of extended STA codes communicated in the course of a standard ADSI session, further functionality is enabled.

FIG. **2**

Referring to FIGS. **1** and **2**, families of ADSI STA extension codes are generally illustrated in tabular form at **40**. These families **40** are an aspect of the present invention and are not specified in the core ADSI protocol. These families **40** are formatted to comply with the requirements for extended STA codes as set forth in the ADSI protocol as defined in Bellcore standard TR-1273 rev.1, which is incorporated herein by reference.

The ADSI protocol standard specifies that all ADSI STA codes have a specific format. Each extended STA code must be a minimum of six bytes in length and a maximum of 255 bytes in length. A first byte (**42**) must represent the value "**151***d*"which is the parameter defined to signify that the SPCS **36** is requesting the CPE **10** to switch to, or in other words commence, an application (Switch To Application—STA). A second byte (**44**) specifies the length of the STA message, which must be between 6 and 255 bytes long. Bytes 3 through 6 (**46**, **48**, **50**, **52**) specify an application code that designates the specific application that the CPE **10** is being requested to commence. In the present embodiment, the application code **46**, **48**, **50**, **52** is the set of ASCII values: "Q" "U" "E" "R", "F" "O" "N" "E", "M" "E" "S" "G", "S" "C" "R" "I", "P" "O" "L" "L", or "R" "E" "M" "I", which conveniently form mnemonics representing respectively; QUERying application **54**, FONE (phone) controlling application **55**, MESsaGing application **56**, SCRIpt application **58**, POLLing application **60**, and REMInder application **61**. As specified in the standard ADSI protocol, an STA code might further include application specific data. In this embodiment, the application specific data includes a one byte action code **62** and a data stream **64** associated with the action code **62**, the data stream **64** having a length of between 1 and 248 bytes and a format dictated by the action code **62**. The action code **62** can be understood to represent a command transmitted by the SPCS **36** to the CPU **10**.

FIG. **3**

Referring to FIGS. **1**,**2** and **3**, there is tabulated a subset, generally illustrated at **66**, of STA messages **54** comprising an ADSI protocol extension QUER for enabling the SPCS **36** to query the CPE **10**. The subset **66** includes a first query **68** that performs the following operation:

"QUERY: What communication mode is the CPE **10** seeking to establish with the SPCS **36**?"

The first query **68** is identified by the action code **62** $81 and passes no data **64**. The first query **68** enables the SPCS **36** to determine the purpose of the call, be it to upload data, download data, update its configuration parameters, or download new scripts.

FIG. **4**

Referring to FIGS. **1**,**2** and **4**, there is tabulated a subset, generally illustrated at **67**, of STA messages **55** comprising an ADSI protocol extension FONE for enabling the SPCS **36** to remotely control the CPR **10**. The subset **67** includes a first command **69** that performs the following operation:

"COMMAND: This marks the end of this connect session. Acknowledge successful session completion with a DTMF "B" and then release the telephone line."

This first command **69** is identified by the action code **62** $01 and passes no data **64**. The first command **69** enables the SPCS **36** to cleanly terminate a connect session.

FIG. **5**

Referring to FIGS. **1**,**2** and **5**, there is tabulated a subset, generally illustrated at **70**, of STA messages **56** comprising an ADSI protocol extension MESG for enabling the SPCS **36** to configure parameters within the CPE **10** that affect: Intelligent Remote Notification (IRN), Calendar Reminder Call (CRC), and Message Sending Retry (MSR) sessions. The subset **70** includes a first command **72** that performs the following operation:

"COMMAND: Parse and store into FLASH **18** the enclosed updated configuration parameters."

This first command **72** is identified by the action code **62** $18 and passes to the CPE **10** data **64** including bytes 1 through 12 which represent the primary message sending server telephone number in binary coded decimal (BCD) format (**74**), byte 13 which is a delimiter $FF (**76**), bytes 14 through 25 which represent the backup message sending server telephone number in BCD format (**78**), byte 26 which is a delimiter FP (**80**), byte 27 which represents in binary format the number of times that the CPE **10** is to retry the SPCS **36** if no successful connection has been achieved (**82**), byte 28 which represents the time delay between consecutive connection retries measured in 10-second increments and stored in binary format (**84**), bytes 29 through 89 which represent a text message in ASCII format to be displayed at the CPE display **25** during an IRN, CRC, or MSR session (**86**), byte 90 which is a delimiter FF (**88**), bytes 91 through 94 which represent the current IRN/CRC/MSR security code

in binary format (**90**), byte 95 is a delimiter $FF (**92**), bytes 96 through 103 represent a new IRN/CRC/MSR security code in binary concatenated twice for confirmation (**94**), and byte 104 is an end of data delimiter $FF (**96**).

These parameter data lengths are maximum lengths. In use, some parameters may have a shorter data length than indicated, in which case the delimiter $FF ensures proper separation. The telephone numbers specified for the primary and backup servers are dialable BCD strings, where the digits 1 through 9 are represented by their respective nibbles. DTMF "*" is represented by nibble $A. DTMF "**0**" is represented by nibble $B. DTMF "#" is represented by nibble $C. Other nibbles are not dialled and terminate the BCD string.

FIG. **6**

Referring to FIGS. **1**,**2** and **6**, there is tabulated a subset, generally illustrated at **98**, of STA messages **58** comprising an ADSI protocol extension SCRI for enabling the SPCS **36** to configure the parameters within the CPE **10** that control the Feature Download Management (FDM) Slot Update (FSU) feature. The subset **98** includes a first command **99** that performs the following operation:

"COMMAND: Commence a non-interactive FDM session within the current Server Display Control (SDC) session."

This first command **99** is identified by the action code **62** $01 and passes data **64** structured as a four byle binary CPE identification code (CPE-ID) **101**.

The FSU subset **98** also includes a second command **100** that performs the following operation:

"COMMAND: Parse and store into FLASH **18** the enclosed updated configuration parameters."

This second command **100** is identified by the action code **62** $06 and passes data **64** structured as follows: bytes 1 through 4 represent the FDM number in binary format (**102**), bytes 5 through 8 represent the FSU security code in binary format (**104**), bytes 9 through 12 represent the time to next call the SPCS **36** in BCD format as month/day/hour/minute (**106**), byte 13 is a delimiter $FF (**108**), bytes 14 through 26 represent the FSU server telephone number in BCD format (**110**), byte 27 is a delimiter $FF (**112**), bytes 28 through 39 represent the backup FSU server telephone number in BCD format (**114**), byte 40 is a delimiter $FF (**116**), byte 41 represents in binary format the number of times that the CPE **10** is to retry the SPCS **36** if no successful connection has been achieved (**118**), byte 42 represents the time delay between consecutive connection retries measured in 10 second increments and stored in binary format (**120**), bytes 43 through 102 represent a text message in ASCII format to be displayed at the CPE display **25** during an FSU session (**122**), and byte 90 is an end of data delimiter $FF (**124**).

These parameter data lengths are maximum lengths. In use, some parameters may have a shorter data length than indicated, in which case the delimiter $FF ensures proper separation. The telephone numbers specified for the primary and backup servers are dialable BCD strings, where the digits 1 through 9 are represented by their respective nibbles. DTMF "*" is represented by nibble $A. DTMP "**0**" is represented by nibble $B. DTMF "#" is represented by nibble $C. Other nibbles are not dialled and terminate the BCD string.

FIG. **7**

Referring to FIGS. **1**,**2** and **7**, there is tabulated a subset, generally illustrated at **126**, of STA messages **60** comprising an ADSI protocol extension POLL for enabling the SPCS **36** Lo configure the parameters within the CPE **10** that affect the Multiple Daily Polling sessions (MDP) and the Sent Page Polling sessions (SPP).

A first command **128** performs the following operation:

"COMMAND: Parse and store into FLASH **18** the enclosed updated MDP configuration parameters."

The first command **128** is identified by the action code **62** $01 and passes data **64** structured as follows: bytes 1 through 12 represent the primary messaging server telephone number in BCD format (**130**), byte 13 is a delimiter $FF (**132**), bytes 14 through 25 represent the backup messaging server telephone number in BCD foxmat (**134**), byte 26 is a delimiter $FF (**136**), bytes 27–28 represent the first daily poll time in BCD stored as hour/minute (**138**), bytes 29–30 represent the second daily poll time in BCD stored as hour/minute (**140**), bytes 31–32 represent the third daily poll time in BCD stored as hour/minute (**142**), bytes 33–34 represent the fourth daily poll time in BCD stored as hour/minute (**144**), bytes 35–36 represent the fifth daily poll time in BCD stored as hour/minute (**146**), byte 37 is a delimiter $FF (**148**), byte 38 represents in binary format the number of times that the CPE **10** is to retry the PCS **36** if no successful connection has been achieved (**150**), byte 39 represents the time delay between consecutive connection retries measured in 10 second increments and stored in binary format (**152**), bytes 40 through 99 represent a text message in ASCII format to be displayed at the CPE display **25** during an MDP session (**154**), and byte 100 is an end of data delimiter $FF (**156**).

The second command STA **158** performs the following operation:

"COMMAND: Parse and store into FLASH **18** the enclosed updated SPP configuration parameters."

The second command **158** is identified by the action code **62** $02 and passes data **64** structured as follows: bytes 1 through 12 represent the primary page response server telephone number in BCD format (**160**), byte 13 is a delimiter $FF (**162**), bytes 14 through 25 represent the backup page response server telephone number in BCD format (**164**), byte 26 is a delimiter $FF (**166**), byte 27 represents the delay measured in 10 second increments and stored in binary format between the time when the CPE **10** originally sent a page and the time that the CPB **10** is to check with the SPCS **36** whether the page has been received (**168**), byte 28 represents the time delay between consecutive connection retries measured in 10 second increments and stored in binary format (**170**). byte 29 represents in binary format the number of times that the CPE **10** is to retry the SPCS **36** if no successful connection has been achieved (**172**), bytes 30 through 89 represent a text message in ASCII format to be displayed at the CPE display **25** during an SPP session (**174**), and byte 90 is an end of data delimiter $FF (**176**).

These parameter data lengths are maximum lengths. In use, some parameters may have a shorter data length than indicated, in which case the delimiter $FF ensures proper separation. The telephone numbers specified for the primary and backup servers are dialable BCD strings, where the digits 1 through 9 are represented by their respective nibbles. DTMF "*" is represented by nibble $A. DTMF "0" is represented by nibble $B. DTMF "#" is represented by nibble $C. Other nibbles are not dialled and terminate the BCD string.

FIG. **8**

Referring to FIGS. **1**,**2** and **8**, there is tabulated a subset, generally illustrated at **127**, of STA messages **61** comprising an ADSI protocol extension REMI for enabling the SPCS **36** to manipulate in CPE FLASH **18** Calendar Reminder Call (CRC) events. The subset **127** includes a first command **129** that performs the following operation: "COMMAND: Parse and store into FLASH **18** the enclosed calendar reminder call event"

This first command **129** is identified by the action code **62** $01 and passes to the CPE **10** data **64** structured as follows: bytes 1 through 4 represent the event time in binary formatted as (MM;DD:HH:MM) **131**, bytes 5 through 8 represent the reminder time in binary formatted as (MM:DD:HH:MM) **133**, byte 9 represents the repeating status of the event (daily, weekly, monthly, yearly, non-repeating) in binary format **135**, bytes 10 through 69 represent the reminder message in ASCII format **137**, byte 70 is a delimiter $FF **139**, and bytes 71 through 82 represent in BCD format the telephone number of the pager that will receive the reminder message **141**.

These parameter data lengths are maximum lengths. In use, some parameters may have a shorter data length than indicated, in which case the delimiter $FF ensures proper separation. The telephone numbers specified for the primary and backup servers are dialable BCD strings, where the digits 1 through 9 are represented by their respective nibbles. DTMF "*" is represented by nibble $A. DTMF "**0**" is represented by nibble $B. DTMF "#" is represented by nibble $C. Other nibbles are not dialled and terminate the BCD string.

Generally, the first command is used to download reminder times and paging messages which the processor compares to the system clock to determine when to transmit an information request message.

FIG. **9**

Referring to FIGS. **1** and **9**, the structure of a segment of the RAM **14** is generally illustrated in tabular form at **178**. The RAM **14** is configured into four main blocks: a block of buffers **180**, a block of clocks and timers **182**, a block of status flags **184**, and a block of counters **186**.

The block of buffers **180** includes a received data queue **185** and a Calling Line Identification Delivery (CLID) buffer **187**.

The received data queue **185** is conventional and is configured to queue ADSI messages as they arrive at the CPE from the SPCS **36**. The ADSI standard data-link layer verifies inbound data and the received data queue **185** is connected to receive only complete and verified ADSI messages. The received data queue **185** maintains separate read and write pointers (not illustrated) to indicate the specific queue location where reading or writing is to next commence.

The CLID buffer **187** receives for storing the most recent CLID data to be received by the CPE **10**.

The block of clocks and timers **182** is a set of RAM **14** locations that store values formatted as either calendar time or raw time. These values are regularly updated by the clock portion of the microprocessor **12**.

A system clock **188** tracks the current date and time. A CPE **10** idle timer **190** monitors the period of time that the CPE **10** has been idle, untouched by its user. A call expiry timer **192** monitors the length of any autonomous call to enforce a maximum duration limit. An initial SPP delay timer **194** measures the delay between the time that the CPE **10** sends a page and the time that it initiates an autonomous call to search for a response to that page. A set of retry timers **196, 198, 200, 202, 204, 206** measures the delay between an unsuccessful autonomous calling session, whether MDP, SPP, CRC, IRN, FSU, or MSR, and the next retry attempt. A set of backup timers **208, 210, 212, 214, 216, 218** measures the delay between an unsuccessful autonomous calling session, whether MDP, SPP, CRC, IRN, is FSU, or MSR, and the next attempt to connect to a backup server.

The block of status flags **184** includes a designated flag **220, 222, 224, 226, 228, 230** for each session type MDP, SPP, CRC, IRN, FSU, and MSR to indicate whether it is time to initiate such a session. Each flag **220, 222, 224, 226, 228, 230** comprises four indicators chat determine whether it is time to originate a session **221**, to retry a session **223**, to try a backup session **225**, or to disable further such sessions **227** pending correction of an unacceptable error level.

The block of counters **186** includes a respective counter **232, 234, 236, 238, 240, 242** to monitor the number of retry attempts for each session type MDP, SPP, CRC, IRN, FSU and a respective counter **244, 246, 248, 250, 252,254** to monitor the number of attempts to connect to a backup server for each type of session MDP, SPP, CRC, IRN, FSU.

FIG. **10**

With reference to FIGS. **1** and **10**, the structure of a segment of the FLASH **18** memory is generally illustrated at **256** in tabular form. The FLASH **19** is arranged into six main blocks: a block storing the configuration for MDP sessions **258**, a block storing the configuration for SPP sessions **260**, a block storing the configuration for IRN/CRC/MSR sessions **262**, a block storing the configuration for FSU sessions **264**, a block storing the global autonomous calling configuration **266**, and a block storing user data **268**.

The MDP configuration block **258** stores a primary message server telephone number **270**, a backup message server telephone number **272**, up to five polling times **274, 276, 278, 280, 282**, a number of times to retry a server in the event of a failed autonomous call **284**, a delay period between retry attempts **286**, and a message to display at the CPE **10** during an MDP session **288**.

The SPP configuration block **260** stores a primary page response server telephone number **290**, a backup page response server telephone number **292**, an initial delay between the time that the CPE **10** transmitted a page and the time when an SPP session is initiated to determine if the page has been received **294**, a delay time period between attempts to retry a server in the event of a failed autonomous call **296**, a maximum number of retry attempts **298**, and a message to display at the CPE **10** during an SPP session **300**.

The IRN/CRC/MSR configuration block **262** stores a primary message sending server telephone number **302**, a backup message sending server telephone number **304**, a maximum number of times to retry a server in the event of a failed autonomous call **306**, a delay period between retry attempts **308**, a message to display at the CPE **10** during an IRN, CRC, or MSR session **310**, and a security code needed to log onto the server **312**.

The FSU configuration block **264** stores a primary FSU server telephone number **314**, a backup FSU server telephone number **316**, a maximum number of times to retry a server in the event of a failed autonomous call **318**, a delay period between retry attempts **320**, the next time to call the server **322**, a message to display at the CPE **10** during an FSU session **324**, and a security code needed to log onto the server **326**.

The global configuration block **266** stores a minimum separation between autonomous calling times **327**, a minimum time that the CPE **10** must be idle before an autonomous call can be made **328**, a maximum autonomous call duration **329**, a maximum number of connection errors after which a particular type of session (MDP, SPP, IRN, CRC, FSU, MSR) will be disabled pending diagnosis **330**, a cumulative number of connection errors for each type of session (MDP, SPP, IRN, CRC, FSU, MSR) **332, 334,336, 338, 340, 342**, a four byte CPE identification number (CPE-ID) **339**, a set of error log records **343** indicating the time and program location of each connection error, and an

array **345** to store any maximum or minimum parameter values specified by telecommunication regulations.

The user data storage block **268** contains two types of data: calendar reminder call (CRC) data packets **344** and intelligent remote notification (IRN) data packets **346**.

Each CRC data packet **344** includes a calendar time of an event **348**, a calendar time to transmit an event reminder **350**, an event repeat status flag **352** that can indicate unique, daily, weekly, monthly, and yearly events, a reminder message to be displayed **354**, and a telephone number of the pager selected to receive the reminder message **349**.

Each IRN data panket **346** includes a telephone number to be matched to Calling Line Identification Delivery data **351**, a telephone number of an associated pager to be called in the event of a match **353**, and a notification message to be displayed at the pager **355**.

FIG. **11**

Referring to FIGS. **1** and **11**, the structure of a segment of the ROM **16** is generally illustrated at **356** in tabular form. The ROM **16** is programmed with codes representing instructions which direct the microprocessor **12** to perform various functions, including functions implementing: a Main Loop **358**, an Autonomous Calling Module **360**, a Timekeeping Interrupt Handler **362**, an Originating Subroutine **364**, a Retry Subroutine **366**, a Backup Subroutine **368**, an MDP Subroutine **370**, a CRC Subroutine **372**, an FSU Subroutine **374**, an SPP Subroutine **376**, an IRN Subroutine **368**, an MSR Subroutine **380**, a Queue Retry/Backup Subroutine **382**, a POLL Subroutine **384**, an MESG Subroutine **386**, an SCRI Subroutine **388**, an REMI Subroutine **389** and a Foreground Task Interrupt Handler **390**.

Operation

FIG. **12**

The operation of the CPE **10** and its interaction with the SPCS **36** will now be described with reference to FIG. **12** which is a flowchart of a Main Program Loop **358** as stored in the ROM **16**.

Referring to FIGS. **1** and **12**, the Main Loop **358** defines an initial state **392** from which the microprocessor **12** begins execution. Blocks **394** and **396** provide conventional code for directing the microprocessor **12** to control a CPE **10**, including instructions: on receiving and reacting to general ADSI instructions from the SPCS **36**, on how to switch between the voice and data modes as defined in the ADSI standard, on handling I/O interrupts from devices such as the keypad **27** and the hook switch sensor **28** and on how to implement certain other functionality not part of the invention, including creating, editing, and viewing messages at the CPE **10**, editing the user data block **268** of FLASH **18**, loading the CLID buffer **187** and updating the system clock **188**. The functionality to receive and react to general ADSI instructions is interrupt driven and fully specified in Bellcore standard TR-1273 wherein only complete and valid ADSI instructions are accepted by the CPE **10**. The microprocessor **12** is further configured to maintain all conventional stacks.

Block **398** defines an Autonomous Calling Module that provides code for directing the microprocessor **12** to perform the autonomous calling functions which are the subject of the present invention. As illustrated in FIG. **10**, the Autonomous Calling Module **398** is located within the Main Loop **358** but does not obstruct other processing defined within the Main Loop **358**.

FIG. **13**

With reference to FIGS. **1** and **13**, the Autonomous Calling Module is generally illustrated at **360** and defines an initial state **392** from which the microprocessor **12** begins execution. The Autonomous Calling Module **360** enables the CPE **10** to call the SPCS **36** upon the occurrence ot pre-specified events.

In overview, upon the occurrence of an event or stimulus that triggers autonomous calling, the CPR **10** attempts to dial a predefined telephone number to call a specified primary server, ie., the messaging platform. If a successful exchange of all required data is completed, the CPE **10** stops. Alternatively, if the call is not successful, the CPE **10** will retry calling the specified server for a specified number of attempts, waiting a specified delay between attempts. If a successful exchange of all required data is completed, the CPE **10** stops. If no successful call is made, the CPE **10** will then attempt to dial a specified backup server a specified number of attempts, waiting a specified delay between attempts. If a successful exchange of all retired data is completed, the CPE **10** stops. If no successful call has been made, then the CPE **10** deems that it has failed and gives up on this particular trigger event. If a sufficient number of failures have been recorded for a particular class of trigger event, then the CPE **10** will disable that class of trigger event until the problem is diagnosed and corrected.

Proceeding now to describe the process in greater detail, the originating Subroutine **364**, the Retry Subroutine **366**, and the Backup Subroutine **368** in parallel direct the microprocessor **12** to determine if an originating, retry, or backup session is due, and if so, to commence one. When the microprocessor **12** has been directed to return from such subroutines, **364**, **366**, **368**, then block **402** directs the microprocessor **12** to return to the Main Loop **358**.

FIG. **14**

With reference to FIGS. **1**, **9**, **10** and **14**, the Timekeeping Interrupt Handler is illustrated generally at **362** and defines an initial state **404** from which the microprocessor **12** begins execution. The Timekeeping Interrupt Handler **362** is invoked on an interrupt basis every time the CPE system clock **188** is incremented by one second. In this way, the Timekeeping Interrupt Handler **362** determines if any event has occurred that would dictate that the CPE **10** call the SPCS **36**. There are four classes of event that dictate such a call.

The first class of events corresponds to the arrival of a specified date and time. Block **406** directs the microprocessor **12** to compare the system clock **188** to the five MDP polling times **274**, **276**, **278**, **280**, **282** and to the reminder times **250** for CRC data packets **344** by block **408**, and to the next FSU poll time **322** by block **410**. If a match is found, then blocks **412**, **414**, **416** respectively direct the microprocessor **12** to set the originating session indicator **221** in the MDP flag **220**, the CRC flag **224**, or the FSU flag **228**. Thus, the CPE is operable to perform the step transmitting an information request message at a predefined time.

The second class of events corresponds to an event in which the CPE **10** checks whether a response to a previously transmitted page has been received at a server. When such a page was originally sent, the microprocessor **12** was directed (not shown) to initialize the SPP initial delay is timer **194** with the initial poll delay **294** configured in FLASH **18**. Block **418** directs the microprocessor **12** to detect when the SPP initial delay timer **194** has expired and hence when an SPP session should be commenced. If such a condition is detected, then block **420** directs the microprocessor **12** to set the originating session indicator **221** in the SPP flag **222**.

The third class of events corresponds to the arrival at the CPE **10** of a telephone call from a telephone number previously stored by the user as CLID data **351** in the user data block **268** of the FLASH **18**. Block **422** directs the microprocessor **12** to compare any new data in the CLID

buffer **180** to the data stored in the CLID data **351** section of FLASH **18**. If a match is detected, then block **424** directs the microprocessor **12** to set the originating session indicator **221** in the IRN flag **226**. Thus, the CPE performs the step of transmitting the information request message in response to CLID information.

The fourth class of events corresponds to the expiry of a timer **196, 200, 204, 198, 202, 206** delaying respectively an MDP, CRC, FSU, SPP, IRN, or MSR retry session or the expiry of a timer **208, 210, 212, 214, 216, 218** delaying respectively an MDP, CRC, FSU, SPP, IRN, or MSR backup session. If such a retry expiry is respectively detected by blocks **426, 430, 434, 428, 432, 436** or such a backup expiry is respectively detected by blocks **438, 440, 442, 444, 446, 448**, then blocks **450, 452, 454, 456, 458, 460, 462, 464, 466, 468, 470, 472** respectively direct the microprocessor **12** to set the retry session indicator **223** in the MDP, CRC, FSU, SPP, IRN, MSR flags **220, 224, 228, 222, 226, 230** or to set the backup session indicator **225** in the MDP, CRC, FSU, SPP, IRN, MSR flags **220, 224, 228, 222, 226, 230**. This, the CPE is directed to perform the step of transmitting the information request message at predefined intervals.

Finally, block **474** directs the microprocessor **12** to terminate the Timekeeping Interrupt Handler **362**.

FIG. **15**

With reference to FIGS. **1, 9** and **15**, the Originating Subroutine **364** is illustrated generally at **364** and defines an initial state **474** from which the microprocessor **12** begins execution. The Originating Subroutine **364** invokes an appropriate originating session after analyzing the originating indicator **221** and the disabled indicator **227** of the set of session flags **220, 222, 224, 226, 228**.

Block **476** directs the microprocessor **12** to determine whether the originating session indicator **221** in the MDP flag **220** has been set and the disabled indicator **227** is clear. If so, then block **478** directs the microprocessor **12** to clear the MDP retry and backup counters **232, 244**, clear the MDP retry and backup timers **196, 208**, and to clear in the MDP flag **220** the originating session indicator **221**, the retry session indicator **223**, and the backup session indicator **225**. Thereafter, the microprocessor **12** is directed to commence the MDP Subroutine **370**.

Similarly blocks **490, 482, 484**, and **486** direct the microprocessor **12** to determine whether the originating session indicator **221** in respectively the CRC, FSU, SPP, and IRN flags **224, 228, 222, 226** have been set and the disabled indicator **227** is clear. If so, then blocks **488, 490, 492, 494** respectively direct the microprocessor **12** to clear the CRC counters **236, 248**, timers **200, 212**, and flag's **224** originating **221**, retry **223**, and backup **225** indicators, to clear the FSU counters **216, 252**, timers **204, 216**, and flags **228** including originating **221**, retry **223**, and backup indicators **225**, to clear the SPP counters **234, 246**, timers **198, 210**, and flags **222** including originating **221**, retry **223**, and backup **225** indicators, and to clear the IRN counters **238, 250**, timers **202, 214**, and flag's **226** originating **221**, retry **223**, and backup **225** indicators. The microprocessor **12** is then respectively directed to commence the CRC Subroutine **372**, the FSU Subroutine **374**, the SPP Subroutine **376**, or the IRN Subroutine **378**.

Finally, block **496** directs the microprocessor **12** to return from the Originating Subroutine **364**.

FIG. **16**

With reference to FIGS. **1, 9** and **16**, the Retry Subroutine is illustrated generally at **366** and defines an initial state **498** from which the microprocessor **12** begins execution. The Retry Subroutine **366** invokes an appropriate retry session after analyzing the retry indicator **223** and the disabled indicator **227** of the set of session flags **220, 222, 224, 226, 228**.

Block **500** directs the microprocessor **12** to determine whether the retry session indicator **223** in the MDP flag **220** has been set and the disabled indicator **227** is clear. If so, then block **502** directs the microprocessor **12** to clear the retry session indicator **223** in the MDP flag **220**.

Thereafter, the microprocessor **12** is directed to commence the MDP Subroutine **370**.

Similarly blocks **504, 506, 508, 510**, and **512** direct the microprocessor **12** to determine whether the retry session indicator **223** in respectively the CRC, FSU, SPP, IRN, or MSR flags **224, 228, 222, 226, 230** has been set and the disabled indicator **227** is clear. If so, then blocks **514, 516, 518, 520, 522** respectively direct the microprocessor **12** to clear the retry session indicator **223** in the CRC, FSU, SPP, IRN, or MSR flags **224, 228, 222, 226, 230**. The microprocessor **12** is then respectively directed to commence the CRC Subroutine **372**, the FSU Subroutine **374**, the SPP Subroutine **376**, the IRN Subroutine **378**, or the MSR Subroutine **380**.

Finally, block **524** directs the microprocessor **12** to return from the Retry Subroutine **466**.

FIG. **17**

With reference to FIGS. **1, 9** and **17**, the Backup Subroutine is illustrated generally at **368** and defines an initial state **526** from which the microprocessor **12** begins execution. The Backup Subroutine **368** directs the CPE **10** to call a backup SPCS **36** if it has experienced repeated failures in establishing a connection with the primary SPCS **36**. In other words, the Backup Subroutine **368** invokes an appropriate backup session after analyzing the backup indicator **225** and the disabled indicator **227** of the set of session flags **220, 222, 224, 226, 228**.

Block **528** directs the microprocessor **12** to determine whether the backup session indicator **225** in the MDP flag **220** has been set and the disabled indicator is clear **227**. If so, then block **530** directs the microprocessor **12** to clear in the MDP flag **220** the backup session indicator **225**.

Thereafter, the microprocessor **12** is directed to commence the MDP Subroutine **370**.

Similarly blocks **532, 534, 536, 538, 540** direct the microprocessor **12** to determine whether the backup session indicator **225** in respectively the CRC, FSU, SPP, IRN, or MSR flags **224, 228, 222, 226, 230** has been set and the disabled indicator is clear **227**. If so, then blocks **542, 544, 546, 548, 550** respectively direct the microprocessor **12** to clear the backup session indicator **225** in the CRC, FSU, SUP, IRN, or MSR flags **224, 228, 222, 226, 230**. The microprocessor **12** is then respectively directed to commence the CRC Subroutine **372**, the FSU Subroutine **374**, the SPP Subroutine **376**, the IRN Subroutine **378**, or the MSR Subroutine **380**.

Finally, block **552** directs the microprocessor **12** to return from the Backup Subroutine **368**.

FIG. **18**

With reference to FIGS. **1, 2, 9, 10** and **18**, the MDP Subroutine is generally illustrated at **370** and defines an initial state **554** from which the microprocessor **12** begins execution. The MDP Subroutine **370** initiates and controls any MDP session.

Block **556** directs the microprocessor **12** to evaluate the polling timers **194, 196, 198, 200, 202, 204, 206, 208, 210, 212, 214, 216, 218** and to evaluate the difference between the CPE System Clock time **188** and the polling calendars **274, 276, 278, 280, 282, 322, 350** to determine whether any

such value is below the poll proximity threshold **327**. If so, then other autonomous calls are scheduled to begin soon and the microprocessor **12** is directed to avoid a scheduling conflict by aborting the current MDP poll in favour of queuing a retry/backup poll session via the Queue

Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the MDP Subroutine **370**.

Alternatively, if no other autonomous call is proximate, block **560** directs the microprocessor **12** to evaluate the CPE idle timer **190** to determine if the CPE **10** has been idle sufficiently long in comparison to the CPE idle time threshold **328** that an autonomous calling session would not be likely to disturb the user who might be using the CPE **10** for conventional telephone purposes. If the CPE **10** has not been sufficiently idle, then the microprocessor **12** is directed to abort the current MDP poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the MDP Subroutine **370**.

Alternatively, if the CPR **10** has been sufficiently idle, then block **562** directs the CPE **10** to call the SPCS **36** to initiate a standard ADSI SDC session, to display the MDP display message **288** on the CPE display **25**, and to poll the received data queue **185** for ADSI messages. If the MDP Subroutine **370** has been called from by the Originating Subroutine **364** or the Retry Subroutine **366**, then the CPE **10** calls the primary message server telephone number **270**. Alternatively, if the MDP Subroutine **370** has been called by the Backup Subroutine **368**, then the CPE **10** calls the backup message server telephone number **272**. Thus, at least one alternating telephone number is dialled when the first predefined telephone number is unavailable. Either way, the CPE **10** polls the received data queue **185** for an ADSI STA **151***d* QUER $81 message **68** from the SPCS **36** so that it can state to the SPCS **36** the purpose of its call.

If the microprocessor **12** determines, under the direction of block **564**, that the CPE **10** has not received a message from the SPCS **36** within the call time-out threshold **329**, then block **566** directs the microprocessor **12** to cause a DTMF "A" to be transmitted to indicate to the SPCS **36** that an unspecified error has occurred and then to cause the CPE **10** to hang up the line **30** and to abort the current MDP poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the MDP Subroutine **370**.

Alternatively, it at block **564** the CPE **10** does receive a message from the SPCS **36** within the call time-out threshold **329**, then block **567** directs the microprocessor **12** to determine if the message is an ADSI STA **151***d* QUER $81 message **68**. If not, then block **568** directs the microprocessor **12** to cause a DTMF "A" to be transmitted to the SPCS **36** to indicate an unspecified error has occurred and to cause the CPE **10** to hang up the line **30**, after which the microprocessor **12** is directed to abort the current MDP poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the MDP Subroutine **370**.

Alternatively, if at block **567** the CPE **10** has received an ADSI STA **151***d* QUER $81 message from the SPCS **36**, then block **570** directs the microprocessor **12** to cause the CPR **10** to transmit a DTMF "1B" to the SPCS **36** to indicate that the call is a CPE **10** initiated daily poll. This indication helps the SPCS **36** decide which ADSI instructions to transmit to the CPE **10**.

Block **572** directs the CPE **10** to poll for further messages from the SPCS **36**. If the microprocessor **12** determines under the direction of block **574** that the CPE **10** has not received a message from the SPCS **36** within the call time-out threshold **329**, then block **576** directs the microprocessor **12** to cause a DTMF "A" signal to be transmitted to indicate to the SPCS **36** that an unspecified error has occurred and then to cause the CPE **10** to hang up the line **30** and to abort the current MDP poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the MDP Subroutine **370**.

Alternatively if at block **574**, the CPE **10** receives a message from the SPCS **36**, then blocks **584**, **586**, **588**, **589**, **590** respectively direct the microprocessor **12** to examine the application name bytes 46, 48, 50, 52 to determine if the message is an ADSI STA **151***d* MESG message **56**, an ADSI STA **151***d* SCRI message **58**, an ADSI STA **151***d* POLL message **60**, an ADSI STA **151***d* REMI message **61** or an ATSI STA **151***d* FONE **55** message.

If at block **584** the message is an ADSI STA **151***d* MESG message **56**, then the microprocessor **12** is directed to the MESG subroutine **386**, upon return from which, the microprocessor **12** is directed back to block **572** to await further messages from the SPCS **36**.

Alternatively, if at block **586** the message is an ADSI STA **151***d* SCRI message **58**, then the microprocessor **12** is directed to the SCRI subroutine **386**, upon return from which, the microprocessor **12** is directed back to block **572** to await further messages from the SPCS **36**.

Alternatively, if at block **588** the message is an ADSI STA **151***d* POLL message **60**, then the microprocessor **12** is directed to the POLL subroutine **386**, upon return from which, the microprocessor **12** is directed back to block **572** to await further messages from the SPCS **36**.

Alternatively, if at block **589** the message is an ADSI STA **151***d* REMI message **61**, then the microprocessor **12** is directed to the REMI subroutine **389**, upon return from which, the microprocessor **12** is directed back to block **572** to await further messages from the SPCS **36**.

Alternatively, if at block **590** the message is an ADSI STA **151***d* FONE $01 message **69** (the microprocessor **12** being directed to examine as well the message's action code byte **62**), then the microprocessor **12** is directed by block **592**: to cause a DTMF "B" to be transmitted to the SPCS **36** to indicate successful completion of the call, to cause the CPE **10** to hang up the line **30**, and to clear all MDP flag indicators **220**, counters **232**, **244**, **332** and timers **196**, **208**. Thereafter, the microprocessor **12** is directed to return from the MDP Subroutine **370**.

Alternatively, if the message does not match, then block **594** directs the microprocessor **12** to cause a DTMF "A" signal to be transmitted to the SPCS **36** to indicate an unspecified error has occurred and to cause the CPE **10** to hang up the line **30** whereafter the CPE **10** will ignore all further ADSI messages. Thereafter, the microprocessor **12** is directed to abort the current MDP poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the MDP Subroutine **370**.

FIG. **19**

With reference to FIGS. **1**, **2**, **9**, **10** and **19**, the CRC Subroutine is generally illustrated at **372** and defines an initial state **596** from which the microprocessor **12** begins execution. The CRC Subroutine **372** initiates and controls any CRC session.

Block **598** directs the microprocessor **12** to evaluate the polling timers **194**, **196**, **198**, **200**, **202**, **204**, **206**, **208**, **210**, **212**, **214**, **216**, **218** and to evaluate the difference between the CPE System Clock time **188** and the polling calendars

274, 276, 278, 280, 282, 322, 350 to determine whether any such value is below the poll proximity threshold 327. If so, then the microprocessor 12 is directed to abort the current CRC poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine 382 before directing the microprocessor 12 to return from the CRC Subroutine 372.

Alternatively, if no other autonomous call is proximate, block 602 directs the microprocessor 12 to evaluate the CPE idle timer 190 to determine if the CPE 10 has been idle sufficiently long in comparison to the CPE idle time threshold 328 that an autonomous calling session would not disturb the user. If not, then the microprocessor 12 is directed to abort the current CRC poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine 382 before directing the microprocessor 12 to return from the CRC Routine 372.

Alternatively, if the CPE 10 has been sufficiently idle, then block 604 directs the CPE 10 to call the SPCS 36 to initiate a standard ADSI SDC session, to display the message sending display message 310 on the CPE display 25, and to poll the received data queue 185 for ADSI messages. If the CRC Subroutine 372 has been called from by the Originating Subroutine 364 or the Retry Subroutine 366, then the CPE 10 calls the primary message sending server telephone number 302. Alternatively, if the CRC Subroutine 372 has been called by the Backup Subroutine 368, then the CPE 10 calls the backup message sending server telephone number 304. Either way, the CPE 10 polls the received data queue 185 for an ADSI STA 151*d* QUER $81 message 68 from the SPCS 36 so that it can state to the SPCS 36 the purpose of its call.

If the microprocessor 12 determines, under the direction of block 606 that the CPE 10 has not received a message from the SPCS 36 within the call time-out threshold 329, then block 608 directs the microprocessor 12 to cause a DTMF "A" signal to be transmitted to indicate to the SPCS 36 that an unspecified error has occurred and then to cause the CPE 10 to hang up the line 30 and to abort the current CRC poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine 382 before directing the microprocessor 12 to return from the CRC Subroutine 372.

Alternatively, if at block 606 if the CPE 10 does receive a message from the SPCS 36 within the call time-out threshold 329 then block 610 directs the microprocessor 12 to determine if the message is an ADSI STA 151*d* QUER $81 message 68. If not, then block 612 directs the microprocessor 12 to cause a DTMF "A" to be transmitted to the SPCS 36 to indicate an unspecified error has occurred and to cause the CPE 10 to hang up the line 30, after which the microprocessor 12 is directed to abort the current CRC poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine 382 before directing the microprocessor 12 to return from the CRC Subroutine 372.

Alternatively, if at block 610 the CPE 10 has received an ADSI STA 151*d* QUER $81 message from the SPCS 36, then block 614 directs the microprocessor 12 to cause the CPE 10 to transmit a DTMF "5B" to the SPCS 36 to indicate that the call is a CPE 10 initiated message, and to send only/non-interactive poll.

Block 616 then directs the CPE 10 to poll for further messages from the SPCS 36. If the microprocessor 12 determines under the direction of block 618 that the CPE 10 has not received a message from the SPCS 36 within the call time-out threshold 329, then block 620 directs the microprocessor 12 to causo the CPE 10 to transmit a DTMF "A" and to hang up the line 30, and thereafter to abort the current CRC poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine 382 before directing the microprocessor 12 to return from the CRC Routine 372.

Alternatively, if at block 618 the CPE 10 receives a message from the SPCS 36, then blocks 622 and 624 respectively direct the microprocessor 12 to examine the message's application name bytes 46, 48, 50, 52 to determine if the message is an ADSI STA 151*d* MESG message or an ADSI STA 151*d* FONE message.

If at block 622 the message is an ADSI STA 151*d* MESG message, then the microprocessor 12 is directed to the MESG subroutine 386, upon return from which, the microprocessor 12 is directed back to block 616 to await further messages from the SPCS 36.

Alternatively, if at block 624 the message is an ADSI STA 151*d* FONE $01 message 69 (the microprocessor 12 being directed to examine the action code byte 62), then the microprocessor 12 is directed by block 626 to cause a DTMF "B" to be transmitted to the SPCS 36 to indicate successful completion of the call, to cause the CPE 10 to hang up the line 30, and to clear all CRC flag indicators 224, counters 236, 248, 338 and timers 200, 212. Thereafter, the microprocessor 12 is directed to return from the CRC Subroutine 372,

Alternatively, if the message does not match, then block 628 directs the microprocessor 12 to cause a DTMF "A" signal to be transmitted to the SPCS 36 to indicate an unspecified error has occurred and to cause the CPE 10 to hang up the line 30. Thereafter, the microprocessor 12 is directed to abort the current CRC poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine 382 before directing the microprocessor 12 to return from the CRC Routine 372.

FIG. 20

With reference to FIGS. 1, 2, 9, 10 and 20, the FSU Subroutine is generally illustrated at 374 and defines an initial state 630 from which the microprocessor 12 begins execution. The FSU Subroutire 374 initiates and controls any FSU session.

Block 632 directs the microprocessor 12 to evaluate the polling timers 194, 196, 198, 200, 202, 204, 206, 208, 210, 212, 214, 216, 218 and to evaluate the difference between the CPE System Clock time 188 and the polling calendars 274, 276, 278, 280, 282, 322, 350 to determine whether any such value is below the poll proximity threshold 327. If so, then the microprocessor 12 is directed to abort the current FSU poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine 382 before directing the microprocessor 12 to return from the FSU Subroutine 374.

Alternatively, if no other autonomous call is proximate, block 636 directs the microprocessor 12 to evaluate the CPE idle timer 190 to determine if the CPE 10 has been idle sufficiently long in comparison to the CPE idle time threshold 328 that an autonomous calling session would not disturb the user. If not, then the microprocessor 12 is directed to abort the current FSU poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine 382 before directing the microprocessor 12 to return from the FSU Subroutine 374.

Alternatively, if at block 636 the CPE 10 has been sufficiently idle, then block 638 directs the CPE 10 to call the SPCS 36 to initiate a standard ADSI SDC session, to display the FSU display message 324 on the CPE display 25, and to poll the received daLa queue 185 for ADSI messages. If the FSU Subroutine 374 has been called from by the Originating Subroutine 364 or the Retry Subroutine 366, then the CPS 10 calls the primary FSU server telephone number 314. Alternatively, if the FSU Subroutine 374 has been called by

the Backup Subroutine **368**, then the CPE **10** calls the backup message sending server telephone number **316**. Either way, the CPE **10** polls the received data queue **185** for an ADSI STA **151***d* QUER $81 message **68** from the SPCS **36** so that it can state to the SPCS **36** the purpose of its call.

If the microprocessor **12** determines under the direction of block **640** that the CPE **10** has not received a message from the SPCS **36** within the call time-out threshold **329**, then block **642** directs the microprocessor **12** to cause the CPE **10** to transmit a DTMF "A" and to hang up the line **30**, thereafter aborting the current FSU poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the FSU Subroutine **374**.

Alternatively, if at block **640** the CPE **10** does receive a message from the SPCS **36** within the call time-out threshold **329** then block **642** directs the microprocessor **12** to determine if the message is an ADSI STA **151***d* QUER $81 message **68**. If not, then block **644** directs the microprocessor **12** to cause a DTMF "A" to be transmitted to the SPCS **36** to indicate an unspecified error has occurred and to cause the CPE **10** to hang up the line, after which the microprocessor **12** is directed to abort the current FSU poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the FSU Subroutine **374**.

Alternatively, if at block **642** the CPE **10** has received an ADSI STA **151***d* QUER $81 message from the SPCS **36**, then block **646** directs the microprocessor **12** to cause the CPE **10** to transmit a DTMF "6B" to the SPCS **36** to indicate that the call is a CPE **10** initiated non-message poll.

Block **648** then directs the CPE **10** to poll for further messages from the SPCS **36**. If the microprocessor **12** determines under the direction of block **650** that the CPE **10** has not received a message from the SPCS **36** within the call time-out threshold **329**, then block **652** directs the microprocessor **12** to cause the CPE **10** to transmit a DTMF "A" and to hang up the line, and thereafter to abort the current FSU poll in favor of quelling a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the FSU Subroutine **374**.

Alternatively, if at block **650** the CPE **10** receives a message from the SPCS **36**, then blocks **654** and **656** respectively direct the microprocessor **12** to examine the application name bytes 46, 48, 50, 52 to determine if the message is an ADSI STA **151***d* SCRI message or an ADSI STA **151***d* FONE message.

If at block **654** the message is an ADSI STA **151***d* SCRI message, then the microprocessor **12** is directed to the SCRI subroutine **388**, upon return from which, the microprocessor **12** is directed back to block **648** to await further messages from the SPCS **36**.

Alternatively, if at block **656** the message is an ADSI STA **151***d* FONE $01 message **69** (the microprocessor **12** being directed to examine the action code byte 62), then the microprocessor **12** is directed by block **658** to cause a DTMF "B" to be transmitted Lu the SPCS **36** to indicate successful completion of the call, to cause the CPE **10** to hang up the line **30**, and to clear all FSU flag indicators **228**, counters **240**, **252**, **340** and timers **204**, **216**. Thereafter, the microprocessor **12** is directed to return from the FSU Subroutine **374**.

Alternatively, if the message does not match, then block **660** directs the microprocessor **12** to cause a DTMF "A" signal to be transmitted to the SPCS **36** to indicate an error and to cause the CPE **10** to hang up the line. Thereafter, the microprocessor **12** is directed to abort the current FSU poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the FSU Routine **374**.

FIG. **21**

With reference to FIGS. **1**, **2**, **9**, **10** and **21**, the SPP Subroutine is generally illustrated at **376** and defines an initial state **662** from which the microprocessor **12** begins execution. The SPP Subroutine **376** initiates and controls any SPP session.

Block **664** directs the microprocessor **12** to evaluate the polling timers **194**, **196**, **198**, **200**, **202**, **204**, **206**, **208**, **210**, **212**, **214**, **216**, **218** and to evaluate the difference between the CPE System Clock time **188** and the polling calendars **274**, **276**, **278**, **280**, **282**, **322**, **350** to determine whether any such value is below the poll proximity threshold **327**. If so, then the microprocessor **12** is directed to abort the current SPP poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the SPP Subroutine **376**.

Alternatively, if at block **664** no other autonomous call is proximate, block **668** directs the microprocessor **12** to evaluate the CPE idle timer **190** to determine if the CPE **10** has been idle sufficiently long in comparison to the CPE idle time threshold **328** such that an autonomous calling session would not disturb the user. If not, then the microprocessor **12** is directed to abort the current SPP poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the SPP Subroutine **376**.

Alternatively, if at block **668** the CPE **10** has been sufficiently idle, then block **670** directs the CPE **10** to call the SPCS **36** to initiate a standard ADSI SDC session, to display the PP message **300** on the CPE display **25**, and to poll the received data queue **185** for ADSI messages. If the SPP Subroutine **376** has been called from by the Originating Subroutine **364** or the Retry Subroutine **366**, then the CPE **10** calls the primary page response server telephone number **290**. Alternatively, if the SPP Subroutine **376** has been called by the Backup Subroutine **368**, then the CPE **10** calls the backup page response server telephone number **292**. Either way, the CPE **10** polls the received data queue **195** for an ADSI STA **151***d* QUER $81 message **68** from the SPCS **36** so that it can state to the SPCS **36** the purpose of its call.

If the microprocessor **12** determines under the direction of block **672** that the CPU **10** has not received a message from the SPCS **36** within the call time-out threshold **329**, then block **674** directs the microprocessor **12** to cause the CPE **10** to transmit a DTMP "A" and to hang up the line, thereafter aborting the current SPP poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the SPP Subroutine **376**.

Alternatively, if at block **672** the CPE **10** does receive a message from the SPCS **36** within the call time-out threshold **329** then block **674** directs the microprocessor **12** to determine if the message is an ADST STA **151***d* QUER $81 message **68**. If not, then block **676** directs the microprocessor **12** to cause a DTMF "A" to be transmitted to the SPCS **36** to indicate an unspecified error has occurred and to cause the CPE **10** to hang up the line, after which the microprocessor **12** is directed to abort the current SPP poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the SPP Subroutine **376**.

Alternatively, if at block **674** the CPE **10** has received an ADSI STA **151***d* QUER $81 message from the SPCS **36**, then block **678** directs the microprocessor **12** to cause the

CPE **10** to transmit a DTMF "4B" to the SPCS **36** to indicate that the call is a CPE **10** initiated page response poll.

Block **680** then directs the CPE **10** to poll for further messages from the SPCS **36**. If the microprocessor **12** determines under the direction of block **682** that the CPE **10** has not received a message from the SPCS **36** within the call time-out threshold **329**, then block **684** directs the microprocessor **12** to cause the CPE **10** to transmit a DTMF "A" and to hang up the line, and thereafter to abort the current SPP poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the SPP Subroutine **376**.

Alternatively, if at block **682** the CPE **10** receives a message from the SPCS **36**, then blocks **686** and **688** respectively direct the microprocessor **12** to examine the message's application name bytes 46, 48, 50, 52 to determine if the message is an ADSI STA **151***d* MESG message or an ADSI STA **151***d* FONE message.

If at block **686** the message is an ADSI STA **151***d* MESG message, then the microprocessor **12** is directed to the MESG subroutine **386**, upon return from which, the microprocessor **12** is directed back to block **680** to await further messages from the SPCS **36**.

Alternatively, if at block **688** the message is an ADSI STA **151***d* FONE $01 message **69** (the microprocessor **12** being directed to examine the action code byte 62), then the microprocessor **12** is directed by block **690** to cause a DTMF "B" to be transmitted to the SPCS **36** to indicate successful completion of the call, to cause the CPE **10** to hang up the line **30**, and to clear all SPP flag indicators **222**, counters **234**, **246**, **334** and timers **198**, **210**. Thereafter, the microprocessor **12** is directed to return from the SPP Subroutine **376**.

Alternatively, if the message does not match, then block **692** directs the microprocessor **12** to cause a DTMF "A" message to be transmitted to the SPCS **36** to indicate an error and to cause the CPE **10** to hang up the line. Thereafter, the microprocessor **12** is directed to abort the current SPP poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the SPP Subroutine **376**.

FIG. **22**

With reference to FIGS. **1**, **2**, **9**, **10** and **22**, the IRN Subroutine is generally illustrated at **378** and defines an initial state **694** from which the microprocessor **12** begins execution. The IRN Subroutine **378** initiates and controls any IRN session.

Block **696** directs the microprocessor **12** to evaluate the polling timers **194**, **196**, **198**, **200**, **202**, **204**, **206**, **208**, **210**, **212**, **214**, **216**, **218** and to evaluate the difference between the CPE System Clock time **188** and the polling calendars **274**, **276**, **278**, **280**, **282**, **322**, **350** to determine whether any such value is below the poll proximity threshold **327**. If so, then the microprocessor **12** is directed to abort the current SPP poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the IRN Subroutine **378**.

Alternatively, if at block **696** no other autonomous call is proximate, block **700** directs the microprocessor **12** to evaluate the CPE idle timer **190** to determine if the CPS **10** has been idle sufficiently long in comparison to the CPE idle time threshold **328** such that an autonomous calling session would not disturb the user. If not, then the microprocessor **12** is directed to abort the current IRN poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the IRN Subroutine **378**.

Alternatively, if at block **700** the CPE **10** has been sufficiently idle, then block **702** directs the CPE **10** to call the SPCS **36** to initiate a standard ADSI SDC session, to display the message sending display message **310** on the CPE display **25**, and to poll the received data queue **185** for ADSI messages. If the IRN Subroutine **378** has been called from by the originating Subroutine **364** or the Retry Subroutine **366**, then the CPE **10** calls the primary message sending server telephone number **302**. Alternatively, if the IRN Subroutine **378** has been called by the Backup Subroutine **368**, then the CPE **10** calls the backup message sending server telephone number **304**. Either way, the CPE **10** polls the received data queue **185** for an ADSI STA **151***d* QUER $81 message **68** from the SPCS **36** so that it can state to the SPCS **36** the purpose of its call.

If the microprocessor **12** determines under the direction of block **704** that the CPE **10** has not received a message from the SPCS **36** within the call time-out threshold **329**, then block **706** directs the microprocessor **12** to cause the CPE **10** to transmit a DTMF "A" and to hang up the line, thereafter aborting the current IRN poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the IRN Subroutine **378**.

Alternatively, if at block **704** the CPE **10** does receive a message from the SPCS **36** within the call time-out threshold **329** then block **708** directs the microprocessor **12** to determine if the message is an ADSI STA **151***d* QUER $81 message **68**. If not, then block **710** directs the microprocessor **12** to cause a DTMF "A" to be transmitted to the SPCS **36** to indicate an unspecified error has occurred and to cause the CPE **10** to hang up the line, after which the microprocessor **12** is directed to abort the current IRN poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the IRN Subroutine **378**.

Alternativeiy, if at block **708** the CPE **10** has received an ADSI STA **151***d* QUER $81 message from the SPCS **36**, then block **712** directs the microprocessor **12** to cause the CPE **10** to transmit a DTMF "5B" to the SPCS **36** to indicate that the call is a CPE **10** initiated message send only/non-interactive poll.

Block **714** then directs the CPE **10** to poll for further messages from the SPCS **36**. if the microprocessor **12** determines under the direction of block **716** that the CPE **10** has not received a message from the SPCS **36** within the call time-out threshold **329**, then block **718** directs the microprocessor **12** to cause the CPE **10** to transmit a DTMF "A" and to hang up the line, and thereafter to abort the current IRN poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the IRN Subroutine **378**.

Alternatively, if at block **716** the CPE **10** receives a message from the SPCS **36**, then blocks **720** and **722** respectively direct the microprocessor **12** to examine the application name bytes 46, 48, 50, 52 to determine if the message is an ADSI STA **151***d* MESG message or an ADSI STA l**51***d* FONE message.

If at block **720** the message is an ADSI STA **151***d* MESG message, then the microprocessor **12** is directed to the MESG subroutine **386**, upon return from which, the microprocessor **12** is directed back to block **714** to await further messages from the SPCS **36**.

Alternatively, if at block **722** the message is an ADSI STA **151***d* FONE $01 message **69** (the microprocessor **12** being directed to examine the action code byte 62), then the microprocessor **12** is directed by block **724** to cause a DTMF

"B" to be transmitted to the SPCS **36** to indicate successful completion of the call, to cause the CPE **10** to hang up the line **30**, and to clear all IRN flag indicators **226**, counters **238**, **250**, **336** and timers **202**, **214**. Thereafter, the microprocessor **12** is directed to return from the IRN Subroutine **378**.

Alternatively, if the message does not match, then block **726** directs the microprocessor **12** to cause a DTMF "A" signal to be transmitted to the SPCS **36** to indicate an error and to cause the CPE **10** to hang up the line. Thereafter, the microprocessor **12** is directed to abort the current IRN poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the IRN Subroutine **378**.

FIG. **23**

With reference to FIGS. **1**, **2**, **9**, **10** and **23**, the MSR Subroutine is generally illustrated at **380** and defines an initial state **728** from which the microprocessor **12** begins execution. The MSR Subroutine **380** iniliates and controls any MSR session. Block **730** directs the microprocessor **12** to evaluate the polling timers **194**, **196**, **198**, **200**, **202**, **204**, **206**, **208**, **210**, **212**, **214**, **216**, **218** and to evaluate the difference between the CPE System Clock time **188** and the polling calendars **274**, **276**, **278**, **280**, **282**, **322**, **350** to determine whether any such value is below the poll proximity threshold **327**. If so, then the microprocessor **12** is directed to abort theo current MSR poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the MSS Subroutine **380**.

Alternatively, it at block **730** no other autonomous call is proximate, block **734** directs the microprocessor **12** to evaluate the CPE idle timer **190** to determine if the CPE **10** has been idle sufficiently long in comparison to the CPE idle time threshold **328** such that an autonomous calling session would not disturb the user. If not, then the microprocessor **12** is directed to abort the current MSR poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the MSR Subroutine **380**.

Alternatively, if at block **734** the CPE **10** has been sufficiently idle, then block **736** directs the CPE **10** to call the SPCS **36** to initiate a standard ADSI SDC session, to display the message sending display message **310** on the CPE display **25**, and to poll the received data queue **185** for ADSI messages. If the MSR Subroutine **380** has been called from by the Originating Subroutine **364** or the Retry Subroutine **366**, then the CPE **10** calls the primary message sending server telephone number **302**. Alternatively, if the MSR Subroutine **380** has been called by the Backup Subroutine **368**, then the CPE **10** calls the backup message sending server telephone number **304**. Either way, the CPE **10** polls the received data queue **185** for an ADSl STA **151***d* QUER $81 message **68** from the SPCS **36** so that it can state to the SPCS **36** the purpose of its call.

If the microprocessor **12** determines under the direction of block **738** that the CPE **10** has not received a message from the SPCS **36** within the call time-out threshold **329**, then block **740** directs the microprocessor **12** to cause the CPE **10** to transmit a DTMF "A" and to hang up the line, thereafter aborting the current IRN poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the MSR Subroutine **380**.

Alternatively, if at block **738** the CPE **10** does receive a message from the SPCS **36** within the call time-out threshold **329** then block **742** directs the microprocessor **12** to determine if the message is an ADSI STA **151***d* QUER $81 message **68**. If not, then block **744** directs the microprocessor **12** to cause a DTMF "A" to be transmitted to the SPCS **36** to indicate an unspecified error has occurred and to cause the CPE **10** to hang up the line, after which the microprocessor **12** i c directed to abort the current MSR poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the MSR Subroutine **380**.

Alternatively, if at block **742** the CPE **10** has received an ADSI STA **151***d* QUER $81 messagc from the SPCS **36**, then block **746** directs the microprocessor **12** to cause the CPE **10** to transmit a DTMF "5B" to the SPCS **36** to indicate that the call is a CPE **10** initiated message send only/non-interactive poll.

Block **748** then directs the CPE **10** to poll for further messages from the SPCS **36**. If the microprocessor **12** determines under the direction of block **750** that the CPE **10** has not received a message from the SPCS **36** within the call time-out threshold **329**, then block **752** directs the microprocessor **12** to cause the CPE **10** to transmit a DTMF "A" and to hang up the line, and thereafter to abort the current MSR poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the MSR Subroutine **380**.

Alternatively, if at block **750** the CPE **10** receives a message from the SPCS **36**, then blocks **754** and **756** respectively direct the microprocessor **12** to examine the message's application name bytes 46, 48, 50, 52 to determine if the message is an ADSI STA **151***d* MESG message or an ADSI STA **151***d* FONE message.

If at block **754** the message is an ADSI STA **151***d* MESG message, then the microprocessor **12** is directed to the MESG subroutine **386**, upon return from which, the microprocessor **12** is directed back to block **746** to await further messages from the SPCS **36**,

Alternatively, if at block **756** the message is an ADSI STA **151***d* FONE $01 message **69** (the microprocessor **12** being directed to examine the action code byte 62), then the microprocessor **12** is directed by block **758** to cause a DTMF "B" to be transmitted to the SPCS **36** to indicate successful completion of the call, to cause the CPE **10** to hang up the line **30**, and to clear all MSR flag indicators **230**, counters **242**, **254**, **342** and timers **206**, **218**. Thereafter, the microprocessor **12** is directed to return from the MSR Subroutine **380**.

Alternatively, it the message does not match, then block **760** directs the microprocessor **12** to cause a DTMF "A" message to be transmitted to the SPCS **36** to indicate an error and to cause the CPE **10** to hang up the line. Thereafter, the microprocessor **12** is directed to abort the

current MSR poll in favor of queuing a retry/backup poll session via the Queue Retry/Backup Subroutine **382** before directing the microprocessor **12** to return from the MSR Subroutine **380**.

FIG. **24**

With reference to FIGS. **1**, **2**, **9**, **10** and **24**, the Queue Retry/Backup Subroutine is generally illustrated at **382** and defines an initial state **762** from which the microprocessor **12** begins execution. The Queue Retry/Backup Subroutine **382** queues up retry and backup attempts to call a SPCS **36** when prior attempts have failed. It also disables types of sessions that have failed too often.

Block **764** directs the microprocessor **12** to cause to be recorded in the failure log **343** the current system time **188**, the subroutine **370**, **372**, **374**, **376**, **378**, **380** that called the Queue Retry/Backup Subroutine **382**, the location in the

calling routine **370, 372, 374, 376, 378, 380** from which the Queue Retry/Backup Subroutine **382** was called, and a record delimiter. The failure log **343** is maintained conventionally as a variable length database, the contents being available to a technician to diagnose problems in the CPE **10** such as conflicting session schedules.

Block **774** then directs the microprocessor **12** to compare the retry counter **232, 234, 236, 238, 240, 242** associated with the calling subroutine **370, 372, 374, 376, 378, 380**, to the maximum number of retry sessions **284, 298, 306, 306, 318, 306** associated with the respective calling subroutine. If the maximum number of retry sessions **284, 298, 306, 306, 318, 306** has not been exceeded, then block **776** directs the microprocessor **12** to increment the appropriate retry counter **232, 234, 236, 238, 240, 242**, and block **778** directs the microprocessor **12** to load the appropriate retry timer **196, 198, 200, 202, 204, 206** with the specified retry delay time **286, 296, 308, 309, 320, 308**. Thereafter, block **772** directs the microprocessor **12** to return to the calling subroutine **370, 372, 374, 376, 378, 380**.

Alternatively, if at block **774** the maximum number of retry attempts has been made, then block **780** directs the microprocessor **12** to compare the backup counter **244, 246, 248, 250, 252, 254** associated with the calling subroutine **370, 372, 374, 376, 378, 380**, to the maximum number of retry sessions **284, 298, 306, 306, 318, 306** associated with the respective calling subroutine. If the maximum number of backup sessions **244, 246, 248, 250, 252, 254** has not been exceeded, then block **782** directs the microprocessor **12** to increment the appropriate backup counter **244, 246, 248, 250, 252, 254**, and block **784** directs the microprocessor **12** to load the appropriate backup timer **209, 210, 212, 214, 216, 218** with the specified retry delay time **286, 296, 308, 308, 320, 308**. Thereafter, block **772** directs the microprocessor **12** to return to the calling subroutine **370, 372, 374, 376, 378, 380**.

Alternatively if at block **780**, the maximum number of backup attempts has been made, then block **766** then directs the microprocessor **12** to increment the failure counter **332, 334, 336, 338, 340, 342** associated with the calling subroutine **370, 372, 374, 376, 378, 380**. Block **768** then directs the microprocessor **12** to compare such failure counter **332, 334, 336, 338, 340, 342** with the feature disable error threshold **330** to determine whether the calling subroutine has such a poor connection record as to warrant disablement pending diagnosis. If so, then block **770** directs the microprocessor **12** to set the disabled indicator **227** in the appropriate status flag **220, 222, 224, 226, 228, 230**. Thereafter, block **772** directs the microprocessor **12** to return to the calling subroutine **370, 372, 374, 376, 378, 380**.

Alternatively, if the disable error threshold **330** has not been exceeded, then block **772** directs the microprocessor **12** to simply return to the calling subroutine **370, 372, 374, 376, 378, 380** and to make no further attempts to complete the instant poll.

FIG. **25**

With reference to FIGS. **1, 2, 9, 10** and **25**, the POLL subroutine is generally illustrated at **384** and defines an initial state **786** from which the microprocessor **12** begins execution. This execution commences within a standard ADSI SDC session. The POLL Subroutine **384** instructs the CPE **10** how to respond to the POLL family **60** of extended STA codes **40** that configure the MDP and SPP sessions.

Block **788** directs the microprocessor **12** to determine if the STA action code **62** of the most recently polled message is a "01" to reconfigure the CPE's **10** MDP parameters. If so, then block **790** directs the microprocessor **12** to parse the STA data **64** into its component parts with reference to the "FF" delimiters **132, 136, 148** and **156** in FIG. **7**. Block **792** then directs the microprocessor **12** to check the parsed data **64** to make sure that it is consistent with the expected data structure **258** in FIG. **10**. If not, block **794** directs the microprocessor **12** to cause a DTMF "5A" to be transmitted to indicate to the SPCS **36** that a parsing error has occurred. Thereafter, block **796** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **792** the parsed data has the expected structure, then block **798** directs the microprocessor **12** to consider whether these new MDP parameters comply with the limits set by telecommunication regulations **345**, for example with respect to repeated automated calling. If not, then block **800** directs the microprocessor **12** to cause a DTMF "1A" to be transmitted to indicate to the SPCS **36** that the new configuration violates local telecommunication regulations. Thereafter, block **796** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **799** the new parameters are permissible, then block **802** directs the microprocessor **12** to present the new parameters to the user via the display **25**. Block **804** directs the microprocessor **12** to poll the user's response at the keypad **27** for a set period of time. If the user rejects the new parameters, then block **806** directs the microprocessor **12** to cause a DTMF "3A" to be transmitted to indicate to the SPCS **36** that the user has rejected the new configuration. Thereafter, block **796** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **804** the user accepts the new parameters or the polling period expires without user input, then block **808** directs the microprocessor **12** to store the parsed data into the MDP configuration block **258** in FLASH **18**. Because this reconfiguration process is autonomous, it is undesirable to make major configuration changes unless the CPE **10** user is known to be present during the process by his acceptance or rejection signal.

Block **810** directs the microprocessor **12** to check the memory transfer, comparing the values stored in RAM **14** to those stored in FLASH **18**. If a memory error has occurred, then block **812** directs the microprocessor **12** to cause a DTMF "4A" to be transmitted to indicate to the SPCS **36** that a memory error has occurred. Thereafter, block **796** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **810** no memory error has occurred, then block **814** directs the microprocessor **12** to ensure that no other error has occurred during this reconfiguration. If another error has occurred, then block **816** directs the microprocessor **12** to cause a DTMF "A" to be transmitted to indicate to the SPCS **36** that an unspecified error has occurred. Thereafter, block **796** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **814** no unspecified error occurred, then block **818** directs the microprocessor **12** to cause a DTMF "B" to be transmitted to indicate to the SPCS **36** that the reconfiguration occurred properly. Thereafter, block **796** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, block **820** directs the microprocessor **12** to determine if the STA action code **62** of the most recently polled message is a "$02" to reconfigure the SPP parameters. If so, then block **822** directs the microprocessor **12** to parse the STA data **64** into its component parts with reference to the "$FF" delimiters **162, 166** and **176** in FIG. **7**.

Block **824** then directs the microprocessor **12** to check the parsed data **64** to make sure that it is consistent with the

expected data structure **260**. If not, block **826** directs the microprocessor **12** to cause a DTMF "5A" to be transmitted to indicate to the SPCS **36** that a parsing error has occurred. Thereafter, block **828** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **824** the parsed data has the expected structure, then block **830** directs the microprocessor **12** to consider whether these new SPP parameters comply with the limits set by telecommunication regulations **345**, for example with respect to repeated automated calling. If not, then block **832** directs the microprocessor **12** to cause a DTMF "1A" to be transmitted to indicate to the SPCS **36** that the new configuration violates local telecommunication regulations. Thereafter, block **828** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **830** the new parameters are permissible, then block **834** directs the microprocessor **12** to present the new parameters to the user via the display **25**. Block **836** directs the microprocessor **12** to poll the user's response at the keypad **27** for a set period of time. If the user rejects the new parameters, then block **838** directs the microprocessor **12** to cause a DTMP "3A" to be transmitted to indicate to the SPCS **36** that the user has rejected the new configuration. Thereafter, block **828** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **836** the user accepts the new parameters or the polling period expires without user input, then block **840** directs the microprocessor **12** to store the parsed data into the SPP configuration block **260** in FLASH **18**.

Block **842** directs the microprocessor **12** to check the memory transfer, comparing the values stored in RAM **14** to those stored in FLASH **18**. If a memory error has occurred, then block **844** directs the microprocessor **12** to cause a DTMF "4A" to be transmitted to indicate to the SPCS **36** that a memory error has occurred. Thereafter, block **828** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **842** no memory error has occurred, then block **846** directs the microprocessor **12** to ensure that no other error has occurred during this reconfiguration. If another error has occurred, then block **848** directs the microprocessor **12** to cause a DTMF "A" to be transmitted to indicate to the SPCS **36** that an unspecified error has occurred. Thereafter, block **828** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **846** no unspecified error occurred, then block **850** directs the microprocessor **12** to cause a DTMF "B" to be transmitted to indicate to the SPCS **36** that the reconfiguration occurred properly. Thereafter, block **828** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **820** the STA action code **62** is not "$02", then block **851** directs the microprocessor **12** to implement further actions related to such configuration but which are left unspecified as not being part of the present invention. Thereafter, block **828** directs the microprocessor **12** to return to the calling subroutine.

FIG. **26**

With reference to FIGS. **1**, **2**, **9**, **10** and **26**, the MESG subroutine is generally illustrated at **386** and defines an initial state **852** from which the microprocessor **12** begins execution. The MESG Subroutine **386** instructs the CPE **10** how to respond to the MESG family **56** of extended STA codes **40** that configure the CRC, IRN and MSR sessions.

Block **854**, not part of the invention, directs the microprocessor **12** through the process of exchanging stored message data such as pages and e-mails between the CPE **10** and the SPCS **36**. This direction is more fully described in the co-pending patent application entitled, "Method, System and Apparatus for Exchanging Stored Information Between a Telecommunications Server and Customer Premises Equipment," which is incorporated herein by reference.

However, block **856** directs the microprocessor **12** to determine whether STA action code **62** of the most recently polled message is a "$18" to reconfigure the IRN/CRC/MSR parameters. If not, then block **858** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if the STA action code **62** is "$18", then block **860** directs the microprocessor **12** to parse the STA data **64** into its components parts with reference to the "$FF" delimiters (**76**, **80**, **88**, **92** and **96** in FIG. **5**). Block **862** then directs the microprocessor **12** to check the parsed data **64** to make sure that it is consistent with the expected data structure **262**. If not, block **864** directs the microprocessor **12** to cause a DTMF "5A" to be transmitted to indicate to the SPCS **36** that a parsing error has occurred Thereafter, block **858** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **862** the parsed data has the expected structure, then block **866** directs the microprocessor **12** to store the parsed data into the IRN/CRC/MSR configuration block **262** in FLASH **18**.

Block **868** directs the microprocessor **12** to check the memory transfer, comparing the values stored in RAM **14** to those stored in FLASH **18**. If a memory error has occurred, then block **870** directs the microprocessor **12** to cause a DTMF "4" to be transmitted to indicate to the SPCS **36** that a memory error has occurred. Thereafter, block **858** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **868** no memory error has occurred, then block **870** directs the microprocessor **12** to ensure that no other error has occurred during this reconfiguration. If another error has occurred, then block **872** directs the microprocessor **12** to cause a DTMF "A" to the transmitted to indicate to the SPCS **36** that an unspecified error has occurred. Thereafter, block **872** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **870** no unspecified error occurred, then block **874** directs the microprocessor **12** to cause a DTMF "B" to be transmitted to indicate to the SPCS **36** that the reconfiguration occurred properly. Thereafter, block **858** directs the microprocessor **12** to return to the calling subroutine.

FIG. **27**

With reference to FIGS. **1**, **2**, **9**, **10** and **27**, the SCRI subroutine is generally illustrated at **388** and defines an initial state **876** from which the microprocessor **12** begins execution. The SCRI Subroutine **388** instructs the CPE **10** how to respond to the SCRI family **58** of extended STA codes **40** that configure the FSU sessions.

Block **878**, not part of the invention, directs the microprocessor **12** through the process of prioritizing, downloading and executing scripts stored within the CPE **10**.

Block **879** directs the microprocessor **12** to determine whether the STA action code **62** of the most recently polled message is a "$01" to commence a non-interactive Feature Downlodd Management (FDM) session within the current Server Display Control (SDC) session. If so, then block **881** directs the microprocessor **12** to determine whether the CPE-ID encapsulated within the data portion of the STA corresponds to the CPE-ID **339** in FLASH **18**. If so, then block **883** directs the microprocessor **12** to cause a DTMF "B" to be transmitted and then to start an FDM session as per

ADSI ref. SR-2461 page 65, Dec. 92 which is incorporated herein by reference, after which block **885** directs the microprocessor **12** to return from the SCRI subroutine **876**. Alternatively if at block **881**, the CPE-IDs do not match, then block **887** directs the microprocessor **12** to cause a DTMF "3A" to be transmitted and then block **885** directs the microprocessor **12** to return directly from the SCRI subroutine **876**.

Alternatively, if at block **879** the action code **62** is not "01", then block **880** directs the microprocessor **12** to determine whether the received STA action code **62** is a "$06" to reconfigure the CPE's **10** FSU parameters. If not, then block **882** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **880** the STA action code **62** is "$06", then block **884** directs the microprocessor **12** to parse the STA data **64** into its component parts with reference to the "$FF" delimiters (**108**, **112**, **116** and **124** in FIG. **6**). Block **886** then directs the microprocessor **12** to check the parsed data **64** to make sure that it is consistent with the expected data structure **264**. If not, block **888** directs the microprocessor **12** to cause a DTMF "5A" to be transmitted to indicate to the SPCS **36** that a parsing error has occurred. Thereafter, block **882** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **886** the parsed data has the expected structure, then block **890** directs the microprocessor **12** to determine whether the FDM number **102** and security code **104** are as expected. If not, then block **892** directs the microprocessor **12** to cause a DTMF "3A" to be transmitted to indicate to the SPCS **36** that a mismatch error has occurred. Thereafter, block **882** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **890** no mismatch has occurred, then block **894** directs the microprocessor **12** to consider whether these new FSU parameters comply with the limits set by telecommunication regulations **345**, for example with respect to repeated automated calling. If not, then block **896** directs the microprocessor **12** to cause a DTMF "1A" to be transmitted to indicate to the SPCS **36** that the new configuration violates local telecommunication regulations.

Thereafter, block **882** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **894** no regulation violation has occurred, then block **898** directs the microprocessor **12** to store the parsed data into the FSU configuration block **264** in FLASH **18**.

Block **900** directs the microprocessor **12** to check the memory transfer, comparing the values stored in RAM **14** to those stored in FLASH **18**. If a memory error has occurred, then block **902** directs the microprocessor **12** to cause a DTMF "4A" to be transmitted to indicate to the SPCS **36** that a memory error has occurred. Thereafter, block **882** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **900** no memory error has occurred, then block **904** directs the microprocessor **12** to ensure that no other error has occurred during this reconfiguration. If another error has occurred, then block **906** directs the microprocessor **12** to cause a DTMF "A" to be transmitted to indicate to the SPCS **36** that an unspecified error has occurred. Thereafter, block **882** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **904** no unspecified error occurred, then block **908** directs the microprocessor **12** to cause a DTMF "B" to be transmitted to indicate to the SPCS **36** that the reconfiguration occurred properly. Thereafter, block **882** directs the microprocessor **12** to return to the calling subroutine.

FIG. **28**

With reference to FIGS. **1**, **2**, **9**, **10** and **28**, the REMI subroutine is generally illustrated at **389** and defines an initial state **910** from which the microprocessor **12** begins execution. The REMI Subroutine **389** instructs the CPE **10** how to respond to the REMI family **56** of extended STA codes **40** that configure the CRC events stored in FLASH **18**.

Block **912**, not part of the invention, directs the microprocessor **12** through the process of manipulating the CRC events **344** in the CPE FLASH **18** under the direction of the SPCS **36**.

However, block **914** directs the microprocessor **12** to determine whether the STA action code **62** of the most recently polled message is a "01" to download a new CRC event **344**. If not, then block **916** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if the STA action code **62** is "$01", then block **918** directs the microprocessor **12** to parse the STA data **64** into its component parts with reference to the "$FF" delimiters (**139** in FIG. **8**). Block **920** then directs the microprocessor **12** to check the parsed data **64** to make sure that it is consistent with the expected data structure **344**. If not, block **922** directs the microprocessor **12** to cause a DTMF "5A" to be transmitted to indicato to the SPCS **36** that a parsing error has occurred. Thereafter, block **916** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **920** the parsed data has the expected structure, then block **924** directs the microprocessor **12** to store the parsed data into the CRC event **344** section of FLASH **18**.

Block **926** directs the microprocessor **12** to check the memory transfer, comparing the values stored in RAM **14**to those stored in FLASH **18**. If a memory error has occurred, then block **928** directs the microprocessor **12** to cause a DTMF "4A" to be transmitted to indicate to the SPCS **36** that a memory error has occurred. Thereafter, block **916** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **926** no memory error has occurred, then block **930** directs the microprocessor **12** to ensure that no other error has occurred during this reconfiguration. If another error has occurred, then block **932** directs the microprocessor **12** to cause a DTMF "A" to be transmitted to indicate to the SPCS **36** that an unspecified error has occurred. Thereafter, block **916** directs the microprocessor **12** to return to the calling subroutine.

Alternatively, if at block **930** no unspecified error occurred, then block **934** directs the microprocessor **12** to cause a DTMF "B" to be transmitted to indicate to the SPCS **36** that the reconfiguration occurred properly. Thereafter, block **916** directs the microprocessor **12** to return to the calling subroutine.

FIG. **29**

With reference to FIGS. **1** and **29**, the Foreground Task Interrupt Handler is generally illustrated at **390** and defines an initial state **936** from which the microprocessor **12** begins execution. The Foreground Task Interrupt Handler is invoked whenever the keypad **27** or the hook-switch are manually engaged by a user, which indicate to the microprocessor **12** that the user wishes to employ the CPE **10** as a conventional telephone.

Block **938** directs the microprocessor **12** to identify the interrupted process from the stack. The microprocessor **12** is then directed to the Queue Retry/Backup Subroutine **382** to

queue a retry/backup poll to replace the interrupted one. The Queue Retry/Backup Subroutine **382** is designed to look beyond the calling Foreground Task Interrupt Handler **390** to identify the interrupted process on the stack, and to operate on the latter's flag, counters, and timers.

Upon returning from the Queue Retry/Backup Subroutine **382**, the block **940** directs the microprocessor **12** to terminate the interrupted process and block **942** directs the microprocessor **12** to end the Foreground Task Interrupt Handler.

From the foregoing, it can be seen that it is advantageous for a CPE **10** to autonomously call a SPCS **36**. Such autonomous calls can be triggered by a regular schedule, calendar events, telephone calls placed to the CPE **10**, or pages transmitted by the CPS **10**. The calls can be for the purpose of exchanging data, configuring parameters, or updating scripts.

While a specific embodiment of the invention has been described and illustrated, such embodiment should be considered illustrative of the invention only and not as limiting the invention as construed in accordance with the accompanying claims.

What is claimed is:

1. A method of automatically receiving information at customer premises equipment (CPE), the method comprising:

a) measuring idle time elapsed since the CPE was last used;

b) transmitting a current information request message from said CPE to a messaging platform when the idle time is greater than a pre-defined idle time threshold; and

c) receiving information from said messaging platform in response to said information request message.

2. A method as claimed in claim **1** further comprising:

a) maintaining a polling calendar identifying a time at which a scheduled information request message is to be sent to said messaging platform;

b) determining the difference in time between a present time and said time at which an information request message is to be sent; and

c) transmitting said current information request message when said idle time is greater than a pre-defined idle threshold and said difference in time is less than a poll proximity threshold.

3. A method as claimed in claim **2** further comprising:

a) measuring time elapsed since transmitting said information request message and continuing to measure time elapsed while said information is being received; and

b) terminating receiving said information when said time elapsed exceeds a pre-defined time duration.

4. A method as claimed in claim **3** further comprising:

a) detecting whether a hook switch at the CPE is in an on-hook state or an off-hook state; and

b) terminating at least one of transmitting and receiving when said hook switch is changed from the on-hook state to the off-hook state.

5. A method as claimed in claim **1** wherein measuring is initiated in response to CLID information.

6. A method as claimed in claim **1**, wherein measuring is initiated at pre-defined time intervals.

7. A method as claimed in claim **1** wherein transmitting includes the step of dialling a first predefined telephone number.

8. A method as claimed in claim **7** further including transmitting said information request using DTMF signals.

9. A method as claimed in claim **8** further including dialling at least one alternate telephone number when a call cannot be completed to a station associated with said first pre-defined telephone number is unavailable.

10. A method as claimed in claim **1** wherein receiving said requested information includes receiving said information according to the ADSI protocol.

11. A method as claimed in claim **1** wherein measuring is initiated at pre-defined times.

12. A method as claimed in claim **11** wherein measuring is further initiated at pre-defined times after said information is received from said messaging platform.

13. A system for automatically receiving information at customer premises equipment (CPE), the apparatus comprising:

(a) a first timer for measuring idle time elapsed since the CPE was last used;

b) a transmitter for transmitting an information request message from said CPE to a messaging platform when the idle time is greater than a pre-defined idle time threshold; and

c) a receiver for receiving information from said messaging platform in response to said information request message.

14. A system as claimed in claim **13** further including a fourth timer for identifying a fourth pre-defined time, said fourth timer being operable to stimulate said transmitter to transmit said information request message at said fourth pre-defined time when the idle time is greater than a pre-defined idle time threshold.

15. A system as claimed in claim **13** further including a fifth timer for identifying pre-defined time intervals, said fifth timer being operable to stimulate said transmitter to transmit said information request message at said pre-defined time intervals when the idle time is greater than a pre-defined idle time threshold.

16. A system as claimed in claim **13** further including a CLID receiver for receiving CLID information and a processor for determining whether or not said CLID information meets pre-defined criteria and if so, stimulating said transmitter to transmit said information request message when the idle time is greater than said pre-defined idle time threshold.

17. An apparatus as claimed in claim **13** wherein said transmitter includes a dialler.

18. An apparatus as claimed in claim **17** wherein said dialler is operable to dial a first predefined telephone number.

19. An apparatus as claimed in claim **17** wherein said dialler is operable to transmit said information request using DTMP signals.

20. A system as claimed in claim **18** wherein said dialler is operable to dial at least one alternate telephone number when a call cannot be completed to a station associated with said first pre-defined telephone number.

21. An apparatus as claimed in claim **13** wherein said receiver is operable to receive said requested information according to the ADSI protocol.

22. A system as claimed in claim **13** further comprising:

a) a storage medium for maintaining a polling calendar identifying times at which a scheduled information request message is to be sent to said messaging platform;

b) a second timer for measuring a proximity time interval between a present time and the time at which a most

proximate scheduled information request message is scheduled to be transmitted from said CPE to said messaging platform; and

c) a processor programmed to cause said transmitter to transmit an information request message when the idle time is greater than said pre-defined idle threshold and said proximity time interval is greater than a pre-defined proximity threshold.

23. An apparatus as claimed in claim **22** further comprising a third timer for measuring time elapsed since transmitting the information request message from said CPE to said messaging platform and for disabling the transmitter and the receiver when the time measured by said third timer exceeds a pre-defined call timeout threshold.

24. A apparatus as claimed in claim **23** further comprising a detector for detecting whether the hook switch is in an on-hook state or an off-hook state and for disabling the transmitter and the receiver in response to the detector detecting that the hook switch is changed from the on-hook state to the off-hook state.

25. A system for providing messaging services at customer premises equipment (CPE), the system comprising:

a) a messaging platform for receiving and storing messages, said messaging platform including a receiver for receiving information request messages and a transmitter for transmitting information; and

b) customer premises equipment including:

i) a first timer for measuring idle time elapsed since the CPE was last used;

ii) a transmitter for transmitting an information request message from said CPE to said messaging platform when the idle time is greater than a pre-defined idle time threshold; and

iii) a receiver for receiving information from said messaging platform in response to said information request message.

26. A system as claimed in claim **25** wherein said information includes a page.

27. A system as claimed in claim **25** wherein said transmitter in said messaging platform includes an FSK transmitter for transmitting said information in an FSK format to said CPE and wherein said receiver in said CPE includes an FSK receiver for receiving said information in an FSK format.

28. A system as claimed in claim **27** wherein said information is provided to said CPE in accordance with an ADSI protocol extension.

29. A system as claimed in claim **25** wherein said transmitter at said CPE includes a DTMF transmitter and said receiver at said messaging platform includes a DTMF receiver, said information request message being transmitted to said messaging platform in DTMF signals.

30. A system as claimed in claim **25** wherein said information includes an e-mail message.

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